

# High Scores, Fragile Targets: Auditing Change in LLM-Annotated Diet-Health Evidence

Anonymous ACL submission

## Abstract

When a benchmark derives both inputs and outcomes from automatic annotations, high discrimination can reflect label composition and representation choices rather than annotation-stable predictive value. We present NutriTURN, a benchmark and audit framework built from 112,453 PubMed claim-record associations for 165 diet-health claims, annotated in full by two LLM pipelines, with a calendar-respecting protocol that separates three phenomena. Composition-driven ranking: an untrained margin rule reaches AUROC 0.881, and an exact pair decomposition places about 91% of a signed logistic score’s estimated +0.067 gain in comparisons across pre-cutoff majority classes. Representation-dependent failure: under a second annotation the score loses to the margin rule, 82% of the loss falls within classes, and replacing its two-branch agreement input by a signed-only one raises its AUROC from 0.758 to 0.940 on common support without changing the outcomes, while the same input establishes no development gain. Annotation-dependent targets: the two annotations share 27 of 96 positive events, and the second agrees less with an annotation-independent numeric anchor. No calendar-cohort gain is established, the score loses on one external domain, and the excess of reversals over era-conditioned controls is confined to the primary annotation and depends on the temporal partition.

## 1 Introduction

Systems that retrieve, extract and verify biomedical evidence are increasingly asked a temporal question: given what has been published so far, is the evidence on a claim likely to change? Review-update prediction (Bashir et al., 2019, 2021; Alharbi and Stevenson, 2026) and living reviews (Elliott et al., 2014; Nevill et al., 2025) motivate the question, and early findings that later attenuate or reverse are documented (Ioannidis, 2005; Ioannidis and Trikalinos, 2005; Prasad et al., 2013). At the scale

of a literature such datasets are built from automatic annotation: a model labels every document, labels before a cutoff become the inputs, and a thresholded aggregate of labels after it becomes the outcome. This construction admits high scores with no predictive value beyond what the labels already carry: under an exchangeable label process the outcome’s probability is a closed-form function of the pre-cutoff counts (Lemma 1), so composition alone ranks it well; a signed input can learn which label class tends to change under one annotator, a class prior with no guarantee of transfer; an input whose definition switches on a threshold can fail when a second annotator moves units across it; a second annotator can define a different set of positive events; and retrospective evaluation lets later outcomes shape the model applied at a cutoff. A benchmark that does not separate them rewards them (Gururangan et al., 2018; Poliak et al., 2018; McCoy et al., 2019).

We present NutriTURN,<sup>1</sup> a benchmark for forecasting evidence-direction change built to test these explanations separately: a longitudinal PubMed stream of 165 exposure-outcome claims, two complete LLM annotations of all 112,453 claim-record associations, an origin-defined calendar cohort, two frozen external domains and null controls, with a protocol that asks whether a gain over an analytic composition reference holds across annotations, in calendar time, within majority classes and under transfer (Section 4). Applying it to composition references, learned scores and prompted LLM forecasters yields four findings (Section 5) that separate three phenomena: (1) label composition explains most of the ranking; (2) the learned gain is concentrated across classes: an exact pair decomposition places about 91% of it in pairs with different pre-cutoff majority classes, a class-rate difference (five

<sup>1</sup>Data, code, and results: <https://anonymous.4open.science/r/NutriTURN-0C33>.

to one) that persists within the reported margin strata; (3) the shipped score’s failure under a second annotation is largely representation-dependent: 82% of its loss falls within classes, and replacing its two-branch agreement input by a signed-only one repairs the loss (0.758 to 0.940 on common support) without changing the outcomes, while the same input establishes no development gain; no gain is established in the calendar cohort and a loss is established on one external domain; (4) repairing the predictor does not repair the target: the two annotations share 27 of 96 positive events, and the one that redefines the positive class most agrees less with an annotation-independent numeric anchor. Our contributions are (i) the benchmark; (ii) a five-check protocol with fixed primary cells and tiers; and (iii) diagnostics that locate where a ranking comes from (composition references, pair decomposition, class-blind and branch-free variants, sign exchanges, null controls, a numeric anchor for both annotations).

## 2 Related Work

**Evidence change and review updating.** Conclusion-change prediction for systematic-review updates (Bashir et al., 2019, 2021) was scaled by Alharbi and Stevenson (2026) to 3,326 Cochrane review-update pairs; it predicts change *given* the new records, whereas our task uses records published before the origin only; conclusiveness assessment (Ban et al., 2026) and living-evidence automation (Song et al., 2026; Harasgama et al., 2026) address the same problem. Trialstreamer (Marshall et al., 2020) and RobotReviewer (Marshall et al., 2016) build evidence sets, Evidence Inference (Lehman et al., 2019; DeYoung et al., 2020) labels effect direction per record, as our label does, and SciFact (Wadden et al., 2020), PUBHEALTH (Kotonya and Toni, 2020) and HealthVer (Sarrouti et al., 2021) decide current support. Our target is a vote-count sign change (Hedges and Olkin, 1980), not a cumulative meta-analytic estimate (Lau et al., 1992; Pogue and Yusuf, 1997; Wetterslev et al., 2008).

**Forecasting and temporal evaluation.** ForecastQA (Jin et al., 2021), Autocast (Zou et al., 2022) and retrieval-augmented LLM forecasters (Halawi et al., 2024) forecast from dated text; retrospective evaluation is vulnerable to temporal leakage (Paleka et al., 2025), and TemporalWiki, StreamingQA and TempLAMA (Jang et al., 2022;

Liska et al., 2022; Dhingra et al., 2022; Lazaridou et al., 2021) evaluate models against knowledge dated after training. Our calendar cohort addresses the same risk for a target in which the annotator, not the world, decides what counts as a change.

**Shortcuts, label quality and LLM annotation.** Hypothesis-only baselines (Gururangan et al., 2018; Poliak et al., 2018), HANS (McCoy et al., 2019), contrast sets (Gardner et al., 2020) and adversarial collection (Nie et al., 2020) expose decision rules that succeed on a benchmark and fail off it (Geirhos et al., 2020), and models can learn the annotators who wrote the data (Geva et al., 2019). Those shortcuts are input artefacts; here the target itself carries little beyond composition, what it carries can be specific to one annotator, and what a score extracts from it can be specific to one feature definition. Annotator disagreement is often signal (Pavlick and Kwiatkowski, 2019; Plank, 2022), distant labels trade noise for scale (Mintz et al., 2009), label corrections can reorder rankings (Northcutt et al., 2021), LLM annotators require task-specific validation (Gilardi et al., 2023; Ziems et al., 2024; Pangakis et al., 2023), and configuration choices change downstream conclusions (Baumann et al., 2025). Measuring inputs and outcome with one instrument is a known source of shared error (Podsakoff et al., 2003); we show its temporal consequence.

## 3 The Benchmark

**Stream.** Each of 165 claims  $c = (e, o)$  pairs an exposure with an outcome and has one fixed PubMed query over MeSH and title/abstract terms, without date or publication-type filters. The archived snapshot holds 112,453 claim-record associations over 96,091 distinct identifiers; the claim list is a curated artefact of 145 candidate claims and 20 documented additions (Appendix A), which contribute 6 of the 71 events in two claims and so do not seed the result. The harvest is bounded to 9-10 September 2026, and the archived identifier sets, not the queries, define the benchmark: re-running the stored queries returns 91.7% of the archived associations; for 19 claims, flagged in the release, the stored query now returns nothing and is not confirmed to be the harvesting query; and 48 claims were truncated by a per-query record cap that removed their early history. Removing either set changes no protocol conclusion (Appendix C.3). Coverage has a concrete cost: of eight claims with

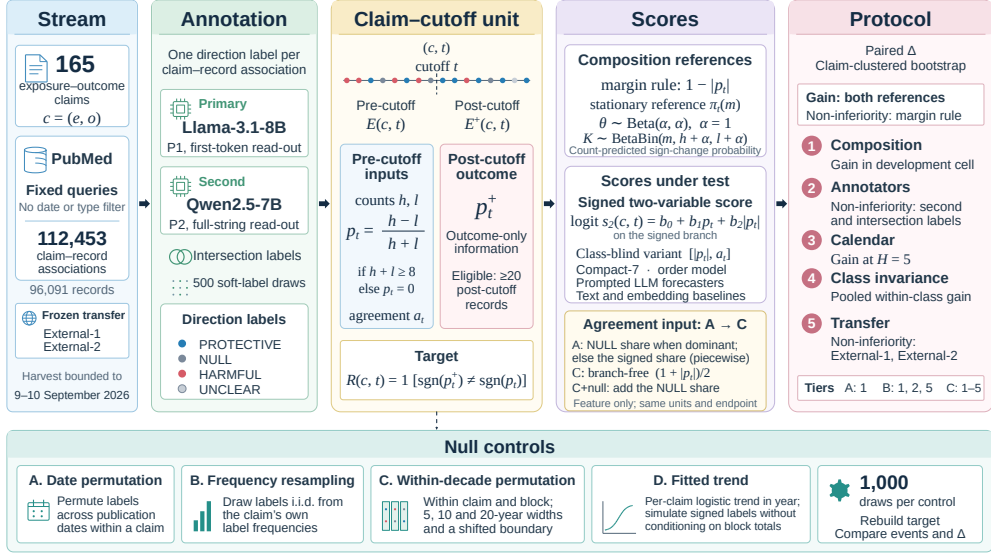


Figure 1: **NutriTURN: construction and evaluation.** A frozen PubMed stream of 165 claims is annotated in full by two configurations that differ in model, prompt and read-out. Each claim-cutoff unit  $(c, t)$  takes its inputs from pre-cutoff labels, its future direction from post-cutoff labels only, and its endpoint  $R(c, t)$  from the comparison of the two signs (red HARMFUL, blue PROTECTIVE, grey NULL). Scores are compared with the composition references by paired  $\Delta$ AUROC; Tier A licenses a retrospective gain over composition, Tier B adds non-inferiority across annotations and domains (at a fixed margin, not robustness), Tier C adds the calendar and class-invariance checks (Section 4). The bottom band shows the four null controls and the A $\rightarrow$ C agreement-input intervention.

widely described reversals, three observable ones have no eligible cutoff before the pivotal trial and no trial report in their archive, so no forecast could have been posed for them (Appendix H). Source records are released by PMID, the external corpora as unit-level label counts only.

**Annotations.** Every claim-record association carries a direction label (PROTECTIVE, NULL, HARMFUL or UNCLEAR) relative to the named exposure and outcome, not to clinical desirability (benefit-valence outcomes invert the label names; removing such claims leaves the result unchanged, Appendix L). The *primary* pipeline reads title and abstract with Llama-3.1-8B-Instruct (Grattafiori et al., 2024) and compares the first-token logits of the four labels. The *second* pipeline re-annotates every association with Qwen2.5-7B-Instruct (Qwen Team, 2024) under the same task framing and text budget, with explicit NULL and UNCLEAR definitions and full label-string probabilities, so the configurations differ in model, prompt wording and read-out (Appendix C.1). *Intersection* labels keep a label only where the pipelines agree (disagreements become UNCLEAR), and 500 *soft-label draws* sample labels from the second pipeline’s probabilities. An LLM-independent numeric anchor agrees with the primary labels on 0.807 [0.782, 0.830] of 1,841 high-

confidence records (1.6% of associations) and with the second pipeline’s on 0.758 [0.733, 0.782]; the anchor is a selected consistency check, not corpus-wide accuracy (Appendix B.2).

**Endpoint.** For a cutoff year  $t$ , let  $E(c, t)$  hold the claim’s records published before  $t$  and  $E^+(c, t)$  those published at or after  $t$ ; the latter never enter an input. With  $h$  and  $l$  the numbers of HARMFUL and PROTECTIVE records in  $E(c, t)$ , the pooled direction is  $p_t = (h - l)/(h + l)$  when  $h + l \geq 8$  and 0 otherwise. The future direction  $p_t^+$  is the same ratio over the signed records of  $E^+(c, t)$  with no minimum count (0 when none follows; every eligible source unit has at least 11), and the endpoint is

$$R(c, t) = \mathbf{1}[\text{sgn}(p_t^+) \neq \text{sgn}(p_t)], \quad (1)$$

with  $\text{sgn}(0) = 0$ . NULL records are excluded from  $p_t$ , so  $|p_t|$  near zero means a balance of signed labels, not the absence of an association. A zero direction on exactly one side is an event, so 11 of the 71 events are such units (seven pre-cutoff exact ties, one sparse history, three post-cutoff ties); nothing bounds the later majority, and the horizon runs to the end of the corpus. The endpoint stays primary because every cell was computed on it; tie-free, minimum-signed-count and fixed-horizon variants reproduce the development ranking (Appendix H).

A unit is eligible when at least 20 records follow the cutoff: 1,012 eligible units in 160 claims with 71 events in 23 claims (Appendix A), with repeated cutoffs of a claim clustered in every interval.

**Evaluation populations.** The *development* cell is a retrospective claim-held-out backtest on the 1,012 eligible units, with training outcomes extending to the end of the corpus. The *calendar cohort* is origin-defined: at origins  $Y \in \{2004, 2007, \dots, 2022\}$  and horizons  $H \in \{3, 5, 10\}$ , a claim is a candidate if it has at least 40 records before  $Y$ , the window  $[Y, Y + H)$  must hold at least ten signed records, training is claim-disjoint and uses only outcomes observable by  $Y$ , and only windows ending by 2025 are evaluated (Appendix E.2); the design is calendar-respecting for documents, not for the annotator. *Frozen transfer* fits once on the source and applies the score, its calibrator and its count model unchanged to External-1 (physical activity, sleep, environment) and External-2 (medication, screening, occupational exposures, clinical risk states).

## 4 Evaluation Protocol

### 4.1 The composition references

The *margin rule* scores a unit by  $1 - |p_t|$  and needs no training. The *stationary reference* gives the endpoint’s probability if a claim’s signed labels were exchangeable: suppose they are conditionally i.i.d. given  $\theta = P(\text{HARMFUL})$  with  $\theta \sim \text{Beta}(\alpha, \alpha)$ , and let  $K$  count HARMFUL labels among the next  $m$  signed records.

**Lemma 1** (stationary sign change). *Given  $(h, l, m)$ ,  $K \sim \text{BetaBin}(m, h + \alpha, l + \alpha)$ , and the probability that the future majority differs from  $\text{sgn}(h - l)$  is*

$$\pi_t(m) = \sum_{k \in \mathcal{K}_t} \text{BetaBin}(k; m, h + \alpha, l + \alpha), \quad (2)$$

with  $\mathcal{K}_t = \{k : \text{sgn}(2k - m) \neq \text{sgn}(h - l)\}$ . For fixed  $n = h + l$ ,  $m$  and  $\alpha$ ,  $\pi_t$  is non-increasing in  $|h - l|/n$  among non-tied histories, so when  $n$ ,  $m$  and  $\alpha$  are common across units the margin rule ranks units as  $\pi_t$  does.

The proof and numerical checks are in Appendix D. Equation (2) is stated for the raw majority  $\text{sgn}(h - l)$ ; the scored reference replaces it by the operational sign  $\text{sgn}(p_t)$  of Equation (1), which differs only on sparse histories, where  $p_t$  is gated to 0 (Appendix D). The analytic reference is

$\pi_t$ ; the margin rule is its untrained surrogate, exact only under common  $(n, m, \alpha)$ , so both are reported. The *realised-count* version inserts the observed future  $m$  (a diagnostic); the *count-predicted* version averages  $\pi_t$  over draws of  $m$  from a count model fitted on the training claims; both use  $\alpha = 1$ .

### 4.2 Scores under test

The *signed two-variable score* is a class-balanced  $L_2$  logistic regression on standardised  $[p_t, a_t]$ , where  $a_t$  is the majority share among signed records, or the NULL share when NULL records dominate. On the signed branch  $a_t = (1 + |p_t|)/2$ , so its logit is

$$\text{logit } s_2(c, t) = b_0 + b_1 p_t + b_2 |p_t|, \quad (3)$$

and  $b_1$  is the only term through which the identity of the majority class enters. Its *class-blind variant* is the same model on  $[|p_t|, a_t]$ . The shipped agreement is *definition A*, the archived reference: the NULL share when that share is at least 0.5, else the signed majority share. The *branch-free variant* uses *definition C*, the signed-only share  $(1 + |p_t|)/2$  on every unit (gated to 0 on sparse histories, as A is; Appendix K); a matched diagnostic intervention that changes the representation and nothing else and inherits none of A’s other results. Richer learned scores (Compact-7 on seven variables, eleven learners on all 55 pre-cutoff variables, order features added to the stationary logit) stay at or below it (Appendices D.2 and N); prompted LLM forecasters read a digest of the pre-cutoff labels (Appendix Q).

Four *null controls* rebuild the target from permuted or simulated labels (1,000 draws each, score refitted; Appendix G): Null A permutes each claim’s labels across its publication dates, Null B draws each label i.i.d. from the claim’s label frequencies, Null C permutes within claim and decade, keeping each claim’s era-level label mix, and Null D draws labels from a per-claim logistic trend of the HARMFUL share on year, keeping the fitted drift without block conditioning.

### 4.3 Checks, cells and tiers

Each check has a primary cell and a pass condition on  $\Delta$ , the paired AUROC difference between a score and a composition reference, with a claim-clustered percentile bootstrap of 2,000 resamples (Efron and Tibshirani, 1993); a difference is *established* when its 95% interval excludes zero.

With 23 positive clusters, percentile intervals can under-cover (Cameron et al., 2008), so “not established” means insufficient evidence, never evidence of absence; a wild cluster bootstrap- $t$  with Webb weights, a clustered-ROC sensitivity and a claim-level bootstrap that refits the score in every resample leave every tested superiority and loss decision unchanged, and one non-inferiority subdecision (External-1) procedure-sensitive (Appendix E.1), which shows stability to those choices and nothing more. A superiority check needs an established gain over both the margin rule and the count-predicted stationary reference; a non-inferiority check is judged against the margin rule. We fixed the cells after observing the baseline results of Section 5, for which they are a post hoc description.

1. **Composition.** In the development cell,  $\Delta$  is established as positive against both references.
2. **Annotators.** Under the second annotation and intersection labels,  $\Delta$  against the margin rule is non-inferior: its lower 95% bound exceeds  $-\delta$ ,  $\delta = 0.02$ , a stated convention below the smallest established difference in Table 2 (0.028).
3. **Calendar.** In the calendar cohort at  $H = 5$ ,  $\Delta$  is established as positive against both references, with no established loss at  $H = 3$  or  $H = 10$ .
4. **Class invariance.** Over pairs of units that share a pre-cutoff majority class ( $p_t = 0$  excluded),  $\Delta$  is established as positive against both references (pooled), and the point difference is non-negative within each class; the class-blind variant is a diagnostic.
5. **Transfer.** In both external domains  $\Delta$  against the margin rule is non-inferior at the same  $\delta$ . Passing all five establishes predictive value beyond composition, class information and one annotator, not a legitimate forecaster. Checks 2 and 5 are non-inferiority conditions because superiority is barely testable there (the margin rule already reaches 0.953 under the second annotation; External-1 has 19 events); the margin  $\delta$  makes a wide interval fail rather than pass. A score reaches *Tier A* (development gain) by passing check 1, *Tier B* (development gain, non-inferior across annotations and domains) by passing checks 1, 2 and 5, and *Tier C* (class-invariant calendar gain) by passing all five with a development  $\Delta$  above the 97.5th percentile of the paired differences the same score attains under Null C. For a method fixed before its outcomes on the released cells are examined, checks 1, 3 and 4 form an intersection-union test at joint

level 5% (Berger, 1982); the baselines here, developed against the released outcomes, forfeit that guarantee. Label-feature and text-reading methods form two *tracks* with different eligible checks (Appendix S).

## 5 Results

### 5.1 Composition explains most of the ranking

Table 1 gives the development cell. The untrained margin rule ranks later change almost as well as any score; the stationary reference with realised counts is indistinguishable from it, as Lemma 1 anticipates, and its count-predicted form gains an established 0.028 from the accrual model, not from Equation (2). The lowest quintile of  $|p_t|$  holds 54 of the 71 events (Appendix H). Richer inputs do not move beyond two variables: Compact-7 matches the signed score to within 0.002, eleven learners on all 55 variables stay at or below it, and the order features add nothing established (Appendix N and Table 19); shared identifiers and single positive claims do not drive the development conclusion (Appendix M).

### 5.2 Most of the learned gain is a majority-class prior

Table 3 locates the gain. An exact decomposition of the signed-minus-margin AUROC difference by pair type places +0.0609 [0.0159, 0.1081], about 91% of the estimated total, in positive-negative pairs whose units have different pre-cutoff majority classes; the additive within-class contribution, +0.0068 [-0.0039, 0.0195], is not established. The incremental gain is therefore concentrated in across-majority-class comparisons, and the class-rate difference persists within the reported margin strata: HARMFUL-majority units change about five times as often as PROTECTIVE-majority units, HARMFUL majorities are not thinner (median  $|p_t|$  0.588 against 0.612), the class odds ratio adjusted for  $|p_t|$  is 7.80 [2.01, 33.82], and no PROTECTIVE-majority event is observed above the lowest margin quintile (Table 33). On the signed branch Equation (3) is linear in  $|p_t|$  within a class, with slope  $b_2 + b_1$  for HARMFUL and  $b_2 - b_1$  for PROTECTIVE majorities; the full-data fit gives  $\hat{b}_2 = -12.2$ , so both slopes are negative ( $-6.9$  and  $-17.5$ , in every leave-one-claim-out fold), and  $\hat{b}_1$  encodes that HARMFUL majorities change more often. The class-rate difference has an era-level correlate: under the primary annotation the HARMFUL share

| Score                               | AUROC [95% CI]    | AUPRC | Brier | Log loss | Recall@10% | $\Delta$ vs. margin [95% CI] |
|-------------------------------------|-------------------|-------|-------|----------|------------|------------------------------|
| Margin rule 1 – $ p_t $ (untrained) | .881 [.824, .936] | .468  | .0515 | .184     | .549       | reference                    |
| Stationary, count-predicted         | .909 [.867, .949] | .497  | .0499 | .200     | .549       | + .028 [.008, .050]          |
| Signed two-variable ( $p_t, a_t$ )  | .948 [.921, .971] | .535  | .0452 | .146     | .676       | + .067 [.017, .120]          |
| Class-blind ( $ p_t , a_t$ )        | .897 [.839, .945] | .445  | –     | –        | –          | + .016 [–.017, .057]         |

Table 1: Development cell: 1,012 eligible units, 71 events in 23 claims (prevalence 0.070), primary annotation, leave-one-claim-out. The margin rule is untrained as a ranking; its Brier and log loss, like those of the learned rows, use a training-only Platt mapping (Platt, 1999) of the score to a probability, whereas the stationary row is uncalibrated (recalibrated: Table 19).  $\Delta$  is paired over 2,000 claim resamples; “–”: not computed. The realised-count stationary diagnostic (.882, +.000 [–.033, .034]) and the branch-free variant (.925, +.044 [–.005, .091]) are in Tables 2 and 18; Compact-7 (.946), the order model (.915) and eleven further learners: Appendices E.1 and N.

| Check         | Cell                                 | Units (ev.) | Margin | Score | $\Delta$ vs margin [95% CI] | Stationary vs margin [95% CI] |
|---------------|--------------------------------------|-------------|--------|-------|-----------------------------|-------------------------------|
| 1 Composition | Development, A                       | 1,012 (71)  | .881   | .948  | + .067 [.017, .120]         | + .028 [.008, .050]           |
|               | Development, C                       | 1,012 (71)  | .881   | .925  | + .044 [–.005, .091]        | + .028 [.008, .050]           |
| 2 Annotators  | Second, A                            | 1,012 (52)  | .953   | .857  | – .096 [–.166, –.037]       | + .010                        |
|               | Second, common supp., A <sup>†</sup> | 1,002 (42)  | .942   | .758  | – .184 [–.268, –.105]       | + .012                        |
|               | Second, common supp., C <sup>‡</sup> | 1,002 (42)  | .942   | .940  | – .002 [–.037, .023]        | –                             |
|               | Second, C                            | 1,012 (52)  | .953   | .952  | – .001 [–.028, .018]        | –                             |
|               | Intersection, A                      | 1,012 (55)  | .928   | .898  | – .030 [–.094, .023]        | + .019                        |
|               | Intersection, C                      | 1,012 (55)  | .928   | .935  | + .007 [–.016, .030]        | –                             |
| 3 Calendar    | H=3                                  | 756 (58)    | .907   | .913  | + .006 [–.012, .023]        | + .013 [–.002, .031]          |
|               | H=5 (primary cell)                   | 625 (45)    | .897   | .920  | + .023 [–.005, .053]        | + .003 [–.021, .030]          |
|               | H=10                                 | 492 (35)    | .867   | .876  | + .008 [–.032, .047]        | – .004 [–.040, .034]          |
| 4 Class inv.  | Within PROT-majority                 | 730 (21)    | .959   | .969  | + .011 [–.001, .024]        | –                             |
|               | Within HARM-majority                 | 274 (42)    | .833   | .863  | + .030 [–.041, .114]        | –                             |
|               | Pooled within-class pairs            | 1,004 (63)  | .909   | .927  | + .018 [–.012, .060]        | –                             |
| 5 Transfer    | External-1, frozen                   | 322 (19)    | .893   | .948  | + .055 [–.052, .106]        | + .013 [–.010, .039]          |
|               | External-2, frozen                   | 1,257 (95)  | .896   | .823  | – .074 [–.123, –.024]       | + .002 [–.014, .017]          |

Table 2: The protocol applied to the signed two-variable score (“Score”; AUROC; bold: established). A: the shipped two-branch agreement input; C: the signed-only (branch-free) input, a matched intervention that changes nothing else (Appendix K). Annotation rows rebuild inputs and targets and refit; calendar rows refit per origin; external rows apply the source-fitted model unchanged. Stationary AUROCs are in Table 18 and Appendix C.1; entries without an interval are point differences of the reported AUROCs; against the stationary reference the shipped score gains +0.039 [0.004, 0.080] in development. Of checks 2 and 5, only the intersection C row passes  $\delta = 0.02$ ; the second-annotation C row misses it by 0.008, the common-support C row by 0.017, and every A row by at least 0.03. <sup>†</sup>At least eight signed pre-cutoff records under both pipelines, refitted (unrefitted: .823, –.119). <sup>‡</sup>The row above with A replaced by C on the same units and outcomes; C – A is +0.182 [0.120, 0.248] (Table 14).

of signed labels falls within claims by  $-0.042$  [–0.061, –0.023] per decade (Appendix C.2), so an early HARMFUL majority is the one most likely to be outvoted later, and a control preserving this drift reproduces the gain (Section 5.5). Whether the drift is the annotator’s is not settled: on anchored records from 2000 onward the within-claim slopes of the HARMFUL share are negative for the anchor and both pipelines ( $-0.053$ ,  $-0.070$ ,  $-0.043$  per decade), but only the primary’s interval excludes zero and the paired slope differences cover zero (Table 9).

### 5.3 The learned gain is not consistent across conditions

Table 2 applies the five checks. The signed score passes check 1 against both references and no other check; the calendar cells establish no gain,

External-2 establishes a loss, and the loss under the second annotation is largely representation-dependent.

*Annotators.* Under the second annotation the ordering against the margin rule reverses: rebuilt and refitted, the shipped score falls below the margin rule by an established margin, while  $\hat{b}_1$  keeps its sign and shrinks (+5.3 to +2.1); the loss survives restriction to the common support and follows the annotator model (Table 15); 82% of it arises within majority classes, so it is not the across-class advantage reversing. Agreement construction contributes substantially to it. On the 1,002-unit cohort with adequate signed support under both pipelines, replacing the two-branch agreement input (A) by the signed-only share (C) raises the second-annotation AUROC

| Quantity                                                                | Value [95% CI]                      |
|-------------------------------------------------------------------------|-------------------------------------|
| <i>Pair decomposition of <math>\Delta\text{AUROC}</math> (Table 22)</i> |                                     |
| Across-class pairs (34,650)                                             | +0.0609 [0.0159, .1081]             |
| Within-class pairs (24,633)                                             | +0.0068 [−.0039, .0195]             |
| Zero-direction pairs (7,528)                                            | −.0010 [−.0032, −.0001]             |
| Across-class share of the gain (.0667)                                  | 91%                                 |
| Within-class loss, 2nd ann. (−.0957)                                    | −.0781 [−.1247, −.0386] (82%)       |
| Event rate, HARM-majority                                               | 42/274 (15.3%)                      |
| Event rate, PROT-majority                                               | 21/730 (2.9%)                       |
| Median $ p_t $ , HARM / PROT                                            | 0.588 / 0.612                       |
| Class odds ratio, adj. for $ p_t $                                      | 7.80 [2.01, 33.82]                  |
| Fitted $\hat{b}_1 / \hat{b}_2$ (Equation (3))                           | +5.3 (+3.0 std.) / −12.2            |
| Slope in $ p_t $ , HARM / PROT                                          | −6.9 / −17.5 (< 0 in 160/160 folds) |
| Signed — class-blind, all pairs                                         | +0.051 [.013, .092]                 |

Table 3: Where the learned gain comes from (development cell, primary annotation; differences against the margin rule are in Table 2). The decomposition partitions the 66,811 positive-negative pairs by whether the two units share a pre-cutoff majority class; contributions are additive and sum to the unrounded totals (+0.0667, −0.0957), intervals are claim-clustered, and the shares are descriptive. HARM, PROT: pre-cutoff majority class ( $p_t \neq 0$ ; 1,004 units, 63 events). Odds ratio: class-balanced logistic regression on class and  $|p_t|$ , claim-clustered (crude 6.11, Mantel-Haenszel 4.35; Table 33); slopes:  $\hat{b}_2 \pm \hat{b}_1$  over leave-one-claim-out folds. Sign-exchange and orientation-clean results: Appendices D.3 and L.

from 0.758 to 0.940, a paired improvement of +0.182 [0.120, 0.248], against a margin rule of 0.942 on the same cohort and outcomes (−0.002 [−0.037, 0.023] against it, which misses  $\delta$  by 0.017), so the point deficit nearly disappears with no outcome changed. The improvement is largest on the NULL-dominant branch, on which  $a_t$  is a NULL share (+0.246 [0.073, 0.405], against +0.041 on the signed branch; Table 14). Refitted on the full cohort, the branch-free score’s difference from the margin rule is −0.001 [−0.028, 0.018] under the second annotation and +0.007 [−0.016, 0.030] under intersection labels: no loss is established in these two annotation comparisons, and non-inferiority at  $\delta$  is established for intersection labels but not for the second annotation (a representation effect within that setting, not a proof that annotation-induced signed membership plays no role). On the primary annotation the branch-free score establishes no development gain, +0.044 [−0.005, 0.091], or +0.051 [−0.006, 0.108] with the NULL share as a third input, but its direct difference from the shipped score is not established either (−0.023 [−0.050, 0.002]; Table 37), so the branch is not shown to carry the primary signal; what is shown is that the only established gain over composition in Table 2 comes from an input

whose definition does not port across annotations. Separately, the second annotation agrees less with the numeric anchor (Section 3): −0.049 [−0.068, −0.030] against the primary, −0.059 [−0.081, −0.036] on the ratio-scale subset; among signed anchors it assigns NULL more often, 9.4% against 4.2%, while opposite-sign assignments have similar frequencies, 7.9% against 7.6% (Table 8): a plausible route into the NULL-dominant branch, not a demonstrated mediation, and no verdict on which pipeline defines the correct claim-level events.

*Calendar.* In the origin-defined cohort no difference is established at any horizon (Appendix E.2); at five years the estimated advantage is +0.023 [−0.005, 0.053], neither a gain nor a loss. No tested text-reading baseline (prompted 70B and 72B models, abstract embeddings) establishes a gain over composition (Appendix R).

*Class invariance and transfer.* Check 4 is not passed: over pairs that share a majority class the gain is not established, in either class or pooled, and the class-blind diagnostic shows no established gain (Table 3). In External-1 the signed score’s estimate, +0.055 [−0.052, 0.106], establishes neither superiority nor non-inferiority; in External-2 it loses by an established margin in AUROC and in AUPRC (Appendix F), so the prior does not transfer frozen; refitted inside External-2 the same two inputs lose only −0.017 [−0.047, 0.009], so the gap is a transfer loss, not an inability of the inputs to rank that domain. The signed score reaches Tier A only; its branch-free form reaches no tier; no tested score reaches Tier B or Tier C.

#### 5.4 Annotation configuration can redefine the positive class

The two annotations agree on most labels yet disagree on most positive events (Table 5): 93.2% of units receive the same endpoint value, mostly shared non-events, while the positive class is largely redefined, on common support, under intersection labels and across soft-label draws alike. A controlled re-annotation that changes one factor at a time (Table 15) attributes most of the redefinition to the model family rather than the prompt or the read-out, and a re-run of the primary configuration with a different batch size alone moves the event set (71 against 73 events, Jaccard 0.846). Agreement on labels does not imply agreement on a thresholded target (Appendix I); the configuration that redefines the positive class most is also the one that agrees less with the numeric anchor (Section 5.3),

| Quantity                        | Obs.  | Null A                        | Null B                        | Null C            | Null D       |
|---------------------------------|-------|-------------------------------|-------------------------------|-------------------|--------------|
| Events, primary                 | 71    | 35.6 [26,46]*                 | 36.5 [20,52]*                 | 61.2 [53,69]*     | 70.1 [54,87] |
| Events, second                  | 52    | 34.3 [24,45]*                 | 36.0 [21,51]*                 | 51.3 [45,57]      | –            |
| Events, intersect.              | 55    | 40.2 [29,51]*                 | 40.9 [25,56]                  | 55.7 [49,62]      | –            |
| Signed AUROC <sup>‡</sup>       | .948  | .948 [.907,.974]              | .947 [.898,.975]              | –                 | –            |
| Margin AUROC <sup>‡</sup>       | .881  | .957 [.929,.977] <sup>†</sup> | .957 [.924,.979] <sup>†</sup> | –                 | –            |
| $\Delta$ , primary <sup>‡</sup> | +.067 | –.009*                        | –.010*                        | +.065 [.047,.083] | +.040        |

Table 4: Observed streams against the null controls (null mean [95% range], 1,000 draws;  $\Delta$ : signed minus margin, paired within draws). \*observed above the 95% null range ( $p < .025$ ; intersection under Null B  $p = .054$ ); <sup>†</sup>below it ( $p \leq .002$ ); <sup>‡</sup>archived run (Null A/B event means 35.3, 36.1); other cells: reimplementation (Appendix G).

| Annotation                 | Events (cl.) | Shared | Jaccard | Sparse | Prim.-trained |
|----------------------------|--------------|--------|---------|--------|---------------|
| Primary (Llama)            | 71 (23)      | 71     | 1.000   | 1      | .948          |
| Second (Qwen)              | 52 (16)      | 27     | .281    | 10     | .878          |
| Intersection               | 55 (19)      | 30     | .313    | 14     | .869          |
| Common support, prim./sec. | 67 / 42      | 23     | .267    | –      | –             |
| Soft-label draws (500)     | 43-57        | –      | –       | –      | –             |

Table 5: Positive events under each annotation of the same 1,012 units (label agreement 67.0%,  $\kappa = 0.52$ ; pooled directions  $r = 0.943$ ; Appendix C.1). “Shared” and Jaccard are relative to the primary annotation; “Sparse”: events on histories with fewer than eight signed records; “Prim.-trained”: the primary-trained score evaluated unchanged against each endpoint.

evidence of a specific annotation discrepancy on the covered records, not of which pipeline defines the correct events; and repairing the predictor’s representation changes none of this, since the 27 shared events are a property of the targets. Annotation configuration changes hypothesis-test outcomes (Baumann et al., 2025); here it changes the positive class.

## 5.5 Excess reversals and the drift behind them

The composition-preserving controls (A and B; Table 4) reproduce the fitted discrimination but not the events: the streams reverse about twice as often, an excess that also appears under the second annotation and, against Null A, under intersection labels. Null C absorbs about 72% of the primary excess and all of it under the other two annotations, and reproduces the signed score’s paired gain, which Nulls A and B do not; the observed +0.067 lies below its 97.5th percentile (0.083), so the Tier C null condition also fails. Null C conditions on claim-decade totals; only 9.6% of units, holding 3 of the 71 primary events, have counts no permutation can change (Appendix G), and what it leaves is partition-dependent: with five-year blocks the observed count falls inside the null range (64.9 [59, 71],  $p = 0.027$ ), with shifted boundaries like-

wise (64.5 [58, 71],  $p = 0.038$ ), with twenty-year blocks the gap widens (56.3 [48, 64],  $p = 0.002$ ), and the second-annotation and intersection counts lie inside the range in all four partitions (Table 31). Null D, a retrospective control that keeps each claim’s fitted HARMFUL-share trend without block conditioning, is not rejected on the count (70.1 [54, 87],  $p = 0.458$ ) or on the gain (+0.040 against +0.067,  $p = 0.099$ ), without being shown to be the generative explanation. The excess over era-conditioned controls is confined to the primary annotation and is a partition-specific contrast, not an identified set of unexplained reversals; across partitions, learned-score differences are evaluated under the primary annotation only.

## 6 Conclusion

On a twice-annotated PubMed stream, an untrained composition rule ranks later change at 0.881; about 91% of the best learned gain lies in comparisons across majority classes; most of that score’s loss under a second annotation is repaired by a signed-only agreement input that leaves the outcomes untouched, while that input establishes no development gain; and the two annotations share 27 of 96 positive events. A forecasting benchmark built from automatic annotations must therefore separate why a score succeeds, why it fails and whether its events are stable; NutriTURN is released as a challenge set for that, on which a method must satisfy each applicable check’s comparison rule on every annotation and report any threshold-switching input in branch-free form. For any benchmark of this construction the audit reduces to five reports: a composition reference beside every learned score, the gain decomposed by pair type, the target rebuilt under a second annotation with event overlap, an annotation-independent check of both, and event counts under composition- and trend-preserving controls.

## Limitations

The target is change in machine-generated direction labels, not expert-verified conclusion change, because literature-scale forecasting data are built from automatic annotation; no human label enters the benchmark, and the released adjudication frame (96 union events, 50 matched non-events) is unexecuted, so the resource is an audit of a construct, not a validated one. The numeric anchor covers 1.6% of associations, is a record-level check that cannot validate the claim-level event set, is weakest on NULL, and its era comparison starts in 2000 with an imprecise anchor trend, so the source of the era drift is not identified. The two annotations differ in model, prompt and read-out at once and span two 7–8B families, so the agreement intervention establishes a representation-dependent contribution to the second-annotation loss, not its whole cause: the branch-free score is non-inferior under neither the second annotation nor common support, and its primary-annotation difference from the shipped score is not established, so the branch is not shown to carry the primary signal. Both annotators were pretrained after most abstracts they label, so the era diagnostics do not separate prior knowledge from change in the literature; Null D is a retrospective parametric control; the partition grid tests counts under three annotations but learned-score differences under one; and the historical-case audit supports no timing claim. With 71 events in 23 claims, comparisons outside the primary cells are uncorrected, the percentile bootstrap can under-cover, the additional procedures show decision stability rather than coverage, the cells were fixed after the baseline results, and the text-reading baselines include no fine-tuned model, so text is not shown to be uninformative. Retrieval is documented, not regenerable: the claim-list assembly is unrecorded, 19 stored queries are unconfirmed, 48 claims are capped, and the external corpora ship as primary-label counts without documents. The analytic reference assumes exchangeable labels and a flat prior, and the calendar cohort conditions on later accrual and is calendar-respecting for documents only. No clinical validity is claimed.

## Ethical Considerations

The benchmark uses public PubMed titles, abstracts and publication-level results, with no individual patient data. Direction labels describe reported associations relative to a named expo-

sure and outcome; they do not establish clinical desirability, causal validity or personalised advice. A high score denotes predicted change in an automatic annotation stream under the stated pipeline, not evidence that an exposure is harmful, and this paper shows that such scores can encode annotation-specific priors. Health communication that uses such signals should retain the cutoff date, the uncertainty, the underlying publications and expert review. The resource is intended for research on evidence streams, not for automated clinical decisions.

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| 989 | Computational Linguistics.                                                    | from the two terms as (“<exposure>”[MeSH                                   | 1042 |
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| 991 | <a href="#">Generalized additive models in Python</a> .                       | AND (“<outcome>”[MeSH Terms] OR                                            | 1044 |
|     |                                                                               | “<outcome>”[Title/Abstract])). Retrieval                                   | 1045 |
|     |                                                                               | uses ESearch with the history server and EFetch                            | 1046 |

in blocks of 500, without date or publication-type filters, so the same search serves every cutoff of a claim. Retrieval stopped at a per-query cap of 800 records for 48 queries, as recorded in the archive metadata; for those claims the archived set is what the cap left (Appendix C.3). The archived snapshot holds the retrieval history as stored for each fixed query: 112,453 claim-record associations over 96,091 distinct identifiers (median 737 records per claim). The dated re-retrieval of Appendix C.3 returns a different identifier set for some claims, so the archived sets, not the queries, define the benchmark. Nothing here implies that a mechanical query recalls every scientifically relevant paper; a record belongs to every claim whose query returns it, and 12,411 identifiers (12.9%) belong to two or more claims (Appendix M). The exact retrieval timestamp is not recorded in the released metadata. Two post hoc observations bound the original retrieval to 2026-09-09 through 2026-09-10 13:53 UTC. The highest archived identifier (42,715,502) entered PubMed on 2026-09-09, and a harvest cannot predate the arrival of a record it contains; the earliest available write of the 165 archived record files is dated 2026-09-10 13:53 UTC, and a copy cannot precede what it copies. The lower bound does not assume that identifiers are assigned in order of entry: it requires only that these records are in the archive. The file times are bulk writes rather than a record-by-record harvest, so they bound the harvest from above without dating it. These bounds constrain the original harvest but do not make the archived identifier sets regenerable, since re-running the stored queries returns a different identifier set (Appendix C.3). The corpus holds 5,818 records dated 2026 and two dated 2027 (ahead-of-print entries), so 2026 is a partially observed year; the fixed-horizon analyses of Appendices H and I treat 2026 as the last calendar year, whereas the origin-defined cohort of Appendix E.2 uses 2025 as the last complete year. The 145-claim candidate list is supplied as a fixed artefact: the procedure that assembled it is not recorded in the released metadata, and the 20 hand-added claims are listed below.

**Cutoffs.** A claim enters with at least 45 records that carry a year and an extractor-assigned direction. With  $y_{(40)}$  the year of its 40th record in date order and  $y_{\max}$  the year of its last record, the grid is the set of distinct integers among seven equally

spaced points from  $y_{(40)} + 2$  to  $y_{\max} - 1$ , provided  $(y_{\max} - 1) - y_{(40)} \geq 4$ . A unit at cutoff  $t$  sees every record with publication year strictly less than  $t$ . Every unit has at least 40 pre-cutoff records; the median unit has 194 pre-cutoff and 287 post-cutoff records; the median pre-cutoff  $I^2$  is 0.935.

**Post-cutoff fields and the target.** For every unit the post-cutoff records give  $n_{\text{post}}$ , the post-cutoff pooled direction  $p_t^+$  over their non-null labels, the post-cutoff agreement, and  $R$  under the sign convention of Equation (1). The fixed horizons recompute the target inside  $[t, t + H)$  for  $H \in \{3, 5, 10\}$  with the same convention, requiring at least ten non-null records in the window and a literature that reaches  $t + H - 1$ .

**Denominators.** The future-eligible sample is the 1,012 units with at least 20 post-cutoff records (160 claims; five claims have no cutoff meeting the threshold), with 71 events in 23 claims. The 145-claim subset (948 units) has 885 future-eligible units with 65 events in 21 claims.

**Hand-added claims.** Twenty claims were added by hand to the 145-claim candidate list: added sugar  $\rightarrow$  diabetes, coffee  $\rightarrow$  cardiovascular disease, dairy  $\rightarrow$  all-cause mortality, dietary cholesterol  $\rightarrow$  cardiovascular disease, dietary pattern  $\rightarrow$  diabetes, eggs  $\rightarrow$  cholesterol, folate  $\rightarrow$  cardiovascular disease, folate  $\rightarrow$  neural tube defects, folic acid and B12  $\rightarrow$  lung cancer, folic acid  $\rightarrow$  colorectal cancer, hormone replacement  $\rightarrow$  cardiovascular disease, legumes  $\rightarrow$  cardiovascular disease, milk  $\rightarrow$  cardiovascular disease, niacin  $\rightarrow$  cardiovascular disease, plant-based diet  $\rightarrow$  cancer, selenium  $\rightarrow$  prostate cancer, sugar-sweetened beverages  $\rightarrow$  obesity, trans fat  $\rightarrow$  cardiovascular disease, vitamin A  $\rightarrow$  cancer and vitamin E  $\rightarrow$  cardiovascular disease. Their entry route is not recoverable, so the claim list is documented rather than regenerable. The analyses repeated on the 145-claim subset are those with an explicit 145-claim row or column in this appendix (the Compact-7 leave-one-claim-out result, the tie-convention table and the rule-comparator table); the operational stationary base-lines, the re-annotation and the common-support sensitivity were run on the full 165-claim corpus only.

**External corpora.** External-1 (physical activity, sleep and environment: 46 claims, 322 units, all future-eligible, 19 events in 7 claims) and External-2 (medication, screening and prevention, occupa-

|                                           | Source                   | External-1 | External-2 |
|-------------------------------------------|--------------------------|------------|------------|
| Claims                                    | 165                      | 46         | 180        |
| Claim-cutoff units                        | 1,078                    | 322        | 1,260      |
| Eligible units ( $\geq 20$ later records) | 1,012                    | 322        | 1,257      |
| Eligible claims                           | 160                      | 46         | 180        |
| Events, primary annotation (claims)       | 71 (23)                  | 19 (7)     | 95 (40)    |
| Event prevalence                          | 0.070                    | 0.059      | 0.076      |
| Calendar cohort, 3 / 5 / 10 y: units      | 756 / 625 / 492 (source) |            |            |
| Calendar cohort, 3 / 5 / 10 y: events     | 58 / 45 / 35 (source)    |            |            |

Table 6: The benchmark under the primary annotation (size of every evaluation population). The source stream holds 112,453 claim-record associations over 96,091 PubMed records; external corpora are released as claim-cutoff units with primary-pipeline labels, for frozen transfer only.

tional exposures and clinical risk states: 180 claims, 1,260 units, 1,257 future-eligible, 95 events in 40 claims under Equation (1)) were built with the same retrieval, extractor, cutoff grid, feature builder and target. No claim is shared with the source benchmark. Table 6 summarises every evaluation population.

## B Direction Extraction and Feature Representation

### B.1 Closed-output extraction prompt

Direction is extracted with Llama-3.1-8B-Instruct in bfloat16, in evaluation mode with no gradient and no generation. The title and abstract are concatenated, truncated to the first 3,000 characters, rendered through the chat template with the system and user turns of Figure 2, tokenised with a 1,600-token limit and right truncation, and scored in a single forward pass with batch size 16 and left padding. The logits at the final position are restricted to the first token of each of the four label strings (PROTECTIVE, HARMFUL, NULL, UNCLEAR); the argmax is the extractor-assigned claim-record direction and the softmax over the four is its confidence. Under the primary tokenizer only NULL is a single token; PROTECTIVE, HARMFUL and UNCLEAR are split, but their first tokens are distinct, so the read-out is a well-defined argmax over four distinct tokens rather than a full-string likelihood. The second pipeline scores complete label strings and is compared with its own first-token read-out in Appendix C.1. Every prompt is checked to contain the claim’s own outcome term before scoring. Because the output space is closed there is no sampling temperature, no decoding and no output parser. UNCLEAR labels are retained for audit and excluded from directional aggregation. Effect

SYSTEM: You classify the direction of association reported between an exposure and an outcome. Answer with exactly one word: PROTECTIVE if the exposure is reported to reduce the outcome, HARMFUL if it increases it, NULL if no association is reported, UNCLEAR if the study does not report this association.

USER: Exposure: {exposure} \n Outcome: {outcome} \n \n Study: {title}. {abstract} \n \n Direction:

Figure 2: The direction-extraction prompt. The four label tokens are scored at the position following Direction::; nothing is generated.

sizes, confidence limits and design indicators are recovered by case-sensitive pattern matching over the abstract text, not by the language model. The feature builder places each parsed interval on a working scale inferred from its limits: when both limits are positive the interval is treated as a ratio measure and represented as  $y = (\ln L + \ln U)/2$  with  $SE = (\ln U - \ln L)/(2 \times 1.96)$ ; otherwise it is treated as a difference measure on the linear scale. The Compact-7 inputs  $\mu_{\text{eff}}$ ,  $\sigma_{\text{eff}}$  and  $n_{\text{eff}}$  (val\_mean, val\_sd, n\_rr in Table 48) are therefore computed over a mixture of log-ratio and additive quantities, not over uniformly commensurate log-ratios; the typed anchor of Appendix B.2 reads the measure from the text instead and is used for validation only.

### B.2 In-domain numeric-anchor validation

We construct an LLM-independent reference directly from the quantitative reporting in the corpus; no language model and no human annotation is used to create it. Every “95% CI” or “95% confidence interval” followed by two decimal limits ( $L, U$ ),  $U > L$ , is a candidate. A candidate is *typed* by the effect-measure token in the 120 characters before it (40 after): ratio measures (odds, risk, hazard, rate and prevalence ratios and their abbreviations) fix the null at 1; additive measures (mean, weighted mean and standardised mean differences, risk differences, regression coefficients and “difference”) fix it at 0. A candidate with no token is EFFECT-TYPE-UNIDENTIFIED, one with tokens of both families EFFECT-TYPE-CONFLICT, and a ratio interval with a non-positive limit INVALID. A typed candidate is *attributed* to the claim when the sentence holding it, or the preceding sentence, names both an exposure term and an outcome term from a per-claim lexicon derived mechanically from the claim query; otherwise it is ANCHOR-AMBIGUOUS-RELATION. The anchor direction of an attributed interval is NULL if it covers the null, HARMFUL

if  $L$  is above it and PROTECTIVE if  $U$  is below it; because the primary prompt defines PROTECTIVE as the exposure reducing the outcome and HARMFUL as increasing it, no judgement about the valence of the outcome is involved, and the anchor measures agreement on the sign relative to the null. A record is *high-confidence* when it has at least one attributed typed interval and all such intervals agree. Of 112,453 claim-record associations, 15,144 carry an interval, 10,650 a typed one, 2,384 an attributed typed one, and 1,841 in 148 claims are high-confidence (1.6% of the corpus; 543 excluded for disagreeing attributed intervals). Table 7 gives the results against the primary labels; every interval is a claim-clustered percentile bootstrap with 2,000 resamples.

**Confusion.** On the high-confidence records (rows anchor, columns extractor PROTECTIVE/HARMFUL/NULL/UNCLEAR): PROTECTIVE 679 / 29 / 27 / 1; HARMFUL 84 / 627 / 35 / 4; NULL 97 / 75 / 179 / 4. Signed anchors are recovered at 0.92 and 0.84; the weak class is NULL (recall 0.50), where the extractor assigns a direction to intervals that cross the null in both directions, consistent with abstracts that state a direction in words while the interval covers the null. Additive measures are rare (167 records) and their HARMFUL recall is 0.27, because “difference” intervals often carry a sign convention (a reduction reported as a positive difference) that the anchor cannot resolve; this is a limitation of the anchor. Read by column, 84 of the 860 records the primary pipeline labels PROTECTIVE carry a HARMFUL anchor (9.8%), and 29 of the 731 it labels HARMFUL carry a PROTECTIVE anchor (4.0%). By design, agreement is 0.79 for synthesis records, 0.81 for observational or other records and 0.81 [0.709, 0.900] for the 89 trial records.

**Both annotations against the anchor.** Table 8 scores the second pipeline and the intersection labels on the same 1,841 records. Paired on identical records, primary minus second is +0.049 [0.030, 0.068] on agreement and +0.030 [0.012, 0.048] on balanced accuracy, both established; on the 1,486 signed anchors, which removes the NULL-rate confound, the gap is +0.074 [0.053, 0.094]. By anchor scale, the paired second-minus-primary agreement difference is -0.059 [-0.081, -0.036] on the 1,691 ratio-scale records, whose anchors carry no sign-convention ambiguity, and -0.010 (point estimate; interval not com-

puted) on the 167 additive-scale records; the two subgroup estimates are not tested against each other. The excess disagreement is a signed-versus-NONE discrepancy rather than sign inversion: among signed anchors the second pipeline assigns NULL to 9.4% against the primary’s 4.2%, whereas opposite-sign assignments have similar frequencies, 7.9% against 7.6% (a descriptive contrast; equality of the opposite-sign rates is not established), and its recall on true NULL anchors rises only from 0.504 to 0.561. NULL is the prompt’s substantive “no association” label, distinct from UNCLEAR, so these are label assignments, not abstentions. Intersection labels retain a signed or NULL label on 1,587 of the 1,841 anchored records (86.2% coverage) and turn the other 254 into UNCLEAR, which the anchor cannot express; the intersection row scores those 254 as disagreements, so it is full-population agreement with UNCLEAR counted as an error, not an accuracy bound of any kind, and on the retained 1,587 records agreement is 0.842 [0.821, 0.861], higher than either pipeline’s because those are the records on which the two pipelines already agree. The second pipeline’s confusion on the 1,841 records (rows anchor, columns PROTECTIVE/HARMFUL/NULL/UNCLEAR, row totals 736/750/355): PROTECTIVE 637 / 30 / 66 / 3; HARMFUL 87 / 559 / 74 / 30; NULL 85 / 61 / 199 / 10; read by column, 87 of the 809 records it labels PROTECTIVE carry a HARMFUL anchor (10.8%, against the primary’s 9.8%) and 30 of the 650 it labels HARMFUL a PROTECTIVE anchor (4.6%, against 4.0%). The anchor is a record-level consistency check on a reporting-selected subset; it does not validate the claim-level event set of either annotation.

**Era-stratified comparison.** Table 9 bins the anchored records by era; the 61 records dated before 2000 form no bin. Agreement with the anchor rises across eras for both pipelines. On the same records the pooled signed HARMFUL share of the anchor, the primary and the second pipeline all decline across the three eras, a descriptive pattern in this sample. The within-claim slopes per decade are -0.053 [-0.125, 0.013], -0.070 [-0.140, -0.002] and -0.043 [-0.112, 0.024]: all three point estimates are negative, but only the primary pipeline’s trend is established, the anchor’s and the second pipeline’s intervals include zero, and the paired slope differences on the same claim resamples, anchor minus primary +0.017 [-0.019, 0.056] and anchor minus second -0.010 [-0.067, 0.038] per

| Evaluation                      | Associations | Claims | Agreement | [95% CI]       | Bal. acc. | Macro-F1 | [95% CI]       | R <sub>PROT</sub> | R <sub>HARM</sub> | R <sub>NULL</sub> |
|---------------------------------|--------------|--------|-----------|----------------|-----------|----------|----------------|-------------------|-------------------|-------------------|
| high-confidence anchor          | 1841         | 148    | .807      | [0.782, 0.830] | .754      | .766     | [0.735, 0.796] | .923              | .836              | .504              |
| exactly one attributed interval | 999          | 133    | .770      | [0.742, 0.796] | .716      | .723     | [0.687, 0.756] | .906              | .818              | .424              |
| excluding extractor-UNCLEAR     | 1832         | 147    | .811      | [0.787, 0.833] | .758      | .768     | [0.735, 0.798] | .924              | .841              | .510              |
| ratio-scale intervals only      | 1691         | 134    | .813      | [0.789, 0.836] | .757      | .768     | [0.732, 0.800] | .929              | .863              | .479              |
| additive-scale intervals only   | 167          | 62     | .737      | [0.650, 0.819] | .655      | .649     | [0.560, 0.726] | .889              | .270              | .806              |
| design: synthesis               | 397          | 122    | .793      | [0.740, 0.840] | .736      | .763     | [0.707, 0.813] | .942              | .667              | .600              |
| design: other                   | 1355         | 130    | .810      | [0.782, 0.835] | .738      | .748     | [0.706, 0.785] | .922              | .858              | .434              |
| design: trial                   | 89           | 47     | .809      | [0.709, 0.900] | .816      | .805     | [0.698, 0.898] | .829              | .895              | .724              |

Table 7: LLM-independent numeric anchor against the primary labels. The effect measure is read from the text next to each 95% interval and the interval is attributed to the claim only when the exposure and outcome terms both occur in the interval’s sentence or in the preceding sentence. Claim-clustered 95% intervals, 2,000 resamples.

| Pipeline                     | Agreement [95% CI] | Bal. acc. [95% CI] | Macro-F1 | R <sub>p</sub> / R <sub>h</sub> / R <sub>n</sub> |
|------------------------------|--------------------|--------------------|----------|--------------------------------------------------|
| Primary                      | .807 [.782, .830]  | .754 [.725, .785]  | .766     | .923 / .836 / .504                               |
| Second                       | .758 [.733, .782]  | .724 [.695, .749]  | .732     | .865 / .745 / .561                               |
| Intersection, all 1,841      | .726 [.698, .753]  | .682 [.650, .712]  | .742     | .845 / .728 / .473                               |
| Intersection, retained 1,587 | .842 [.821, .861]  | .792 [.763, .820]  | .805     | .950 / .860 / .566                               |

Table 8: The numeric anchor against each annotation on the same 1,841 high-confidence records in 148 claims (claim-clustered 95% intervals, 2,000 resamples). Paired primary minus second: +0.049 [0.030, 0.068] on agreement, +0.074 [0.053, 0.094] on the 1,486 signed anchors, +0.059 [0.036, 0.081] on the 1,691 ratio-scale records. Among signed anchors: NULL assigned by primary / second 4.2% / 9.4%, opposite sign 7.6% / 7.9%. Intersection: the first row scores the 254 UNCLEAR records as disagreements, the second scores the 1,587 records that retain a label, a selected subset on which both pipelines agree.

| Era   | <i>n</i> | Agreement primary / second | HARMFUL share anchor / primary / second |
|-------|----------|----------------------------|-----------------------------------------|
| 2000s | 394      | .751 / .678                | .619 / .565 / .548                      |
| 2010s | 646      | .800 / .762                | .502 / .463 / .448                      |
| 2020s | 740      | .841 / .796                | .449 / .392 / .382                      |

Table 9: Anchored records by era (61 records before 2000 form no bin). Within-claim slopes of the HARMFUL share per decade on these records: anchor −0.053 [-0.125, 0.013], primary −0.070 [-0.140, -0.002], second −0.043 [-0.112, 0.024]; paired differences, anchor minus primary +0.017 [-0.019, 0.056], anchor minus second −0.010 [-0.067, 0.038].

decade, both cover zero: the anchor’s rate of drift is not distinguishable from either pipeline’s, and the comparison establishes neither a common within-claim drift process nor equal rates. The primary pipeline’s over-HARMFUL rate (records it labels HARMFUL whose anchor is NULL or PROTECTIVE) falls from 0.098 in the 2000s to 0.035 in the 2020s, so the annotator’s bias relative to the anchor is era-dependent and shrinks. The anchored subset is reporting-selected, and on it the primary slope is steeper than corpus-wide (−0.042; Appendix C.2).

**Scope.** The anchor covers a small, reporting-selected subset, cannot express UNCLEAR, and measures consistency with reported effect estimates on that subset, not the accuracy of the full label set.

### B.3 The full pre-cutoff representation

The frozen feature builder computes 55 pre-cutoff columns per unit from the records published before the cutoff (Table 48), organised in six blocks: direction (11), of which the four trajectory variables are a subset; heterogeneity (6); decline (4); design (8); volume (7); and 19 derived variables on numeric reporting and effect estimates. Three groups of columns are numerically identical by construction (*n\_rr*, *n\_rr\_studies*, *n\_numeric*; *n\_pre\_t*, *n\_records*; *spread*, *val\_spread*), so the 55 columns span 51 distinct-valued variables, and *decline\_sufficient* is constant at zero. The representation is what every unit of the released benchmark carries, what every rule comparator and single signal reads, and the input of the learned baselines marked “all pre-cutoff variables”. Compact-7 reads seven of the columns, marked \* in the inventory; the set was fixed before any external evaluation, contains no trajectory, heterogeneity, decline or design variable, and is tested rather than assumed (Appendix N). The inventory is reproduced at the end of the appendix as a data dictionary.

### B.4 Specification chronology and freezing

Table 10 records, for each fixed component, how it was selected, what data were visible when it was fixed and when it was frozen. The seven Compact-7 variables were written as a single set operationalising static direction-and-effect aggregation and were never varied; they are not the output of any variable search, and the logistic specification is the library default (standardised inputs,  $L_2$ ,  $C = 1$ ,

class balancing), whose regularisation strength was deliberately not tuned because a sweep scored on the same claims would be optimistic. Both were fixed while the source benchmark, including its subsequent-stream target column, existed, and before any Compact-7-versus-later-change result had been computed, so they are treated as a frozen source specification rather than a prospectively pre-registered feature set. The source method was then frozen in a hash-signed configuration record before the External-1 and External-2 corpora were built, and no external outcome was used to alter it. The two-variable score was defined after Compact-7 as a diagnostic of its inputs and acquires no retrospective preregistration status.

## C Second Annotation, Snapshot Versioning and Notation

### C.1 Second annotation pipeline and cross-pipeline sensitivity

Every claim-record association with a primary annotation and a publication year (112,453) was re-annotated with Qwen2.5-7B-Instruct in bfloat16 using the same task framing, the same 3,000-character title-plus-abstract budget and the same 1,600-token truncation as the primary extractor, with the NULL and UNCLEAR definitions made explicit (Figure 3); the primary prompt (Figure 2) is preserved verbatim, including its historical wording, and the two executed configurations therefore differ in prompt wording, model family and read-out. Instead of the first-token read-out, the probability of each label is the teacher-forced probability of the complete label string normalised over the four labels; the first-token read-out and the share of next-token mass on the four first tokens (mean 0.999) are stored alongside, and the two read-outs agree on 99.2% of associations and define the same 52 events. Agreement is reported with its denominator: all 112,453 associations, the associations both pipelines resolve (non-UNCLEAR), and the 56,423 both call signed. Agreement measures how stable the target is across annotators, not its accuracy. Unit-level statistics rebuild the pre-cutoff inputs and the endpoint of the 1,012 future-eligible units under the second pipeline, under label intersection (associations on which the pipelines disagree become UNCLEAR) and under 500 soft-label draws (one label per claim-record association per draw from the second pipeline’s label probabilities, reused across every cutoff of the claim; a record

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SYSTEM: You classify the direction of association reported between an exposure and an outcome. Answer with exactly one word: PROTECTIVE if the exposure is reported to reduce the outcome, HARMFUL if it increases it, NULL if the abstract directly reports no association for the named exposure and outcome, UNCLEAR if the pair is absent from the abstract or its direction cannot be determined.  
 USER: Exposure: {exposure} \n Outcome: {outcome} \n \n Study: {title}. {abstract} \n\n Direction:

---

Figure 3: The executed second-pipeline prompt (Qwen2.5-7B-Instruct, chat template, bfloat16). The four complete label strings are scored by teacher forcing after Direction:; nothing is generated.

shared by two claims is drawn independently per claim), and refit the two-variable score, the margin rule and the count-predicted baseline leave-one-claim-out each time (Table 11). Two comparisons are distinguished: a score *rebuilt and refitted* on an annotation’s inputs and endpoint, and the *primary* score (primary inputs, trained on the primary endpoint) evaluated against another annotation’s endpoint. Soft-label ranges are variation across annotation draws with no claim-bootstrap layer. The NULL-dominant agreement branch holds 42 units under the primary pipeline and 332 under the second; event retention by branch is given in the table. Table 14 shows that the branch change in the agreement input contributes substantially to the reversed ordering within the second-annotation setting; it does not show that changed signed membership or the imputed ties play no role in the primary-to-second difference.

**Common-support sensitivity.** Table 12 restricts the full-cohort out-of-fold predictions to the 1002 units (160 claims) with at least eight signed pre-cutoff records under both pipelines; exact ties with adequate support are retained. Low event overlap persists (Jaccard 0.267) and the method ordering under the second pipeline is unchanged (signed minus margin  $-0.119$   $[-0.215, -0.045]$ ; signed minus stationary  $-0.131$   $[-0.236, -0.052]$ ). The reversal persists when sparse histories are excluded from the evaluation population, and it strengthens when the signed score is refitted leave-one-claim-out on the 1002 units: the signed score falls to 0.758 and signed minus margin to  $-0.184$   $[-0.268, -0.105]$ . Sparse histories therefore produce the reversal neither through evaluation nor through training.

**Within-class comparison.** Table 13 pools pairs of units that share a pre-cutoff majority class ( $p_t \neq 0$ ) under each annotation, using the leave-

| Component                                                                                                                 | Selection basis                                                                                                   | Visible when fixed                                                                                                      | Frozen                                                           | Later analyses (sensitivity only)                                               |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Compact-7 variables ( $p$ , $a$ , $n_{\text{dir}}$ , $n_{\text{pre}}$ , $\mu_{\log RR}$ , $\sigma_{\log RR}$ , $n_{RR}$ ) | Hand-composed as one set operationalising static direction-and-effect aggregation; never varied; no subset search | Source pre-cutoff variables and the subsequent-stream target column; no Compact-7 later-change result; no external data | Hash-signed source-method record, before External-1/2 were built | Forward and backward selection, nested $L_1$ selection, learned models (App. N) |
| Logistic specification ( $L_2$ , $C = 1$ , class balancing)                                                               | Library default; $C$ not tuned (a same-claim sweep declined as optimistic)                                        | As above: source pre-cutoff variables and the subsequent-stream target column; no Compact-7 later-change result         | Same record                                                      | Learned-model comparisons (App. N)                                              |

Table 10: Specification chronology and freezing. “Visible” lists the data that existed when the component was fixed; “Frozen” names the stage at which it was locked.

| Primary (rows) \ second (columns) | PROTECTIVE | HARMFUL | NULL   | UNCLEAR |
|-----------------------------------|------------|---------|--------|---------|
| PROTECTIVE                        | 36,782     | 1,421   | 12,374 | 473     |
| HARMFUL                           | 2,726      | 15,494  | 8,775  | 2,290   |
| NULL                              | 2,329      | 402     | 22,177 | 672     |
| UNCLEAR                           | 581        | 174     | 4,900  | 883     |

| Agreement branch of the unit | Units | Primary events | Second events | Both |
|------------------------------|-------|----------------|---------------|------|
| Primary: signed branch       | 969   | 59             | 42            | 22   |
| Primary: NULL-dominant       | 42    | 11             | 9             | 4    |
| Primary: fewer than 8 signed | 1     | 1              | 1             | 1    |
| Second: signed branch        | 670   | 28             | 24            | 10   |
| Second: NULL-dominant        | 332   | 39             | 18            | 13   |
| Second: fewer than 8 signed  | 10    | 4              | 10            | 4    |

| Endpoint cross-tabulation | Second: event | Second: non-event | Total |
|---------------------------|---------------|-------------------|-------|
| Primary: event            | 27            | 44                | 71    |
| Primary: non-event        | 25            | 916               | 941   |
| Total                     | 52            | 960               | 1012  |

| Pipeline     | Events on sparse histories (< 8 signed) | Primary score on this endpoint, AUROC |
|--------------|-----------------------------------------|---------------------------------------|
| Primary      | 1                                       | .948                                  |
| Second       | 10                                      | .878                                  |
| Intersection | 14                                      | .869                                  |

Table 11: Cross-pipeline sensitivity, detail. Claim-record association-level agreement with its denominators: 67.0% of all 112,453 associations ( $\kappa = 0.52$ ), 72.7% of associations both pipelines resolve ( $\kappa = 0.58$ ), 92.7% of the 56,423 both call signed ( $\kappa = 0.83$ ); agreement is not accuracy. 21,149 primary signed annotations become NULL and 4,147 switch sign (NULL share 22.7% against 42.9%). Exact ties with at least eight signed records: 7 units under the primary pipeline (all events), 0 under the second. The last block reports the primary-trained score (primary inputs, trained on the primary endpoint) evaluated unchanged against each endpoint, as in Table 5; the rebuilt-and-refitted scores (.857, .898) are those of Table 2. Soft-label draws (500; variation across annotation draws, no claim-bootstrap layer): 43-57 events, signed score .779-.880, margin rule .938-.964. The Qwen first-token read-out defines the same 52 events as the full-string read-out.

| Support / training scope                                                                                                                      | Pipeline | Units / claims | Events / positive claims | Shared events | Jaccard | Margin AUROC / AUPRC | Stationary AUROC / AUPRC | Signed AUROC / AUPRC | Signed – margin [95% CI] |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------|--------------------------|---------------|---------|----------------------|--------------------------|----------------------|--------------------------|
| <i>Full archived cohort, out-of-fold scores</i>                                                                                               |          |                |                          |               |         |                      |                          |                      |                          |
| Full cohort                                                                                                                                   | Primary  | 1,012 / 160    | 71 / 23                  | 71            | 1.000   | .881 / .468          | .909 / .497              | .948 / .535          | + .067 [.017, .120]      |
| Full cohort                                                                                                                                   | Second   | 1,012 / 160    | 52 / 16                  | 27            | .281    | .953 / .611          | .963 / .626              | .857 / .369          | – .096 [–.166, –.037]    |
| <i>Common support (<math>\geq 8</math> signed pre-cutoff records under both pipelines), full-cohort-trained out-of-fold scores restricted</i> |          |                |                          |               |         |                      |                          |                      |                          |
| Common support, restricted                                                                                                                    | Primary  | 1002 / 160     | 67 / 23                  | 23            | .267    | .889 / .460          | .915 / .487              | .948 / .556          | + .059 [.016, .103]      |
| Common support, restricted                                                                                                                    | Second   | 1002 / 160     | 42 / 14                  | 23            | .267    | .942 / .422          | .954 / .425              | .823 / .137          | –.119 [–.215, –.045]     |
| <i>Common support, signed score refitted leave-one-claim-out on the 1002 units</i>                                                            |          |                |                          |               |         |                      |                          |                      |                          |
| Common support, refitted                                                                                                                      | Primary  | 1002 / 160     | 67 / 23                  | 23            | .267    | .889 / .460          | –                        | –                    | + .059 [.016, .104]      |
| Common support, refitted                                                                                                                      | Second   | 1002 / 160     | 42 / 14                  | 23            | .267    | .942 / .422          | –                        | .758 / –             | –.184 [–.268, –.105]     |

Table 12: Common-support sensitivity. The support mask (hash in the released manifest) keeps units with at least eight signed pre-cutoff records under both pipelines (1002 units; 10 excluded, of which 10 sparse under the second pipeline and 1 under the primary pipeline; the primary sparse unit is among the second pipeline’s sparse units). Each pipeline keeps its own operational endpoint; in the restricted rows, scores are the full-cohort leave-one-claim-out predictions restricted to the mask; in the refitted row, the signed score is refitted on the mask. All 2,000 claim-clustered resamples were evaluable for both pipeline comparisons.

one-claim-out predictions of the development cell. Under the primary annotation the signed score’s

within-class gain over the margin rule is not established, and the class-blind variant matches it;

| Annotation                    | Margin | Signed | Class-blind | Signed – margin [95% CI] |
|-------------------------------|--------|--------|-------------|--------------------------|
| Primary, pooled               | .909   | .927   | .926        | +.018 [−.012, .060]      |
| HARMFUL-majority (274, 42)    | .833   | .863   | –           | +.030 [−.041, .114]      |
| PROTECTIVE-majority (730, 21) | .959   | .969   | –           | +.011 [−.001, .024]      |
| Second, pooled                | .929   | .631   | .929        | −.298 [−.538, −.147]     |
| HARMFUL-majority              | .906   | .478   | –           | −.428 [−.666, −.216]     |
| PROTECTIVE-majority           | .955   | .810   | –           | −.145 [−.373, .011]      |
| Intersection, pooled          | .897   | .801   | .889        | −.095 [−.191, −.016]     |

Table 13: Within-class AUROC, development cell, each annotation rebuilt and refitted: pooled over pairs sharing a majority class, and within each class (units, events in parentheses). Intervals: claim-clustered bootstrap, 2,000 resamples.

under the second annotation and intersection labels the signed score falls below the margin rule within classes by an established margin. The class-blind variant’s within-class values (0.926, 0.929, 0.889) lie within 0.017 of the margin rule’s (0.909, 0.929, 0.897) under every annotation, and no within-class gain of the class-blind variant is established. Under the second annotation the signed score falls to 0.478 within HARMFUL-majority units, an established loss; this comparison includes NULL-dominant units, whose agreement input changes definition, and Table 14 attributes the loss to that change.

#### Branch-free agreement under every annotation.

Table 14 substitutes the agreement definitions of Appendix K upstream of the two-variable score under each annotation and refits leave-one-claim-out. Under the shipped two-branch definition A the score gains +0.067 over the margin rule on the primary annotation and loses −0.096 on the second; under the branch-free definition C the primary gain is +0.044 [−0.005, 0.091], not established, and the second-annotation difference is −0.001 [−0.028, 0.018]; adding the NULL share as a third input (C+null) gives +0.051 [−0.006, 0.108] and +0.006 [−0.022, 0.032]. Stratifying the shipped score by branch under the second annotation locates the loss: on the 670 signed-branch units the difference is −0.041 with the full-cohort fit restricted and −0.004 refitted within the branch, whereas on the 332 NULL-dominant units it is −0.231 [−0.481, −0.017] restricted and +0.047 refitted.

#### The matched intervention on common support.

The headline intervention holds the cohort and outcomes fixed and changes only the agreement input: on the 1,002 common-support units (Table 12), refitting the two-variable score with definition C instead of A raises its second-annotation AUROC from 0.758 to 0.940, a paired improvement (C

minus A) of +0.182 [0.120, 0.248]; the margin rule reaches 0.942 on the same units, and C minus margin is −0.002 [−0.037, 0.023] on the same 2,000 claim resamples, whose lower bound misses the non-inferiority margin  $\delta = 0.02$  by 0.017. In full-cohort slices the improvement is +0.246 [0.073, 0.405] on the NULL-dominant branch and +0.041 on the signed branch; the difference between the two branch effects is not tested, so “largest on NULL-dominant units” is descriptive. Across annotations 296 units move from the primary signed branch into the second pipeline’s NULL-dominant branch and none move in the reverse direction. The full-cohort slices and the common-support refit are complementary analyses, not an exact decomposition of one another. The intervention establishes a substantial representation-dependent contribution to the second-annotation loss within that setting; it does not show that the branch is its only cause, it does not show that annotation-induced signed membership plays no role in the primary-to-second difference, and it does not change the 27 shared events of Table 5. On the primary annotation the branch-free score’s development gain is not established (+0.044), but neither is its difference from the shipped score (−0.023 [−0.050, 0.002]; Table 37), so the result is that the branch-free model does not establish a development gain, not that removing the branch eliminates the primary signal. The branch-free score’s full-cohort AUROCs are 0.952 under the second annotation and 0.935 under intersection labels.

#### C.2 Controlled annotation and era effects

To separate the factors that distinguish the two archived configurations, every association was re-annotated in cells that change one factor at a time: the primary configuration re-run with different batching, Llama-3.1-8B with the primary prompt and full-string read-out, Llama with the second prompt, and Qwen2.5-7B with each prompt. Table 15 compares neighbouring cells. Changing the model moves the event set most (Jaccard 0.312 under either prompt) and shifts the NULL share most (+0.154 and +0.315); changing the prompt moves it less (0.520 and 0.538); changing the read-out (0.899) stays close to the run-to-run level (0.846). The method ordering follows the model family: the signed score falls below the margin rule in both Qwen cells (−0.092 and −0.176) and in neither Llama cell, consistent with the representation effect that Table 14 demonstrates for the archived

| <i>(a) Signed two-variable score – margin rule by agreement definition (full cohort, refitted)</i> |                       |                      |                        |
|----------------------------------------------------------------------------------------------------|-----------------------|----------------------|------------------------|
| Annotation                                                                                         | Shipped A             | Branch-free C        | C + NULL share         |
| Primary                                                                                            | +0.067 [.017, .120]   | +0.044 [−.005, .091] | +0.051 [−.006, .108]   |
| Second                                                                                             | −0.096 [−.166, −.037] | −0.001 [−.028, .018] | +0.006 [−.022, .032]   |
| Intersection                                                                                       | −0.030 [−.094, .023]  | +0.007 [−.016, .030] | +0.013 [−.032, .051]   |
| <i>(b) Shipped score – margin rule by agreement branch, second annotation, full cohort</i>         |                       |                      |                        |
| Branch                                                                                             | Units                 | Restricted           | Refitted within branch |
| Signed                                                                                             | 670                   | −.041                | −.004                  |
| NULL-dominant                                                                                      | 332                   | −.231 [−.481, −.017] | +0.047                 |
| <i>(c) Intervention A → C, second annotation, outcomes fixed</i>                                   |                       |                      |                        |
| Cohort                                                                                             | Units (events)        | AUROC A → C          | C − A [95% CI]         |
| Common support (margin .942; C − margin −0.002 [−.037, .023])                                      | 1,002 (42)            | .758 → .940          | +0.182 [.120, .248]    |
| Full cohort, NULL-dominant branch                                                                  | 332                   | −                    | +0.246 [.073, .405]    |
| Full cohort, signed branch                                                                         | 670                   | −                    | +0.041                 |

Table 14: The agreement-input intervention. (a) Paired differences from the margin rule under each agreement definition and annotation (leave-one-claim-out, claim-clustered 95% intervals). (b) The shipped score stratified by agreement branch under the second annotation (“restricted”: full-cohort out-of-fold predictions restricted to the branch; “refitted”: refit within the branch). (c) The matched intervention on the same cohort and outcomes, agreement input replaced and score refitted; C − margin on common support is −0.002 [−0.037, 0.023]. Across annotations 296 units move from the primary signed branch into the second pipeline’s NULL-dominant branch and none move in the reverse direction.

| Change   | Between                | $\kappa$ | Jaccard | NULL shift | Events |
|----------|------------------------|----------|---------|------------|--------|
| Batching | primary re-run         | .980     | .846    | .000       | 71→73  |
| Read-out | Llama P1: token→string | .986     | .899    | +0.006     | 73→77  |
| Prompt   | Llama: P1→P2           | .633     | .520    | −.117      | 77→75  |
| Prompt   | Qwen: P1→P2            | .823     | .538    | +0.044     | 49→51  |
| Model    | P1: Llama→Qwen         | .561     | .312    | +0.154     | 77→49  |
| Model    | P2: Llama→Qwen         | .402     | .312    | +0.315     | 75→51  |

Table 15: Controlled annotation, one factor per row. P1, P2: the primary and second prompts (Figures 2 and 3); cells other than the re-run use full-string read-out. The signed score falls below the margin rule in both Qwen cells (−0.092, −0.176) and in neither Llama cell.

second configuration and with the larger NULL-dominant branch that the Qwen configurations produce. Across the archived annotations the fitted class coefficient keeps its sign but shrinks: raw  $\hat{b}_1$  is +5.331 (primary), +2.138 (second) and +1.912 (intersection), standardised +3.010, +1.298 and +1.223; the primary full-data fit has  $\hat{b}_0 = +3.467$ ,  $\hat{b}_1 = +5.331$  and  $\hat{b}_2 = -12.193$ .

**Run-to-run variation.** The re-run keeps the checkpoint, prompt, tokenisation, bfloat16 inference and first-token read-out of the archived primary configuration and changes only the batch size (the archived run used 16, with left padding). The label changes are consistent with bfloat16 numerical variation: a different batch size changes the padded tensor shapes and, with them, the order of floating-point reductions, which moves the logits slightly and flips labels whose two highest first-

token logits nearly tie. At the label level the effect is small ( $\kappa = 0.980$ ), yet it moves the event set (Jaccard 0.846; 71 against 73 events), because the thresholded endpoint turns small label changes near the majority boundary into changes of the positive class. Released labels are fixed by hash, so the benchmark itself does not vary; the variation concerns re-annotation.

**Era effects.** Cross-pipeline agreement rises from  $\kappa = 0.411$  for records from the 1960s to 0.549 for the 2000s. Pooled over claims, the primary pipeline’s HARMFUL share falls from 0.463 to 0.225 across decades while its PROTECTIVE share rises from 0.289 to 0.503; these pooled shares mix claims. With claim fixed effects the drift is smaller but remains under the primary annotation: the within-claim slope of the HARMFUL share is −0.042 [−0.061, −0.023] per decade against a pooled slope of −0.118 [−0.170, −0.060], so claim composition accounts for about two thirds of the pooled shift; under the second annotation the within-claim slope is −0.014 [−0.034, 0.004] against −0.101 [−0.171, −0.022]. Across the 114 claims that have records in both the 1990s and the 2020s, the primary HARMFUL share falls from 0.401 to 0.351 (−0.050 [−0.092, −0.008]). Re-annotating a random sample of 2,000 associations with the exposure and outcome names masked changes 25.1% of primary labels: 35.3% for records before 2000 against 24.0% after (differ-

ence +0.114 [0.038, 0.190]). The probe removes textual cues as well as named-claim knowledge, so it shows that label behaviour differs by era without identifying why; on the anchored subset from 2000 onward the anchor’s own HARMFUL-share slope has a similarly signed point estimate, with the stated uncertainty (Table 9).

### C.3 Snapshot versioning and dataset card

The stored corpus (version 1) was assembled without recording the retrieval timestamp, the per-query PubMed count or the query translation, so its identifier sets cannot be regenerated; it is released as a *non-regenerable derived snapshot* whose files are fixed by SHA-256 hashes in the manifest. Post hoc provenance evidence nevertheless constrains the original harvest to a narrow interval: the highest archived PubMed identifier (42,715,502) entered PubMed on 2026-09-09, establishing a lower bound, and the earliest available write of the 165 archived record files occurred on 2026-09-10 13:53 UTC, establishing an upper bound. The exact original retrieval timestamp was not stored, so 2026-09-09 through 2026-09-10 13:53 UTC is the interval the surviving evidence supports and not a recovered value; the bounds say when the harvest ran and do not make its identifier sets regenerable. On 2026-09-22 (UTC) every stored query string was re-run through PubMed ESearch at one timestamp, storing the count, the query translation, the complete identifier list and its hash (version 2 identifier set; abstracts were not re-fetched, the benchmark was not rebuilt, and no performance figure derives from it). Table 16 compares the versions at the association level (claim-identifier pairs) and at the level of distinct identifiers: 103,175 of 112,453 archived associations (91.7%) and 88,649 of 96,091 distinct archived identifiers (92.3%) are returned again. Fifty-four claims have per-claim Jaccard overlap below 0.8, through two disjoint mechanisms. Nineteen stored queries return no records at re-retrieval; they are exactly the nineteen claims that appear to lose more than 10% of their archived identifiers. Why they return nothing is not recoverable from the surviving metadata: an edited query string, a changed automatic term translation or a query that never ran as stored are all consistent with the evidence; the release flags these 19 claims (manifest field `query_unconfirmed`, true for the 19 and false for the other 146), whose stored query is not confirmed to be the harvesting query, and states the discrepancy without assigning a mechanism. The

other 35 return more records than were archived and lose none; their low overlap reflects the per-query cap described below (the archive holds 56 to 4,888 records per claim). Neither mechanism is change in the literature. Removing all 54 claims changes no conclusion in any of the twelve protocol cells.

**Retrieval cap.** The archive metadata records, for each query, whether retrieval stopped at the per-query cap: 48 queries did, and their archives hold 780-800 records each. Forty of the 48 are among the 54 low-overlap claims above (the 35 whose stored queries now return more records than were archived, and 5 of the 19 whose stored queries now return none), and 8 overlap with today’s retrieval at 0.8 or more. The cap removed early history. For 43 capped claims with a re-retrieved set to compare against, the archive begins later than the earliest record available today in 42 (median gap 45 years), covers none of the oldest quartile of today’s records for the median claim, and covers less than half of today’s records for 36 claims; the beta-carotene-cancer claim, for example, holds 797 archived records from 2012-2026 against a literature that begins in 1977. Capped claims accordingly have later cutoffs (median 2021 against 2015). Removing all 48 capped claims leaves 715 units. The signed score’s gain over the margin rule under the primary annotation remains established, +0.070 [0.009, 0.131] against +0.067 [0.017, 0.120] on the full benchmark, and so does its loss under the second annotation, -0.128 [-0.203, -0.049] against -0.096 [-0.166, -0.037]. The streams most plausibly affected by the cap therefore do not produce either central comparison. Rebuild levels: released predictions and outcomes support metric and interval reproduction; association-level labels, identifiers and years support the directional rebuilds listed in the manifest, including counts, operational direction, agreement, endpoints and stationary probabilities, using the documented count-model outputs or fitting procedure; recreating text-derived numeric and design features and rerunning annotation requires the original abstract text and the documented pipeline configurations, and these stages are not reproducible from the released package alone. The dated query-identifier audit does not recreate the original text snapshot. The 145-claim curated list is a fixed artefact whose assembly procedure is not recoverable; the 20 later additions are documented individually. The external cor-

| Quantity                                       | Value                              |
|------------------------------------------------|------------------------------------|
| Archived snapshot (v1): claim queries          | 165                                |
| Archived claim-record associations             | 112,453                            |
| Archived records dated 2026 (partial year)     | 5,818                              |
| Original snapshot retrieval window             | 2026-09-09 to 2026-09-10 13:53 UTC |
| Exact original retrieval timestamp             | not recorded                       |
| Re-retrieval timestamp (UTC)                   | 2026-09-22 06:01:37                |
| Associations returned at re-retrieval (v2)     | 153,428                            |
| Archived associations returned again           | 103,175 (91.7%)                    |
| Distinct archived identifiers / returned again | 96,091 / 88,649 (92.3%)            |
| Distinct identifiers at re-retrieval           | 132,629                            |
| Archived associations no longer returned       | 9,278                              |
| Associations new at re-retrieval               | 50,253                             |
| Per-claim Jaccard, median [min, max]           | 0.992 [0.000, 1.000]               |
| Claims losing > 10% of stored identifiers      | 19                                 |
| External corpora re-retrievable                | no (unit-level files only)         |

Table 16: Snapshot versioning. Version 1 is the stored, non-regenerable derived snapshot on which every experiment runs; version 2 is the timestamped ESearch re-retrieval of the same query strings (identifier lists and hashes only). Per-query counts, translations and identifier hashes are in the released manifest.

pora were shipped as unit-level feature files without query strings and cannot be re-retrieved; their selection (disjoint exposure domains chosen before any transfer result) and freeze (feature builder and extractor revision identical to the source) are recorded in the dataset card.

#### C.4 Notation and terminology

Table 17 lists the notation and terminology used throughout the paper.

### D Stationary Baseline, Order Features and Sign Invariance

#### D.1 Derivation and checks

**Setting.** Fix a claim and a cutoff with  $h = n_t^+$  HARMFUL and  $l = n_t^-$  PROTECTIVE signed pre-cutoff annotations,  $n = h + l > 0$ , and  $m \geq 1$  signed records in the future window. Under the model of Section 4.1 the posterior is  $\theta \mid h, l \sim \text{Beta}(h + \alpha, l + \alpha)$  and the number  $K$  of future HARMFUL labels is Beta-Binomial,  $\text{BetaBin}(k; m, a, b) = \binom{m}{k} B(k + a, m - k + b) / B(a, b)$  with  $a = h + \alpha$ ,  $b = l + \alpha$ . The future pooled direction is  $p_t^+ = (2K - m) / m$ . For the theoretical raw-majority event,  $R_{\text{raw}} = 1$  iff  $\text{sgn}(2K - m) \neq \text{sgn}(h - l)$ , which gives Equation (2). The archived operational event instead compares the future sign with  $s_t = \text{sgn}(p_t)$ ; its probability is  $\pi_t^{\text{op}}(m) = \sum_k \mathbf{1}[\text{sgn}(2k - m) \neq s_t] \text{BetaBin}(k; m, h + \alpha, l + \alpha)$ , with the same posterior, and is the scored stationary reference throughout. The formula is symmetric under  $h \leftrightarrow l$  (checked numerically to  $10^{-15}$ ), finite and in  $[0, 1]$  on the tested domain.

| Term or symbol                                             | Meaning                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PubMed record                                              | One title-abstract-year-identifier tuple returned by a fixed query; called a record in manuscript prose. The historical executed extraction prompt uses the word “Study” verbatim.                                                                                                                                                                                                                         |
| automatic annotation                                       | The extractor-assigned direction of a claim-record association (PROTECTIVE $-1$ , NULL $0$ , HARMFUL $+1$ , UNCLEAR).                                                                                                                                                                                                                                                                                      |
| claim-cutoff unit $(c, t)$                                 | A claim and a calendar year; inputs from records with $\tau_i < t$ .                                                                                                                                                                                                                                                                                                                                       |
| future-eligible historical unit $n_t^+, n_t^-, n_t^0, n_t$ | A unit with at least 20 later records; the main evaluation population (1,012 source units).<br>Pre-cutoff counts of HARMFUL, PROTECTIVE, NULL and signed records.                                                                                                                                                                                                                                          |
| $h, l, n = h + l$                                          | Actual HARMFUL, PROTECTIVE and signed pre-cutoff counts; never replaced by zeros on sparse histories.                                                                                                                                                                                                                                                                                                      |
| $r_t$                                                      | Raw signed mean $(h - l) / n$ , defined for $n > 0$ .                                                                                                                                                                                                                                                                                                                                                      |
| $p_t,  p_t $                                               | Archived operational feature: $r_t$ when $n \geq 8$ , and 0 otherwise; $ p_t $ its absolute margin.                                                                                                                                                                                                                                                                                                        |
| $s_t$                                                      | Operational current sign $\text{sgn}(p_t)$ ; sparse evidence and a supported exact tie both give 0 but are identified separately.                                                                                                                                                                                                                                                                          |
| $R_{\text{raw}} / \pi_t^{\text{raw}}(m)$                   | Theoretical raw-majority event (future raw majority differs from $\text{sgn}(h - l)$ ) and its probability, Equation (2); a labelled diagnostic.                                                                                                                                                                                                                                                           |
| $R / \pi_t^{\text{op}}(m)$                                 | Archived operational event (future sign differs from $s_t$ ; no future gate) and its probability with the same posterior; the scored stationary reference.                                                                                                                                                                                                                                                 |
| uncalibrated / recalibrated $a_t$                          | $\pi_t^{\text{op}}$ without Platt scaling (still operational) versus the fitted logistic recalibration of logit $\pi_t^{\text{op}}$ in Table 19.                                                                                                                                                                                                                                                           |
| $p_t^+, m$                                                 | Agreement: majority share among signed records, or the NULL share when NULL dominates (definition A); the branch-free variant uses the signed-only share on every unit (definition C).<br>Pooled direction and signed count of the subsequent stream.                                                                                                                                                      |
| subsequent-stream sign change $R$                          | Primary endpoint, Equation (1).                                                                                                                                                                                                                                                                                                                                                                            |
| cumulative updated-direction change $R_{\text{cum}}$       | Sign change of the pooled direction over all records after updating.                                                                                                                                                                                                                                                                                                                                       |
| margin rule stationary baseline                            | The untrained score $1 -  p_t $ ; the reference of every $\Delta$ .<br>Count-predicted operational Beta-Binomial probability $\pi_t^{\text{op}}$ (uncalibrated).                                                                                                                                                                                                                                           |
| order features $z_t$                                       | Seven sign-invariant features of the pre-cutoff label order (Appendix D.2).                                                                                                                                                                                                                                                                                                                                |
| retrospective claim-held-out backtest                      | Leave-one-claim-out fitting with training outcomes to the end of the corpus.                                                                                                                                                                                                                                                                                                                               |
| origin-defined calendar cohort $\Delta$                    | Cohort defined at origin $Y$ from pre-origin eligibility; accrual reported as censoring; claim-disjoint training. Each table states its reference and direction: ranking differences are model minus reference (margin rule unless stated); proper-score improvement columns use reference loss minus model loss, so positive values indicate improvement; all intervals use the stated paired resampling. |
| realised-count / count-predicted                           | $\pi_t^{\text{op}}$ with the observed future signed count (diagnostic) or averaged over a training-fitted count model (deployable); $\pi_t^{\text{raw}}$ is reported only as a diagnostic.                                                                                                                                                                                                                 |
| rebuilt vs. primary score                                  | A score refitted on an annotation’s own inputs and endpoint, or the primary score evaluated against another annotation’s endpoint.                                                                                                                                                                                                                                                                         |
| interval types                                             | Claim-bootstrap CI; wild cluster bootstrap- $t$ CI; null range over 1,000 control draws; flip range over 200 recodings; annotation-draw range over 500 soft-label draws.                                                                                                                                                                                                                                   |

Table 17: Notation and terminology used throughout the paper.

**Full statement.** Write  $\pi_t^{\text{raw}}(m)$  for  $\pi_t(m)$  of Lemma 1. (a) For fixed  $n, m$  and  $\alpha$ ,  $\pi_t^{\text{raw}}(m)$  is non-increasing in the raw absolute margin  $|h - l| / n$  among histories with a non-zero raw majority; (b) as  $m \rightarrow \infty$ ,  $\pi_t^{\text{raw}}(m) \rightarrow I_{1/2}(\max(h, l) + \alpha, \min(h, l) + \alpha)$  for  $h \neq l$  and  $\rightarrow 1$  for  $h = l$ , where  $I_x(a, b)$  is the regularised incomplete Beta function; for  $h = l$  and finite  $m$  the probability is

1 for odd  $m$  and  $1 - P(K = m/2 \mid h, l)$  for even  $m$ ; (c) when  $n, m$  and  $\alpha$  are common across units, the margin rule is a ranking surrogate for  $\pi_t^{\text{raw}}$  on the non-tied subset. [Lemma 1](#) states (a) and (c).

*Proof.* (a) Let  $n, m$  and  $\alpha$  be fixed and  $h > l$  (the case  $h < l$  is symmetric). Moving one record from  $l$  to  $h$  replaces the posterior  $\text{Beta}(a, b)$  by  $\text{Beta}(a + 1, b - 1)$ , which is larger in the likelihood-ratio order and therefore stochastically larger. A Beta-Binomial variable is a mixture of Binomial( $m, \theta$ ) laws, each stochastically increasing in  $\theta$ , so  $K$  is stochastically larger under the larger posterior and the lower tail  $P(2K - m \leq 0)$ , the change event for a HARMFUL raw majority, is non-increasing in  $h$ , that is in  $|h - l|/n$ . The argument uses a non-zero raw majority: the change event of a tied history is  $\{2K \neq m\}$ , and the ordering does not extend across the tie ( $\pi^{\text{raw}}(30, 30; 2) = 32/63 \approx 0.508$  but  $\pi^{\text{raw}}(31, 29; 2) = 475/651 \approx 0.730$  with  $\alpha = 1$ ). (b) For  $h > l$  the change event is  $\{2K - m \leq 0\}$ . Conditionally on  $\theta$ ,  $K/m \rightarrow \theta$  almost surely as  $m \rightarrow \infty$ ; the posterior is continuous, so  $P(\theta = 1/2 \mid h, l) = 0$ , and hence  $\pi_t^{\text{raw}}(m) \rightarrow P(\theta < 1/2 \mid h, l) = I_{1/2}(h + \alpha, l + \alpha)$ . For  $h < l$  the change event is  $\{2K - m \geq 0\}$ , giving  $\pi_t^{\text{raw}}(m) \rightarrow P(\theta > 1/2 \mid h, l) = 1 - I_{1/2}(h + \alpha, l + \alpha) = I_{1/2}(l + \alpha, h + \alpha)$ . The two cases give  $I_{1/2}(\max(h, l) + \alpha, \min(h, l) + \alpha)$ . For  $h = l$ , change occurs whenever the future sample is not tied; it therefore has probability 1 for odd  $m$  and  $1 - P(K = m/2 \mid h, l)$  for even  $m$ , and because the posterior has no atom at  $1/2$  the probability of a future exact tie vanishes and the change probability tends to 1. (c) If  $n, m$  and  $\alpha$  are common across units,  $\pi_t^{\text{raw}}$  is a function of  $h - l$  alone, non-increasing in  $|h - l|/n$  by (a) on the non-tied subset and symmetric under  $h \leftrightarrow l$ ; ranking by  $\pi_t^{\text{raw}}$  on that subset is therefore the ranking by  $1 - |h - l|/n$ , with ties within equal margins. The ranking equivalence is restricted to non-tied raw-majority histories; no general ordering of tied histories relative to non-tied histories is asserted.  $\square$

**Executed endpoint contract, ties and sparse evidence.** The endpoint uses  $\text{sgn}(0) = 0$ . The frozen builder returns  $p_t = r_t = (h - l)/(h + l)$  when  $h + l \geq 8$  and  $p_t = 0$  otherwise; the operational current sign is  $s_t = \text{sgn}(p_t)$ , so  $s_t = 0$  for an exact tie with at least eight signed records and for a sparse history. The future side uses the raw signed mean of any non-empty signed sample and 0 when no signed record follows; no future gate is applied,

and no future-eligible source unit has an empty signed future sample. The archived event equals the reproduced operational event on all 1,012 units (0 mismatches). The scored baselines use the operational indicator  $\pi_t^{\text{op}}$ ; the raw-majority indicator  $\pi_t^{\text{raw}}$  differs on sparse histories (one unit under the primary pipeline, ten under the second, fourteen under intersection labels), where  $s_t = 0$  makes any non-zero future sign a change while the posterior still uses the actual counts (for  $h = 7, l = 0$  the posterior is  $\text{Beta}(8, 1)$  and the operational event has probability 1 for odd  $m$ ). Seven source units under the primary pipeline have an exact tie with at least eight signed records; all seven are events. The raw-majority scores are retained as labelled diagnostic columns in the released per-unit files; aligning the indicator changed the count-predicted score on 1 source units (maximum absolute change 0.959). In the count-predicted implementation the drawn future count is floored at one, so the  $m = 0$  branch never occurs; in the Dirichlet-multinomial draws a future window without signed records is a change iff  $s_t \neq 0$ , the endpoint's rule.

**Count-predicted implementation.** The realised  $m$  is unavailable at the cutoff. On the training claims of each fold we fit  $\log(1 + m) = b_0 + b_1 \log(1 + n) + b_2 \log(1 + \text{accrual}_t) + \varepsilon$  by least squares, where accrual is the number of signed records per year over the five years before  $t$  and  $\varepsilon$  is Gaussian with the residual standard deviation; 400 draws of  $m$  are rounded to integers and floored at one; for each distinct drawn count the implementation evaluates the operational probability  $\pi_t^{\text{op}}(m)$  with the current sign  $s_t$  and averages these values with the draw multiplicities (no Monte Carlo sampling of labels). The posterior retains the actual pre-cutoff signed counts. The raw-majority probabilities of [Equation \(2\)](#) are retained separately as diagnostic outputs. The held-out claim is excluded from the count-model fit. An earlier reference draws the ratio  $m/n$  from the training claims instead (*ratio* row of [Table 18](#)). The Dirichlet-multinomial version places  $\text{Dir}(h + 1, l + 1, n_t^0 + 1)$  on the three resolved labels, draws the future total count from the same count model fitted to signed-plus-NUL counts, and compares the sign of the future signed majority with the current sign; a future window with no signed record counts as a change when the current sign is non-zero.

**Numerical checks.** The implementation passes the boundary tests of the corrected proposition (ex-

act 32/63 and 475/651 at  $\alpha = 1$ ; symmetry under  $h \leftrightarrow l$ ; probability 1 for tied histories and odd  $m$ ;  $\pi^{\text{raw}}(8, 0; m) \rightarrow 1/512$  as  $m$  grows; non-increasing in the raw margin on non-tied histories; probability mass sums to one). Only finite sums enter any prediction; the limiting expression is not used. A posterior-predictive Monte Carlo check draws  $\theta^{(b)} \sim \text{Beta}(h + 1, l + 1)$  and  $K^{(b)} \sim \text{Binomial}(m, \theta^{(b)})$ ,  $b = 1, \dots, 2,000$ , for every source unit with its realised  $m$  and the operational current sign: the mean absolute difference between the empirical change frequency and  $\pi_t^{\text{op}}$  is 0.0012 (maximum 0.025, mean Monte Carlo standard error 0.0015; 98.8% of units within two standard errors). A separate plug-in simulation fixes  $\theta = h/n$  and draws 1,000 future samples of the realised size for the operational event; it is a plug-in Binomial comparison, not a posterior draw, and Figure 4A reports it as such: the plug-in Binomial probability matches its own simulation at  $r = 0.999$ , and the flat-prior operational probability correlates with the same simulation at  $r = 0.949$  with a higher expected event total (39.9 against 26.6) because the prior places mass toward balance. Expected event totals are run-specific: the operational realised-count probability sums to 39.9 on the 1,012 units, the raw-majority probability of Equation (2) to 38.9 (the sparse unit contributes the difference), and the raw-majority plug-in to 25.6. The within-claim permutation of this check (1,000 draws that re-draw the pre-cutoff labels as well, operational indicator) gives 35.4 expected events and a per-unit correlation of  $r = 0.44$  with the count-conditional operational probability; the coherent Null A run of Appendix G (1,000 draws, Table 30) has mean 35.3 events and is a separate run, not the same quantity. Non-tied monotonicity holds in all 624 tested  $(n, m)$  configurations.

## D.2 Order features and the offset diagnostic

Let  $z_1, \dots, z_n$  be the signed pre-cutoff labels in publication order (records with equal years keep the archived file order), multiplied by the sign of the current majority so that +1 agrees with the majority and -1 contradicts it, and let  $h = \lfloor n/2 \rfloor$ . The seven features are: (1) *recent minus historical dissent*, the share of -1 among records dated within six years before  $t$  minus the share among earlier records; (2) *recency-weighted margin*,  $\sum_i w_i z_i / \sum_i w_i$  with  $w_i = \exp(-(t - 1 - \tau_i)/5)$ ; (3) *cumulative-margin slope*, the running mean of  $z$  at  $n$  minus its value at  $h$ ; (4)

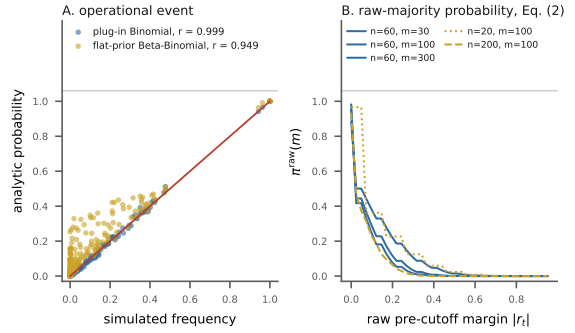


Figure 4: Comparison with a plug-in Binomial simulation. A: per-unit analytic change probability against the change frequency in 1,000 draws with  $\theta$  fixed at the pre-cutoff proportion and the realised future count; the plug-in Binomial matches this simulation and the flat-prior Beta-Binomial adds prior mass toward balance. B: Equation (2) as a function of the pre-cutoff margin for several pre-cutoff and future sample sizes on non-tied histories.

*opposing run length*, the number of consecutive -1 labels at the end of the sequence; (5) *contraction*,  $|\bar{z}_{1:h}| - |\bar{z}_{h+1:n}|$ ; (6) *CUSUM range*,  $(\max_j S_j - \min_j S_j)/n$  with  $S_j = \sum_{i \leq j} (z_i - \bar{z})$ ; (7) *acceleration of contradiction*, the dissent share in the last quarter of the sequence minus the share in the preceding quarter. Units with fewer than four signed records receive zeros. All seven are unchanged when every signed label of the claim is exchanged. The offset model is fitted by unweighted penalised likelihood with the offset fixed; the ridge strength  $C \in \{0.1, 1, 10\}$  is chosen inside each training fold by grouped five-fold log loss over the training claims. Two reference fits use the same offset with composition variables ( $|p_t|$ ,  $a_t$ ,  $\log(1 + n)$ ) and with composition plus order features (Table 20).

**Matched recalibration comparison.** Because the offset model’s intercept recalibrates the uncalibrated operational stationary probability, its proper-score improvement over that offset does not isolate order information. Table 19 therefore compares, with identical estimator (unweighted  $L_2$  logistic,  $C = 1$ ), identical leave-one-claim-out folds and identical bootstrap resamples, a logistic regression on  $\text{logit } \pi_t^{\text{op}}$  alone (slope and intercept recalibration) and on  $\text{logit } \pi_t^{\text{op}}$  plus the seven order features. This is a different estimand from the fixed-offset model of Table 20, whose offset slope is fixed at one.

| Score                                                   | AUROC [95% CI]    | AUPRC [95% CI]    | Log loss | Brier | Cal. slope | ECE  | $\Delta$ AUROC vs. margin | $\Delta$ AUROC vs. two-variable |
|---------------------------------------------------------|-------------------|-------------------|----------|-------|------------|------|---------------------------|---------------------------------|
| Beta-Binomial, realised future count (diagnostic)       | .882 [.816, .942] | .490 [.374, .619] | .272     | .0491 | .31        | .033 | +.000 [−.033, .034]       | −.066 [−.125, −.013]            |
| Stationary Beta-Binomial, count-predicted (operational) | .909 [.867, .949] | .497 [.379, .618] | .200     | .0499 | .53        | .033 | +.028 [.008, .050]        | −.039 [−.080, −.004]            |
| Beta-Binomial, training ratio of counts                 | .877 [.804, .938] | .467 [.337, .600] | .194     | .0498 | .64        | .025 | −.004 [−.055, .044]       | −.071 [−.133, −.016]            |
| Dirichlet-multinomial, realised count (diagnostic)      | .849 [.768, .929] | .483 [.368, .605] | .297     | .0489 | .28        | .031 | −.032 [−.080, .009]       | −.099 [−.176, −.025]            |
| Dirichlet-multinomial, count-predicted                  | .893 [.840, .943] | .492 [.379, .615] | .223     | .0497 | .40        | .030 | +.011 [−.011, .033]       | −.055 [−.100, −.011]            |
| Margin rule 1 − $ p_t $ , untrained                     | .881 [.824, .936] | .468 [.359, .589] | .184     | .0515 | .92        | .012 | reference                 | −                               |
| Two-variable logistic ( $p, a$ )                        | .948 [.921, .971] | .535 [.391, .676] | .146     | .0452 | .94        | .027 | +.067 [.017, .120]        | reference                       |
| Compact-7 logistic                                      | .946 [.918, .969] | .527 [.378, .669] | .149     | .0460 | .94        | .014 | +.064 [.011, .122]        | −.002 [−.011, .006]             |

Table 18: Stationary posterior-predictive references for the archived operational endpoint on the source cohort (1,012 units, 71 events); all rows evaluate the archived operational endpoint  $\pi_t^{\text{op}}$ . The stationary posterior-predictive rows use uncalibrated probabilities; probability metrics for the margin, two-variable and Compact-7 scores use training-only Platt calibration (Table 1 and Appendix E.1). Realised-count rows use observed future counts and are diagnostic rather than origin-available forecasts; count-predicted rows integrate over the training-fitted count model of Appendix D; the ratio row draws the future/past count ratio from the training claims. The raw-majority probability of Equation (2) is a separately labelled diagnostic in the released per-unit files (columns \*\_rawsign), not a row of this table. A prevalence-only forecast has log loss .254 and Brier .0652. The calibrated two-variable score improves on the deployable baseline by .0047 in Brier [−.004, .014] and .054 in log loss [−.001, .123] (paired, positive = better).

| Metric      | Recalibrated OS | + order features | Improvement [95% CI] |
|-------------|-----------------|------------------|----------------------|
| AUROC       | .899            | .916             | +.017 [−.012, .045]  |
| AUPRC       | .454            | .520             | +.066 [−.065, .195]  |
| Log loss    | .167            | .158             | +.009 [−.010, .032]  |
| Brier score | .0487           | .0458            | +.003 [−.004, .011]  |

Table 19: Matched recalibration comparison on the source cohort. The reference is a fitted logistic recalibration of the uncalibrated operational stationary logit  $\pi_t^{\text{op}}$  (a different procedure from the uncalibrated baseline of Table 18 and from the fixed-offset model of Table 20); the augmented model additionally uses the seven order features. Both use unweighted  $L_2$  logistic regression with  $C = 1$ , the same leave-one-claim-out folds and the same claim-bootstrap resamples. Positive values favour the augmented model: augmented minus reference for AUROC and AUPRC, reference minus augmented for log loss and Brier score, computed from unrounded paired predictions. None of the paired intervals excludes zero. **OS: operational stationary**

### D.3 Sign-flip experiments

Direction-coding sensitivity on the source cohort; the invariant model and the augmented models have not been evaluated under transfer.

### D.4 Exact pair decomposition of the AUROC difference

The AUROC of a score is the share of positive-negative unit pairs it orders correctly (ties counted one half), so a paired AUROC difference between two scores is a mean over the same pairs of the difference in their pairwise ordering indicators, and it partitions exactly by pair type. We partition the pairs of the development cell by whether the positive and the negative unit share a pre-cutoff majority class (within-class), have different classes (across-class), or involve at least one zero-direction

unit ( $p_t = 0$ ). In the development cell the 71 positives (42 HARMFUL-majority, 21 PROTECTIVE-majority, 8 zero-direction) and 941 negatives (232 and 709; no zero-direction unit is a negative) form  $71 \times 941 = 66,811$  pairs:  $42 \times 709 + 21 \times 232 = 34,650$  across-class,  $42 \times 232 + 21 \times 709 = 24,633$  within-class and  $8 \times 941 = 7,528$  zero-direction, and each component’s contribution is the sum of its pairs’ indicator differences divided by 66,811. Table 22 reports the signed-minus-margin difference by component under both annotations with claim-clustered intervals. Under the primary annotation the across-class contribution is the only established positive component; the within-class contribution is additive and has a different denominator from the within-class AUROC difference of Table 13, which is normalised over within-class pairs alone; the zero-direction contribution is small and established negative,  $-0.0010$  [−0.0032, −0.0001], which the margin rule’s perfect ranking of the eight  $p_t = 0$  events (all scored  $1 - |p_t| = 1$ ) produces; the components sum to the unrounded total  $+0.0667$ . The percentage shares are descriptive proportions of the estimated total, not interval-bearing quantities. Under the second annotation the within-class component is  $-0.0781$  [−0.1247, −0.0386], 82% of the total  $-0.0957$  and established, the across-class component  $-0.0175$  [−0.0543, 0.0132] is not established, and the zero-direction component is exactly zero because the ten sparse-history events are ranked above every negative by both scores; the loss is therefore not the primary across-class advantage reversing, and the branch-free intervention of Table 14 acts on the within-class component. The margin-stratified rates of Table 33 show the

| Model                              | AUROC [95% CI]    | AUPRC | Log loss | $\Delta$ AUROC vs. offset | $\Delta$ AUPRC vs. offset | $\Delta$ log loss vs. offset (positive = better) |
|------------------------------------|-------------------|-------|----------|---------------------------|---------------------------|--------------------------------------------------|
| Stationary offset only ( $\pi_t$ ) | .909 [.867, .949] | .497  | .200     | reference                 | reference                 | .000 [.000, .000]                                |
| Offset + order features            | .915 [.877, .952] | .514  | .167     | +0.006 [−.016, .028]      | +0.017 [−.063, .097]      | .034 [−.009, .083]                               |
| Offset + composition               | .911 [.871, .946] | .442  | .167     | +0.002 [−.015, .021]      | −.055 [−.108, −.007]      | .033 [−.006, .078]                               |
| Offset + composition + order       | .914 [.877, .949] | .508  | .166     | +0.005 [−.019, .031]      | +0.011 [−.098, .117]      | .035 [−.013, .089]                               |
| Order features only, no offset     | .898 [.851, .942] | .468  | .169     | −.011 [−.036, .015]       | −.028 [−.185, .132]       | .031 [−.021, .092]                               |
| Two-variable logistic              | .948 [.921, .971] | .535  | .146     | +0.039 [.004, .080]       | +0.038 [−.085, .155]      | −                                                |
| Margin rule                        | .881 [.824, .936] | .468  | .184     | −.028 [−.050, −.008]      | −.029 [−.057, −.002]      | −                                                |

Table 20: Fixed-offset diagnostic: incremental value of temporal order beyond the uncalibrated operational stationary probability (source cohort, leave-one-claim-out, nested ridge selection). The offset is the uncalibrated operational stationary probability. Standardised coefficients of the full-data fit: dissent recent vs hist -0.01, recency weighted margin +0.97, cum margin slope -0.29, opp run len +0.01, contraction +0.79, cusum max +0.11, contra accel +0.37.

| Model                              | AUROC [95% CI]    | AUPRC | $\Delta$ AUROC vs. margin | $\Delta$ AUROC vs. signed two-variable | AUROC under claim-wise flips, mean [min, max] |
|------------------------------------|-------------------|-------|---------------------------|----------------------------------------|-----------------------------------------------|
| Margin rule                        | .881 [.824, .936] | .468  | reference                 | −.067 [−.120, −.017]                   | −                                             |
| Two-variable, signed               | .948 [.921, .971] | .535  | +0.067 [.017, .120]       | reference                              | .896 [.886, .917]                             |
| Compact-7, signed                  | .946 [.918, .969] | .527  | +0.064 [.011, .122]       | −.002 [−.011, .006]                    | .897 [.885, .921]                             |
| Sign-invariant ( $ p_t $ , $a_t$ ) | .897 [.839, .945] | .445  | +0.016 [−.017, .057]      | −.051 [−.092, −.013]                   | unchanged                                     |
| Sign-invariant, full               | .913 [.872, .951] | .466  | +0.032 [−.022, .091]      | −.035 [−.068, −.005]                   | unchanged                                     |
| Two-variable, flip-augmented       | .901 [.844, .947] | .436  | −                         | −.047 [−.089, −.011]                   | .901 (full flip)                              |
| Compact-7, flip-augmented          | .900 [.846, .947] | .441  | −                         | −.048 [−.090, −.012]                   | .900 (full flip)                              |

Table 21: Sign-invariance experiments on the source cohort. Claim-wise flips exchange the signed labels of exactly half of the claims (all cutoffs of a claim together) in each of 200 draws and refit the signed models leave-one-claim-out; a full exchange of every claim leaves the signed scores unchanged because the fitted sign of  $\beta$  reverses. Direction strata of the signed two-variable score: PROTECTIVE-majority units (730 units, 21 events) AUROC .969 (margin .959, sign-invariant .925); HARMFUL-majority units (274, 42) .863 (margin .833, sign-invariant .905).

| Component                                                                         | Pairs  | Contribution [95% CI]   | Share |
|-----------------------------------------------------------------------------------|--------|-------------------------|-------|
| <i>Primary, shipped A (<math>\Delta = +.0667</math> [.017, .120])</i>             |        |                         |       |
| Across-class                                                                      | 34,650 | +0.0609 [.0159, .1081]  | 91%   |
| Within-class                                                                      | 24,633 | +0.0068 [−.0039, .0195] | 10%   |
| Zero-direction                                                                    | 7,528  | −.0010 [−.0032, −.0001] | −2%   |
| <i>Second annotation, shipped A (<math>\Delta = −.0957</math> [−.166, −.037])</i> |        |                         |       |
| Within-class                                                                      | −      | −.0781 [−.1247, −.0386] | 82%   |
| Across-class                                                                      | −      | −.0175 [−.0543, .0132]  | 18%   |
| Zero-direction                                                                    | −      | 0 [0, 0]                | 0%    |

Table 22: Pair decomposition of the paired signed-minus-margin AUROC difference (development cell, leave-one-claim-out predictions; claim-clustered 95% intervals over the same 2,000 resamples). Pair counts follow from the class split of Table 3 and were verified against the decomposition script; contributions are additive and sum to the unrounded totals. Second-annotation pair counts are in the release.

class-rate difference read by strata: no PROTECTIVE-majority event is observed above the lowest quintile, among units that repeat claim cutoffs and are not independent.

## E Origin-Defined Cohort and Full Metric Suite

### E.1 Full metric suite

Tables 24 to 26 report AUROC, AUPRC, log loss, Brier score, calibration slope and intercept, expected calibration error, and precision and recall at 5, 10 and 20% review budgets for the model-cohort combinations evaluated. The offset and

sign-invariant diagnostics are source-only. Stationary rows use uncalibrated operational probabilities; the score-calibration treatments are specified in the captions (training-only Platt scaling: leave-one-claim-out on the source, per-origin training folds in the calendar cohort, the frozen source calibrator on the external corpora). Table 19 is a separate matched-recalibration experiment. Pooled review-budget metrics are retrospective ranking summaries and rank units by the uncalibrated score.

**Inference sensitivity.** Table 23 recomputes the paired  $\Delta$ AUROC intervals of Table 2 under procedures suited to few positive clusters: per-unit DeLong placement contributions to  $\Delta$  are aggregated to claim level and a wild cluster bootstrap- $t$  with Webb six-point weights (9,999 draws) and a leave-one-claim-out jackknife are applied to seven cells (the development, second-annotation, intersection and three calendar contrasts of Table 2, plus signed minus class-blind); a clustered-ROC sensitivity treats claims as the sampling unit of the ROC estimates themselves: the per-unit DeLong placement contributions of both scores are aggregated to claim level, and the variance of the paired difference is the claim-clustered variance of the per-claim contributions to  $\hat{A}_1 - \hat{A}_2$ , so that the covariance between the two AUROC estimates enters through the paired contributions ( $\text{Var}(\hat{A}_1 - \hat{A}_2) = \text{Var} \hat{A}_1 + \text{Var} \hat{A}_2 - 2 \text{Cov}(\hat{A}_1, \hat{A}_2)$ ) rather than be-

ing dropped; it was run on those seven cells and on the pooled within-class and two external cells of Table 2; and a claim-level bootstrap that refits the score inside every resample, rather than resampling fixed out-of-fold predictions, was run for the four source-cohort cells (the calendar cells would require rebuilding the per-origin training grid inside every replicate and were not run) (2,000 resamples; every duplicate copy of a held-out claim drawn by the resampling is removed from that claim’s training folds; no resample is discarded, and a fold whose training side holds a single class assigns that class’s mean). Table 23 names the procedure behind each column and marks the cells each covered. The tested superiority and loss decisions against zero are unchanged: the development gain remains established ( $[0.018, 0.115]$  under the wild cluster bootstrap,  $[0.016, 0.117]$  under the clustered ROC variance,  $[0.014, 0.117]$  under refitting), the second-annotation loss remains established ( $[-0.158, -0.033]$  and  $[-0.242, -0.039]$ ), the External-2 loss remains established under the clustered variance ( $[-0.122, -0.025]$ ), and no non-established cell becomes established against zero. One non-inferiority subdecision is procedure-sensitive: External-1 is not established as non-inferior under the primary percentile interval (lower bound  $-0.052$ ) but is under the clustered-ROC interval (lower bound  $-0.012$ , above  $-\delta = -0.02$ ); the protocol specifies the percentile interval as primary, so the Table 2 verdict stands, and the complete transfer check and the tier assignments are unchanged because External-2 fails. The refitting interval for the intersection contrast,  $[-0.177, 0.020]$ , is about 1.7 times as wide as the percentile interval ( $0.197$  against  $0.117$ ), an observed widening on a cell with 19 positive claims rather than an established relationship. The procedures do not give numerically identical intervals. Their agreement on the decisions against zero supports the stability of those decisions to the tested inference choices; it does not establish nominal coverage, remove the retrospective specification of the cells, or bear on annotation validity. The two single-class within-class cells of Table 2 received the percentile bootstrap only (under the second annotation the same-class-pairs contrast of Table 13,  $-0.298$ , has clustered interval  $[-0.461, -0.135]$ ), and the A-to-C intervention of Table 14 received the percentile bootstrap for C minus A and C minus margin only.

## E.2 Origin-defined calendar cohort

Table 27 is the census of the cohort of Section 3: every claim with at least 40 records before the origin is a candidate; accrual is at least ten signed records in the window; windows ending after 2025 are excluded rather than evaluated on a partially observed year (the stored corpus holds 5,818 records dated 2026). The accrual model is a class-balanced logistic regression on  $\log(1 + n_t)$ ,  $\log(1 + \text{accrual}_t)$  and  $\log(1 + n_{\text{pre}})$  fitted on all earlier cohort units of the other claims. Training for the change model uses every earlier unit with an observable outcome on a two-year cutoff grid, with the test claim’s own units removed; one fit per test claim and origin. Figure 5 shows the per-origin values.

**Accrual-weighted sensitivity.** Accrued and censored candidates differ on pre-origin features, most in recent accrual (up to 1.13 standard deviations). Weighting each accrued unit by the inverse of its predicted probability of accrual, estimated from pre-origin features on claim-disjoint units, moves no paired difference by more than 0.014, and every weighted comparison reaches the same conclusion as Table 2. At ten years no candidate is censored, so no weighting is needed.

## F Frozen External Transfer and the Two-Variable Score

Refitting a balanced logistic model on eight named signals separately in each domain keeps the signs of agreement (negative), heterogeneity  $I^2$  (negative) and volume (positive) in all three domains, while recent dissent, design mix and the decline signal change sign; a refitted coefficient’s sign is weak evidence about the transport of a frozen score, so this is recorded only as context for Table 28 (the full coefficient table is in the released results).

**The two-variable score.** It is the same class-balanced  $L_2$  logistic regression as Compact-7, fitted on standardised  $[p, a]_t$  only. Table 29 reports it leave-one-claim-out on the source under each agreement definition of Appendix K, with paired differences from Compact-7, and frozen (fitted once on the 1,012 source units, never refitted) on External-1 and External-2, where agreement is rebuilt from the released label counts under the same definition. Under the shipped definition A it gives 0.948 against 0.946 on the source ( $\Delta +0.002 [-0.006, 0.011]$ ), 0.948 against 0.951 on External-1 ( $-0.004 [-0.017, 0.002]$ ) and 0.823 against

| Cell                              | $\Delta$ | Percentile     | Wild cluster- $t$ | Jackknife      | Clustered ROC  | Refitting      | Pos. cl. |
|-----------------------------------|----------|----------------|-------------------|----------------|----------------|----------------|----------|
| Prim., signed — margin            | +0.067   | [.017, .120]   | [.018, .115]      | [.012, .121]   | [.016, .117]   | [.014, .117]   | 23/160   |
| Prim., signed — class-blind       | +0.051   | [.013, .092]   | [.013, .088]      | [.010, .091]   | [.013, .088]   | [.015, .120]   | 23/160   |
| Second, signed — margin           | −0.096   | [−.166, −.037] | [−.157, −.034]    | [−.163, −.028] | [−.158, −.033] | [−.242, −.039] | 16/160   |
| Intersect., signed — margin       | −0.030   | [−.094, .023]  | [−.082, .022]     | [−.094, .034]  | [−.089, .029]  | [−.177, .020]  | 19/160   |
| Calendar $H=3$                    | +0.006   | [−.012, .023]  | [−.011, .023]     | [−.012, .024]  | [−.011, .023]  | —              | 25/158   |
| Calendar $H=5$                    | +0.023   | [−.005, .053]  | [−.003, .050]     | [−.007, .053]  | [−.005, .051]  | —              | 19/142   |
| Calendar $H=10$                   | +0.008   | [−.032, .047]  | [−.028, .044]     | [−.033, .049]  | [−.029, .046]  | —              | 17/122   |
| Pooled within-class pairs (prim.) | +0.018   | [−.012, .060]  | —                 | —              | [−.012, .049]  | —              | 23/160   |
| External-1, frozen                | +0.055   | [−.052, .106]  | —                 | —              | [−.012, .121]  | —              | 7/46     |
| External-2, frozen                | −0.074   | [−.123, −.024] | —                 | —              | [−.122, −.025] | —              | 40/180   |

Table 23: Paired  $\Delta$ AUROC intervals under four inference procedures beyond the claim-clustered percentile bootstrap of the main text (2,000 resamples): a wild cluster bootstrap- $t$  over claims with Webb six-point weights (9,999 draws), a leave-one-claim-out jackknife, a clustered-ROC sensitivity (claim-level DeLong variance of each AUROC; run for all ten cells), and a claim-level bootstrap with the score refitted in every resample (2,000 resamples, seed 3; run for the four source-cohort cells). “—”: not run. “Pos. cl.”: claims with at least one event / claims in the cell. The two single-class within-class cells of Table 2 received the percentile bootstrap only.

| Score                                                                         | AUROC | AUPRC | Log loss | Brier | Cal. slope | Cal. int. | ECE  | P@5  | R@5  | P@10 | R@10 | P@20 | R@20 | NNR@10 | Share@R.8 |
|-------------------------------------------------------------------------------|-------|-------|----------|-------|------------|-----------|------|------|------|------|------|------|------|--------|-----------|
| <i>Source, claim-held-out (1,012 units, 71 events), 1012 units, 71 events</i> |       |       |          |       |            |           |      |      |      |      |      |      |      |        |           |
| Margin rule 1 — $ p_t $ , untrained                                           | .881  | .468  | .184     | .0515 | .924       | −.169     | .012 | .490 | .352 | .386 | .549 | .267 | .761 | 2.590  | .222      |
| Stationary Beta-Binomial, count-predicted (operational)                       | .909  | .497  | .200     | .0499 | .534       | .002      | .033 | .510 | .366 | .386 | .549 | .282 | .803 | 2.590  | .179      |
| Beta-Binomial, realised future count (diagnostic)                             | .882  | .490  | .272     | .0491 | .314       | −.515     | .033 | .549 | .394 | .406 | .577 | .272 | .775 | 2.463  | .215      |
| Dirichlet-multinomial, count-predicted                                        | .893  | .492  | .223     | .0497 | .395       | −.346     | .030 | .510 | .366 | .386 | .549 | .277 | .789 | 2.590  | .205      |
| Two-variable logistic ( $p, \alpha$ )                                         | .948  | .535  | .146     | .0452 | .936       | −.090     | .027 | .529 | .380 | .475 | .676 | .327 | .930 | 2.104  | .139      |
| Compact-7 logistic                                                            | .946  | .527  | .149     | .0460 | .936       | −.092     | .014 | .549 | .394 | .455 | .648 | .327 | .930 | 2.196  | .141      |
| Random forest (all 55)                                                        | .938  | .545  | .149     | .0452 | .949       | −.059     | .014 | .529 | .380 | .475 | .676 | .302 | .859 | 2.104  | .151      |
| XGBoost (all 55)                                                              | .932  | .543  | .152     | .0440 | .958       | −.057     | .013 | .608 | .437 | .485 | .690 | .302 | .859 | 2.061  | .151      |
| Stationary offset + order features                                            | .915  | .514  | .167     | .0477 | .668       | −.461     | .022 | .549 | .394 | .436 | .620 | .282 | .803 | 2.295  | .181      |
| Sign-invariant logistic                                                       | .913  | .466  | .168     | .0495 | .952       | −.075     | .014 | .510 | .366 | .426 | .606 | .287 | .817 | 2.349  | .177      |
| Two-variable, flip-augmented                                                  | .901  | .436  | .176     | .0513 | .932       | −.109     | .006 | .490 | .352 | .396 | .563 | .277 | .789 | 2.525  | .202      |

Table 24: Full metric suite, source cohort(s). P@ $b$  and R@ $b$ : precision and recall when the  $b\%$  highest-scored units are reviewed; NNR@10: units reviewed per event found at the 10% budget; Share@R.8: share of units that must be reviewed to reach recall 0.8 (workload reduction is one minus this share). Calibration slope and intercept are from a logistic regression of the outcome on the logit of the calibrated probability; ECE uses ten equal-width bins.

| Score                                                   | AUROC | AUPRC | Log loss | Brier | Cal. slope | Cal. int. | ECE  | P@5  | R@5  | P@10 | R@10 | P@20 | R@20 | NNR@10 | Share@R.8 |
|---------------------------------------------------------|-------|-------|----------|-------|------------|-----------|------|------|------|------|------|------|------|--------|-----------|
| <i>Calendar cohort, 3 y, 756 units, 58 events</i>       |       |       |          |       |            |           |      |      |      |      |      |      |      |        |           |
| Margin rule 1 — $ p_t $ , untrained                     | .907  | .405  | .193     | .0573 | 1.151      | .272      | .026 | .447 | .293 | .368 | .483 | .298 | .776 | 2.714  | .208      |
| Stationary Beta-Binomial, count-predicted (operational) | .920  | .448  | .184     | .0549 | .674       | .145      | .028 | .500 | .328 | .408 | .534 | .318 | .828 | 2.452  | .185      |
| Two-variable logistic ( $p, \alpha$ )                   | .913  | .351  | .179     | .0554 | 1.121      | .117      | .015 | .395 | .259 | .408 | .534 | .305 | .793 | 2.452  | .206      |
| Compact-7 logistic                                      | .890  | .385  | .203     | .0575 | .778       | −.332     | .020 | .474 | .310 | .382 | .500 | .305 | .793 | 2.621  | .218      |
| <i>Calendar cohort, 5 y, 625 units, 45 events</i>       |       |       |          |       |            |           |      |      |      |      |      |      |      |        |           |
| Margin rule 1 — $ p_t $ , untrained                     | .897  | .372  | .195     | .0572 | .912       | −.021     | .022 | .419 | .289 | .323 | .444 | .256 | .711 | 3.100  | .246      |
| Stationary Beta-Binomial, count-predicted (operational) | .901  | .394  | .208     | .0550 | .485       | −.172     | .031 | .484 | .333 | .355 | .489 | .256 | .711 | 2.818  | .227      |
| Two-variable logistic ( $p, \alpha$ )                   | .920  | .380  | .169     | .0522 | 1.168      | .385      | .014 | .387 | .267 | .403 | .556 | .288 | .800 | 2.480  | .190      |
| Compact-7 logistic                                      | .893  | .364  | .198     | .0590 | .674       | −.629     | .041 | .323 | .222 | .387 | .533 | .256 | .711 | 2.583  | .218      |
| <i>Calendar cohort, 10 y, 492 units, 35 events</i>      |       |       |          |       |            |           |      |      |      |      |      |      |      |        |           |
| Margin rule 1 — $ p_t $ , untrained                     | .867  | .285  | .210     | .0617 | .749       | −.386     | .028 | .320 | .229 | .286 | .400 | .224 | .629 | 3.500  | .256      |
| Stationary Beta-Binomial, count-predicted (operational) | .863  | .309  | .298     | .0598 | .314       | −.468     | .041 | .440 | .314 | .286 | .400 | .235 | .657 | 3.500  | .293      |
| Two-variable logistic ( $p, \alpha$ )                   | .876  | .303  | .190     | .0564 | .947       | −.068     | .010 | .360 | .257 | .347 | .486 | .255 | .714 | 2.882  | .234      |
| Compact-7 logistic                                      | .797  | .218  | .381     | .0942 | .241       | −1.820    | .090 | .240 | .171 | .265 | .371 | .184 | .514 | 3.769  | .392      |

Table 25: Full metric suite, forward cohort(s). P@ $b$  and R@ $b$ : precision and recall when the  $b\%$  highest-scored units are reviewed; NNR@10: units reviewed per event found at the 10% budget; Share@R.8: share of units that must be reviewed to reach recall 0.8 (workload reduction is one minus this share). Calibration slope and intercept are from a logistic regression of the outcome on the logit of the calibrated probability; ECE uses ten equal-width bins.

0.819 on External-2 (+0.004 [−0.004, 0.016]); on External-2 the unfitted  $1 - |p_t|$  (0.896) is above both frozen scores in AUROC (−0.074 [−0.123, −0.024] for the two-variable score) and in AUPRC (−0.220 [−0.308, −0.104]), and the frozen scores are above it on External-1 without an established gain (+0.055 [−0.052, 0.106]), so dropping the count and effect variables does not remove the transfer gap. The gap is a transfer loss rather than an inability of

the two inputs to rank the domain: refitted inside External-2 leave-one-claim-out, the two-variable score’s difference from the margin rule is −0.017 [−0.047, 0.009] (External-1: +0.054 [−0.068, 0.116]), so letting the model adapt to the target domain recovers most of what the frozen check measures.

| Score                                                   | AUROC | AUPRC | Log loss | Brier | Cal. slope | Cal. int. | ECE  | P@5  | R@5  | P@10 | R@10 | P@20 | R@20 | NNR@10 | Share@R.8 |
|---------------------------------------------------------|-------|-------|----------|-------|------------|-----------|------|------|------|------|------|------|------|--------|-----------|
| <i>External-1, frozen, 322 units, 19 events</i>         |       |       |          |       |            |           |      |      |      |      |      |      |      |        |           |
| Margin rule 1 – $ p_e $ , untrained                     | .893  | .505  | .157     | .0415 | 1.192      | -.076     | .033 | .438 | .368 | .281 | .474 | .219 | .737 | 3.556  | .261      |
| Stationary Beta-Binomial, count-predicted (operational) | .907  | .507  | .160     | .0442 | .609       | -.397     | .045 | .438 | .368 | .281 | .474 | .234 | .789 | 3.556  | .211      |
| Two-variable logistic ( $p, \alpha$ )                   | .948  | .679  | .114     | .0330 | 1.073      | .036      | .019 | .563 | .474 | .469 | .789 | .281 | .947 | 2.133  | .109      |
| Compact-7 logistic                                      | .951  | .662  | .112     | .0326 | 1.188      | .030      | .016 | .563 | .474 | .438 | .737 | .281 | .947 | 2.286  | .109      |
| <i>External-2, frozen, 1257 units, 95 events</i>        |       |       |          |       |            |           |      |      |      |      |      |      |      |        |           |
| Margin rule 1 – $ p_e $ , untrained                     | .896  | .487  | .183     | .0522 | 1.252      | .512      | .017 | .524 | .347 | .421 | .558 | .283 | .747 | 2.377  | .231      |
| Stationary Beta-Binomial, count-predicted (operational) | .898  | .486  | .207     | .0543 | .626       | .148      | .035 | .540 | .358 | .429 | .568 | .267 | .705 | 2.333  | .242      |
| Two-variable logistic ( $p, \alpha$ )                   | .823  | .267  | .265     | .0742 | .439       | -1.343    | .052 | .333 | .221 | .341 | .453 | .243 | .642 | 2.930  | .345      |
| Compact-7 logistic                                      | .819  | .257  | .260     | .0739 | .470       | -1.312    | .051 | .302 | .200 | .333 | .442 | .223 | .589 | 3.000  | .353      |

Table 26: Full metric suite, external cohort(s). P@ $b$  and R@ $b$ : precision and recall when the  $b\%$  highest-scored units are reviewed; NNR@10: units reviewed per event found at the 10% budget; Share@R.8: share of units that must be reviewed to reach recall 0.8 (workload reduction is one minus this share). Calibration slope and intercept are from a logistic regression of the outcome on the logit of the calibrated probability; ECE uses ten equal-width bins.

| Origin           | Window       | Candidates | Accrued | Censored | Events | Rate | AUROC among accrued: stationary / margin / two-var / Compact-7 |      |      |      | Status                                  |
|------------------|--------------|------------|---------|----------|--------|------|----------------------------------------------------------------|------|------|------|-----------------------------------------|
| Horizon 3 years  |              |            |         |          |        |      |                                                                |      |      |      |                                         |
| 2004             | [2004, 2007] | 68         | 65      | 3        | 4      | .062 | .975                                                           | .939 | .943 | .930 | complete                                |
| 2007             | [2007, 2010] | 88         | 82      | 6        | 10     | .122 | .926                                                           | .926 | .933 | .924 | complete                                |
| 2010             | [2010, 2013] | 99         | 92      | 7        | 7      | .076 | .940                                                           | .916 | .887 | .903 | complete                                |
| 2013             | [2013, 2016] | 115        | 112     | 3        | 11     | .098 | .908                                                           | .904 | .897 | .820 | complete                                |
| 2016             | [2016, 2019] | 122        | 117     | 5        | 7      | .060 | .916                                                           | .929 | .933 | .935 | complete                                |
| 2019             | [2019, 2022] | 143        | 136     | 7        | 9      | .066 | .907                                                           | .877 | .901 | .893 | complete                                |
| 2022             | [2022, 2025] | 165        | 152     | 13       | 10     | .066 | .928                                                           | .914 | .934 | .929 | complete                                |
| Horizon 5 years  |              |            |         |          |        |      |                                                                |      |      |      |                                         |
| 2004             | [2004, 2009] | 68         | 66      | 2        | 4      | .061 | .964                                                           | .927 | .932 | .875 | complete                                |
| 2007             | [2007, 2012] | 88         | 86      | 2        | 10     | .116 | .882                                                           | .890 | .897 | .842 | complete                                |
| 2010             | [2010, 2015] | 99         | 97      | 2        | 8      | .083 | .898                                                           | .882 | .930 | .892 | complete                                |
| 2013             | [2013, 2018] | 115        | 113     | 2        | 9      | .080 | .933                                                           | .928 | .937 | .922 | complete                                |
| 2016             | [2016, 2021] | 122        | 121     | 1        | 7      | .058 | .902                                                           | .925 | .946 | .917 | complete                                |
| 2019             | [2019, 2024] | 143        | 142     | 1        | 7      | .049 | .892                                                           | .889 | .921 | .941 | complete                                |
| 2022             | [2022, 2027] | 165        | –       | –        | –      | –    | –                                                              | –    | –    | –    | window extends past 2025: not evaluated |
| Horizon 10 years |              |            |         |          |        |      |                                                                |      |      |      |                                         |
| 2004             | [2004, 2014] | 68         | 68      | 0        | 5      | .074 | .924                                                           | .844 | .740 | .816 | complete                                |
| 2007             | [2007, 2017] | 88         | 88      | 0        | 8      | .091 | .875                                                           | .905 | .900 | .798 | complete                                |
| 2010             | [2010, 2020] | 99         | 99      | 0        | 5      | .051 | .832                                                           | .860 | .892 | .753 | complete                                |
| 2013             | [2013, 2023] | 115        | 115     | 0        | 8      | .070 | .832                                                           | .838 | .851 | .771 | complete                                |
| 2016             | [2016, 2026] | 122        | 122     | 0        | 9      | .074 | .884                                                           | .883 | .920 | .898 | complete                                |
| 2019             | [2019, 2029] | 143        | –       | –        | –      | –    | –                                                              | –    | –    | –    | window extends past 2025: not evaluated |
| 2022             | [2022, 2032] | 165        | –       | –        | –      | –    | –                                                              | –    | –    | –    | window extends past 2025: not evaluated |

Table 27: Census of the origin-defined calendar cohort. Candidates are claims with at least 40 records before the origin; censored units have fewer than ten signed records in the window; the rate is the event rate among accrued units. Per-origin AUROCs are claim-disjoint and rest on 4-11 events. Accrual and events are not reported for windows that extend past 2025.

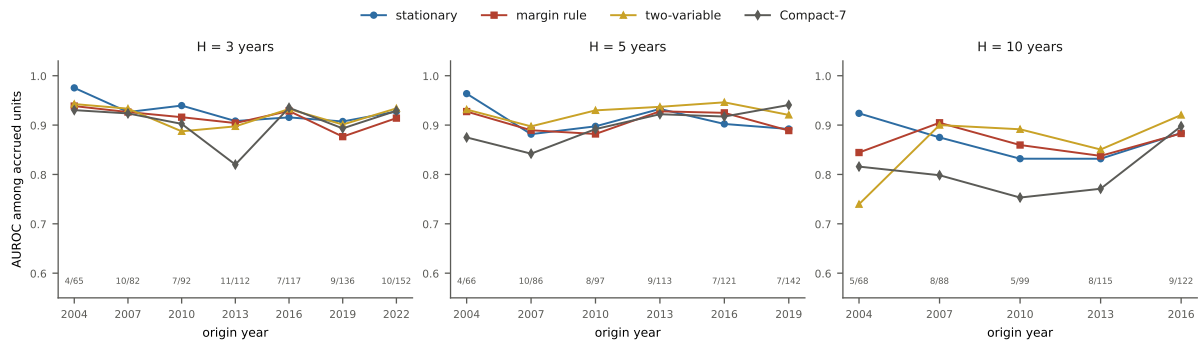


Figure 5: Origin-defined calendar cohort: AUROC among accrued units by origin and horizon for the stationary Beta-Binomial baseline, the margin rule, the two-variable score and Compact-7 (claim-disjoint training, complete windows only). Labels give events/accrued units per origin; per-origin values rest on 4-11 events and are indicative only.

## G Stationary Null Models

Two null models test whether the observed ranking could arise from a claim’s stationary label composition without any temporal organisation.

Null A permutes each claim’s direction labels across its records (publication years fixed); Null B draws each record’s label i.i.d. from the claim’s own label frequencies, so Null A preserves each claim’s realised label counts exactly while Null

| Domain                                 | Score             | AUROC [95% CI]           | AUPRC [95% CI]           |
|----------------------------------------|-------------------|--------------------------|--------------------------|
| External-1<br>(322 units, 19 events)   | Compact-7, frozen | <b>.951</b> [.906, .993] | <b>.662</b> [.417, .874] |
|                                        | $1 -  p_t $       | .893 [.838, .977]        | .505 [.309, .811]        |
|                                        | Low agreement     | .856 [.796, .965]        | .437 [.200, .693]        |
|                                        | Recent dissent    | .737 [.371, .963]        | .341 [.052, .662]        |
| External-2<br>(1,257 units, 95 events) | Compact-7, frozen | .819 [.758, .871]        | .257 [.167, .387]        |
|                                        | $1 -  p_t $       | <b>.896</b> [.856, .932] | <b>.487</b> [.358, .599] |
|                                        | Low agreement     | .882 [.839, .916]        | .392 [.257, .529]        |
|                                        | Recent dissent    | .887 [.852, .917]        | .349 [.253, .454]        |

Table 28: Frozen external categorical validation under the tie convention of Equation (1). Compact-7 is fitted once on the 1,012 source future-eligible units and never refitted on target-domain labels; the three unfitted signals need no fitting. Intervals are claim-clustered over the external claims.

B preserves the generating probabilities and expected frequencies, not the exact counts of every draw. For every draw the direction block, Compact-7’s direction inputs and the target are rebuilt from the permuted or simulated stream, the text-derived effect-estimate inputs are kept, and the full leave-one-claim-out pipeline is rerun for Compact-7 and the two-variable model; the pipeline reproduces the observed 0.946 exactly when run on the real stream. Each null is run for 1,000 draws split over eight seeds. All 1,000 draws of each null were evaluable. Null A: Compact-7 null mean 0.936 (SD 0.023, 95% null interval [0.882, 0.967]), two-variable 0.948 [0.907, 0.974], unfitted  $1 - |p_t|$  0.957 [0.929, 0.977], Compact-7 AUPRC 0.351 [0.192, 0.519], events 35.3 [24, 47], prevalence 0.035. Null B: 0.935 [0.873, 0.970], 0.947 [0.898, 0.975], 0.957 [0.924, 0.979], AUPRC 0.362 [0.171, 0.570], events 36.1 [21, 52], prevalence 0.036. Observed: 0.946, 0.948, 0.881, 0.527, 71, 0.070. Empirical upper-tail  $p$ -values: Compact-7 AUROC 0.36 and 0.40, two-variable AUROC 0.57 and 0.55, Compact-7 AUPRC 0.024 and 0.067, events and prevalence 0.001 under both nulls; the unfitted  $1 - |p_t|$  is *below* its null (lower-tail  $p = 0.002$  and 0.001). The observed fitted-score AUROCs are not distinguishable from these null distributions, which is not an equivalence test and does not show that the histories contain no temporal information; it shows that high AUROC alone does not establish incremental temporal-order information. The nulls are rejected on the number of events: the observed annotation streams exhibit sign-change events about twice as often as a stationary process with the same composition, and those changes are less tied to  $|p_t|$ , which is why the margin rule ranks the observed outcomes (0.881) worse than the null outcomes

(0.957) while the fitted scores do not. Table 30 and Figure 6 report the null distributions, the observed values and the empirical upper-tail  $p$ -values,  $(\#\{\text{draws} \geq \text{observed}\} + 1)/(n + 1)$ . Because the nulls keep every claim’s label composition, a rejected null would show only that the ranking depends on the temporal ordering of the labels, not that a causal temporal mechanism exists. The quantities that are distinguishable are the event count and the prevalence. The difference between the two-variable and margin-rule null means is  $-0.009$  (Null A) and  $-0.010$  (Null B), against  $+0.067$  observed; paired within each draw, the observed difference lies above the null distribution ( $p = 0.001$ ). AUPRC is treated separately: it depends on both ranking and prevalence, the prevalence differs between observed and null streams, one Compact-7 contrast ( $p = 0.067$ ) and both two-variable contrasts (0.112 and 0.189) do not cross 0.05, and no common-prevalence comparison is available, so no temporal-discrimination gain is inferred from AUPRC.

#### Every annotation, and drift-preserving controls.

A reimplementation that rebuilds the target and reruns the leave-one-claim-out two-variable model (not Compact-7) extends both nulls to the second annotation and intersection labels, and adds Null C, which permutes labels within claim and decade so that the era-level label mix of every claim is preserved, 1,000 draws each (Table 32). On the primary annotation it gives 35.6 [26, 46] and 36.5 [20, 52] null events, against the archived 35.3 and 36.1. The excess of events over Nulls A and B holds under the second annotation for both nulls and under intersection labels for Null A only (Null B:  $p = 0.054$ ). Null C absorbs about 72% of the primary excess (71 against 61.2 [53, 69],  $p = 0.008$ ); under the second annotation and intersection labels the observed counts fall inside its range ( $p = 0.475$  and 0.627 in this reimplementation run; 0.489 and 0.641 in the independent partition-grid run of Table 31, a Monte Carlo difference between two draws of 1,000). Null C also reproduces the signed score’s paired gain over the margin rule under the primary annotation ( $+0.067$  observed against a null mean of  $+0.065$  [0.047, 0.083],  $p = 0.427$ ), whereas under Nulls A and B the same difference has  $p = 0.001$ : preserving the era-level label mix suffices to reproduce the gain under this scheme, which does not identify the drift’s contribution. The paired signed-

| Def. | Domain     | Units | Events | Two-var AUROC [95% CI] | AUPRC | C-7, same def. | $\Delta$ vs C-7 [95% CI] | $1 -  p_t $ |
|------|------------|-------|--------|------------------------|-------|----------------|--------------------------|-------------|
| A    | source     | 1012  | 71     | .948 [.921, .971]      | .535  | .946           | +.002 (no interval)      | .881        |
| A    | External-1 | 322   | 19     | .948 [.892, .994]      | .679  | .951           | -.004 [-.017, .002]      | .893        |
| A    | External-2 | 1257  | 95     | .823 [.766, .874]      | .267  | .819           | +.004 [-.004, .016]      | .896        |
| B    | source     | 1012  | 71     | .890 [.830, .939]      | .387  | .899           | -.009 (no interval)      | .881        |
| B    | External-1 | 322   | 19     | .895 [.812, .990]      | .478  | .893           | +.002 [-.020, .025]      | .893        |
| B    | External-2 | 1257  | 95     | .670 [.608, .733]      | .115  | .680           | -.010 [-.032, .015]      | .896        |
| C    | source     | 1012  | 71     | .925 [.890, .959]      | .500  | .925           | -.001 (no interval)      | .881        |
| C    | External-1 | 322   | 19     | .964 [.932, .994]      | .657  | .961           | +.003 [-.004, .008]      | .893        |
| C    | External-2 | 1257  | 95     | .841 [.785, .894]      | .398  | .833           | +.009 [-.006, .029]      | .896        |
| D    | source     | 1012  | 71     | .849 [.758, .922]      | .349  | .858           | -.009 (no interval)      | .881        |
| D    | External-1 | 322   | 19     | .895 [.794, .994]      | .514  | .876           | +.018 [-.023, .065]      | .893        |
| D    | External-2 | 1257  | 95     | .601 [.540, .667]      | .085  | .646           | -.045 [-.081, -.006]     | .896        |

Table 29: Two-variable score under each agreement definition (A shipped; B three-way maximum share; C signed-only share; D entropy consensus), leave-one-claim-out on the source and frozen on the external corpora with agreement rebuilt from the released label counts. The comparator is Compact-7 rebuilt under the *same* definition (external rows: paired interval on identical claim resamples; source rows: point difference only). The paired source differences of each definition’s two-variable score from the definition-A score ( $-0.058$ ,  $-0.023$ ,  $-0.099$  for B, C, D) are in Table 37. The margin rule is definition-independent. On External-2 the branch-free score (0.841) remains below the margin rule (0.896).

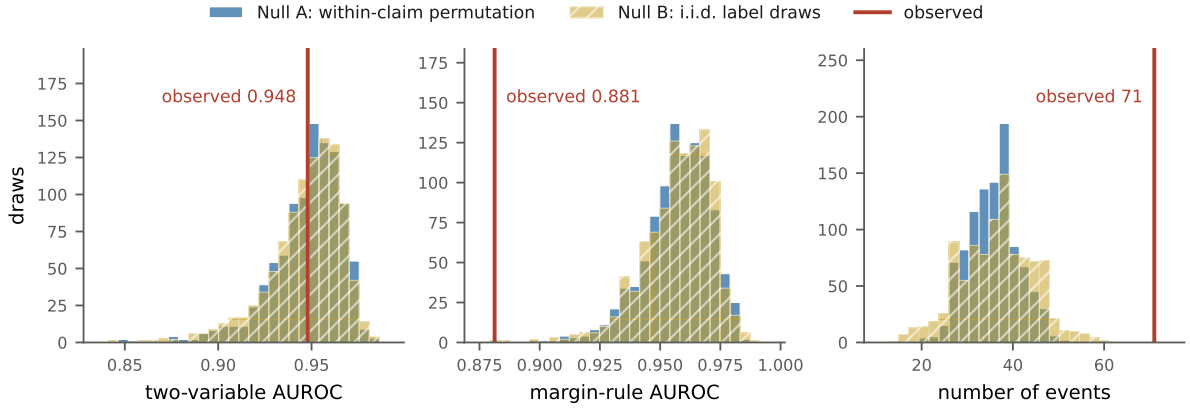


Figure 6: Null A and Null B distributions under the primary annotation for the two-variable AUROC, the margin-rule AUROC and the event count (1,000 draws per null), with the observed values marked. Nulls C and D and the second and intersection annotations are summarised in Tables 4, 31 and 32.

| Null                             | Metric           | Observed | Null mean | Null 95% interval | $p$ (observed $\geq$ null) | Draws |
|----------------------------------|------------------|----------|-----------|-------------------|----------------------------|-------|
| Null A: within-claim permutation | Compact-7 AUROC  | .946     | .936      | [0.882, 0.967]    | .365                       | 1000  |
| Null A: within-claim permutation | Compact-7 AUPRC  | .527     | .351      | [0.192, 0.519]    | .024                       | 1000  |
| Null A: within-claim permutation | 2-variable AUROC | .948     | .948      | [0.907, 0.974]    | .569                       | 1000  |
| Null A: within-claim permutation | 2-variable AUPRC | .535     | .424      | [0.252, 0.600]    | .112                       | 1000  |
| Null A: within-claim permutation | 1-tp, tl AUROC   | .881     | .957      | [0.929, 0.977]    | .999                       | 1000  |
| Null A: within-claim permutation | event prevalence | .070     | .035      | [0.024, 0.046]    | .001                       | 1000  |
| Null A: within-claim permutation | events           | 71.000   | 35.299    | [24.000, 47.000]  | .001                       | 1000  |
| Null B: stationary voting        | Compact-7 AUROC  | .946     | .955      | [0.873, 0.970]    | .396                       | 1000  |
| Null B: stationary voting        | Compact-7 AUPRC  | .527     | .362      | [0.171, 0.570]    | .067                       | 1000  |
| Null B: stationary voting        | 2-variable AUROC | .948     | .947      | [0.898, 0.975]    | .547                       | 1000  |
| Null B: stationary voting        | 2-variable AUPRC | .535     | .432      | [0.223, 0.634]    | .189                       | 1000  |
| Null B: stationary voting        | 1-tp, tl AUROC   | .881     | .957      | [0.924, 0.979]    | 1.000                      | 1000  |
| Null B: stationary voting        | event prevalence | .070     | .036      | [0.021, 0.051]    | .001                       | 1000  |
| Null B: stationary voting        | events           | 71.000   | 36.099    | [21.000, 52.000]  | .001                       | 1000  |

Table 30: Stationary-voting and permutation nulls. Null A permutes each claim’s direction labels across its publication years; Null B draws labels i.i.d. from each claim’s stationary label distribution over the observed publication times. Every draw rebuilds the benchmark and reruns the leave-one-claim-out pipeline.

minus-margin difference follows the same ordering where it is negative: under the second annotation the observed  $-0.096$  lies above its Null A mean of  $-0.204$  ( $p = 0.004$ ), so the signed score loses to the margin rule, but by less than on exchangeable

streams.

**What Null C conditions on.** Null C permutes the labels of a claim’s records, of every label type including NULL and UNCLEAR, among the record positions that fall in the same claim-block cell, so that each cell keeps its label totals and every record keeps its date. Labels in complete blocks on either side of a cutoff therefore cannot cross it; only the allocation within the block that straddles the cutoff can change a unit’s pre- and post-cutoff counts, and a cutoff on a block boundary leaves them, and hence the endpoint and the two count-derived inputs, unchanged. The control tests what remains after conditioning on the observed claim-block label totals and does not identify the causal contribution of temporal drift. An invariance audit counts the units whose invariance is guaranteed by the date partition alone: every record of the straddling

block, whatever its label, lies on one side of the cutoff, or the cutoff falls on a block boundary (a condition on signed records alone would not suffice, since a permutation can move a signed label onto the position of a NULL or UNCLEAR record on the other side). The condition is sufficient, not necessary: a straddling block whose records all carry one label type, or in which the same label appears on both sides of the cutoff with no other label present, also cannot change its counts under any permutation, and such homogeneous crossing blocks are not counted, so the reported shares are lower bounds on the fraction of units whose counts cannot change. Under the ten-year partition 9.6% of units, holding 3 of the 71 primary events, are invariant by the partition alone, so the reproduction of the event count is not largely guaranteed by that invariance; five-year blocks condition on more temporal structure and fix 31.2% of units and 20 events. A unit that is not invariant may still keep its endpoint in most draws, so “not invariant” is not “substantially perturbed”.

**Partition sensitivity.** The decision convention throughout, as in Table 4, is the central 95% null range: an observed value is outside it when its upper-tail probability is below 0.025, not 0.05. Table 31 varies the block for the primary annotation: with five-year blocks the observed 71 events fall inside the null range (64.9 [59, 71],  $p = 0.027$ ), as they do with decade boundaries shifted by five years (64.5 [58, 71],  $p = 0.038$ ), both probabilities below 0.05 but above the 0.025 threshold, while twenty-year blocks widen the gap (56.3 [48, 64],  $p = 0.002$ ). Under the second annotation and intersection labels the observed counts (52 and 55) lie inside the null range in every one of the four partitions. The excess over era-conditioned controls is therefore confined to the primary annotation and depends on the partition in both its magnitude and its decision at the 0.025 threshold; this restriction concerns the era-conditioned comparisons and does not remove the excess over Nulls A and B that the second and intersection annotations also show (Table 32). The grid evaluates event counts under all three annotations; the paired signed-minus-margin difference was evaluated across partitions under the primary annotation only, where it is reproduced under every block ( $p$  from 0.087 to 0.427), and no partition robustness of the learned advantage is claimed under the other annotations.

**Null D.** Null D replaces the block conditioning by a parametric trend. For each claim a logistic regression of the probability that a signed record is HARMFUL on publication year is fitted to that claim’s signed labels over the archived history by maximum likelihood, with a numerical ridge of  $10^{-6}$  added to the information matrix as optimizer damping only, not as a penalty in the objective. For a claim with a single signed class or complete separation the unpenalised likelihood has no finite maximiser, and the fit used is the numerically truncated iterate at the optimizer’s stopping rule, whose fitted probabilities lie near 0 or 1; those stopping choices therefore help define the generated probabilities for such claims, and the control is not a penalised estimator. The fitted parameters are then held fixed, and in each of 1,000 draws every signed record’s label is redrawn from the fitted probability at its own date while NULL and UNCLEAR records keep their labels and positions, after which the target is rebuilt and the score refitted. The control therefore carries no parameter uncertainty, is fitted on the archived history rather than on origin-available data, and is a retrospective diagnostic of whether a smooth per-claim drift in the signed label mix accounts for the observed quantities, not an origin-available reference and not a generative model of conclusion change. It is not rejected on the event count (70.1 [54, 87],  $p = 0.458$ ) or on the gain (+0.040 against +0.067,  $p = 0.099$ ); a failure to reject does not establish it as the true generating process. The residual of about ten reversals reported for the ten-year Null C is therefore a conditional, partition-specific contrast, not an identified set of unexplained reversals, and is not carried in the main text.

## H Complete Target Robustness

**Zero-direction convention.** Eleven future-eligible source units have a zero direction on exactly one side of the target: eight with  $p_t = 0$ , of which seven are supported exact ties (15/15, 16/16, 18/18, 13/13 and 57/57 signed pre-cutoff records among them) and one is the sparse-history fallback (fewer than eight signed records), and three with  $p_t^+ = 0$  (47/47, 18/18 and 135/135 signed post-cutoff records). Under Equation (1) a unit with a zero direction on exactly one side is an event, because a resolved direction differs from no direction, whereas a unit with zero direction on both sides would not be; all eleven observed

| Control                                             | Events [range] | $p$  | Paired $\Delta$ | $p$  |
|-----------------------------------------------------|----------------|------|-----------------|------|
| C, 10-year blocks                                   | 61.2 [53, 69]  | .008 | +.065           | .427 |
| C, 5-year blocks                                    | 64.9 [59, 71]  | .027 | +.065           | .427 |
| C, shifted 5 y                                      | 64.5 [58, 71]  | .038 | +.059           | .131 |
| C, 20-year blocks                                   | 56.3 [48, 64]  | .002 | +.051           | .087 |
| D, fitted trend                                     | 70.1 [54, 87]  | .458 | +.040           | .099 |
| <i>Second annotation (52 events; counts only)</i>   |                |      |                 |      |
| C, 10-year blocks                                   | 51.3 [45, 57]  | .489 | –               | –    |
| C, 5-year blocks                                    | 52.3 [48, 57]  | .645 | –               | –    |
| C, shifted 5 y                                      | 52.5 [47, 58]  | .633 | –               | –    |
| C, 20-year blocks                                   | 48.4 [41, 55]  | .202 | –               | –    |
| <i>Intersection labels (55 events; counts only)</i> |                |      |                 |      |
| C, 10-year blocks                                   | 55.7 [49, 62]  | .641 | –               | –    |
| C, 5-year blocks                                    | 54.7 [50, 60]  | .541 | –               | –    |
| C, shifted 5 y                                      | 53.8 [48, 60]  | .401 | –               | –    |
| C, 20-year blocks                                   | 55.9 [48, 64]  | .661 | –               | –    |

Table 31: Partition sensitivity of Null C and the trend-preserving Null D (1,000 draws each;  $p$ : upper-tail probability of the observed count or paired gain; an observed value lies outside the central 95% null range when  $p < 0.025$ ). Primary annotation: 71 events,  $\Delta = +0.067$ ; invariant units / events under the all-records condition of the text: 9.6% / 3 (10-year), 31.2% / 20 (5-year), lower bounds that exclude homogeneous crossing blocks; not computed for the shifted and 20-year partitions. Learned-score differences were evaluated across partitions under the primary annotation only.

| Annotation   | Observed | Null A        | Null B        | Null C        |
|--------------|----------|---------------|---------------|---------------|
| Primary      | 71       | 35.6 [26, 46] | 36.5 [20, 52] | 61.2 [53, 69] |
| Second       | 52       | 34.3 [24, 45] | 36.0 [21, 51] | 51.3 [45, 57] |
| Intersection | 55       | 40.2 [29, 51] | 40.9 [25, 56] | 55.7 [49, 62] |

Table 32: Events under the exchangeable controls for every annotation (null mean [95% range]; reimplementation, 1,000 draws per null). The archived primary values of Table 30 are 35.3 and 36.1.

zero-direction units are zero on one side only and are therefore events: the source benchmark has 71 events in 23 claims, the 145-claim subset 65 in 21, External-1 19 in 7 and External-2 95 in 40. Under an alternative binary tie mapping that assigns a tie to the protective sign, the source count is 63 events in 19 claims; the tie-free sensitivity (Table 35) avoids dependence on either mapping.

**Decisive reversals.** Requiring  $|p_t|$  and  $|p_t^+|$  to exceed a margin removes indecisive units and events alike. At  $\delta = 0.2$  only seven events remain; the signed score’s difference from the margin rule moves to +0.194 under the primary annotation and to −0.466 under the second, each with an interval too wide to rank methods, so the endpoint amplifies each annotation’s direction without deciding anything on this benchmark.

**Margins and quintiles.** Table 35 gives the full margin sweep. The lowest quintile of  $|p_t|$  ( $|p_t| \leq$

| $ p_t $ quintile | HARM-majority $n$ / events / rate | PROT-majority $n$ / events / rate |
|------------------|-----------------------------------|-----------------------------------|
| Q1 (lowest)      | 59 / 26 / .441                    | 137 / 21 / .153                   |
| Q2               | 49 / 10 / .204                    | 152 / 0 / .000                    |
| Q3               | 72 / 3 / .042                     | 130 / 0 / .000                    |
| Q4               | 68 / 2 / .029                     | 134 / 0 / .000                    |
| Q5 (highest)     | 26 / 1 / .038                     | 177 / 0 / .000                    |

Table 33: Event rate by pre-cutoff majority class within  $|p_t|$  quintiles (1,004 units with  $p_t \neq 0$ ). Median  $|p_t|$ : 0.588 for HARMFUL majorities, 0.612 for PROTECTIVE.

0.31) holds 54 of the 71 events at a rate of 0.267 [0.164, 0.376]; the next quintiles hold 11, 3, 2 and 1 events at rates 0.054, 0.015, 0.010 and 0.005. Table 33 splits each quintile by pre-cutoff majority class: PROTECTIVE-majority events occur only in the lowest quintile, HARMFUL-majority events in every quintile, and the class effect holds within strata (crude odds ratio 6.11, Mantel-Haenszel over quintiles 4.35, adjusted for  $|p_t|$  in a claim-clustered class-balanced logistic regression 7.80 [2.01, 33.82]).

**Coverage of selected historical cases.** Table 34 lists eight claims among the 165 whose evidence is generally described as having reversed, with the year and primary source of the trial or guidance usually cited for the reversal, the first year of the archived stream, and whether the subsequent-stream endpoint of Equation (1) is positive at any cutoff under the primary annotation. The table describes archive coverage and endpoint values; it supports no claim about detection timing. The endpoint compares the pooled direction of all records before a cutoff with that of all records at or after it, so the cutoff is the origin at which the forecast is posed, not the date at which any change occurs: a positive endpoint at the 1990 cutoff of the folic acid claim is compatible with a change driven by evidence published in 2007, and no lag can be read from the difference between a cutoff year and a pivotal year. Seven of the eight claims have archives that begin before the pivotal year; the beta-carotene archive begins in 2012 because of the retrieval cap (Appendix C.3) and is unobservable. An archive that begins before the pivotal year does not by itself show that an eligible cutoff with adequate pre-pivotal evidence exists or that the pivotal publication is among the archived records, so the table gives both: the two hormone-replacement claims and the vitamin E claim have no eligible cutoff before their pivotal trials and their archives do not contain the trial reports, so no forecast could have been posed for them; the selenium, folic acid,

niacin and dietary cholesterol claims have two to four eligible cutoffs before their pivotal year, and the three trial-anchored ones contain the trial reports; and hormone replacement and cardiovascular disease holds  $p_t$  between  $-0.04$  and  $-0.09$  at every one of its seven cutoffs with no positive endpoint. A timing claim would require a time-localised change statistic, which this benchmark does not define.

**Zero-direction and signed-future-evidence sensitivities.** Excluding the eleven zero-direction units and refitting gives  $0.938$  [ $0.907, 0.964$ ] and AUPRC  $0.417$  [ $0.263, 0.570$ ] on 60 events in 19 claims; the primary model restricted to the same units gives  $0.938$  (paired  $\Delta -0.001$  [ $-0.007, 0.006$ ]). On the zero-direction-free units Compact-7 stays clearly above the unfitted signals ( $+0.076$  [ $0.030, 0.142$ ] over  $1 - |p_t|$ ,  $+0.043$  [ $0.005, 0.089$ ] over low agreement,  $+0.054$  [ $-0.001, 0.114$ ] over recent dissent). Requiring at least 20 signed post-cutoff records to define the target leaves 1,000 units and all 71 events, with  $0.945$  [ $0.917, 0.970$ ]: the future target remains predictable when it must be supported by a minimum number of signed post-cutoff observations.

**Fixed horizons.** Within fixed windows the ranking is  $0.908$  [ $0.871, 0.940$ ] at three years (827 units, 57 events in 26 claims),  $0.916$  [ $0.874, 0.948$ ] at five years (783 units, 51 events) and  $0.916$  [ $0.870, 0.955$ ] at ten years (526 units, 39 events), with AUPRC  $0.391$ ,  $0.409$  and  $0.479$ . These rows require at least ten *non-null* records in the window. The horizon rows of [Appendix I](#) (821/56, 777/50 and 526/39 units/events;  $0.908$ ,  $0.930$  and  $0.916$ ) instead require at least ten *signed* records in the window, which is why the three- and five-year cohorts and their AUROCs differ between the two tables; they are distinct eligibility rules, not the same analysis.

## I Cumulative and Three-State Endpoints

With  $p$  the mean of the  $-1/0/+1$  codes over signed records and  $\text{sgn}(0) = 0$ , the primary endpoint is  $R_{\text{sub}} = \mathbf{1}[\text{sgn}(p_{\text{post}}) \neq \text{sgn}(p_{\text{pre}})]$ , the cumulative endpoint  $R_{\text{cum}} = \mathbf{1}[\text{sgn}(p_{\text{cum}}) \neq \text{sgn}(p_{\text{pre}})]$  with  $p_{\text{cum}}$  over all records of the claim, and the three-state endpoint counts a change of the state  $\{-1, 0, +1\}$  with 0 for  $|p| < \delta$ , for the subsequent and the cumulative comparison. Horizon variants use the window  $[t, t + H)$  with at least ten signed records in the window and a claim last year of

at least  $t + H - 1$ . [Table 36](#) gives the endpoint variants on the source future-eligible population, with the fixed-horizon rows subject to their stated horizon-specific eligibility, and the frozen models refitted leave-one-claim-out. The event sets nest: 34 units are events under both  $R_{\text{sub}}$  and  $R_{\text{cum}}$ , 37 under  $R_{\text{sub}}$  only (in twelve claims, e.g. `eggs_cvd_events` at 1988:  $p_{\text{pre}} = 0.87$  on 30 signed records,  $p_{\text{post}} = -0.01$  on 539,  $p_{\text{cum}} = 0.04$ ), and none under  $R_{\text{cum}}$  only. The cumulative endpoint is rarer; on it Compact-7 gives  $0.921$  [ $0.862, 0.984$ ] ( $0.9205$  before rounding), the two-variable score  $0.941$  and the unfitted margin rule  $0.932$ , so the untrained rule exceeds Compact-7's point estimate and no score “improves” on the cumulative endpoint in an established sense. On the cumulative fixed-horizon variants the margin rule alone reaches  $0.992$ ,  $0.991$  and  $0.971$  (14, 17 and 20 events): some endpoint constructions are largely predictable from the initial margin, and these are rare events on different samples, not learned-model performance. The three-state endpoint at  $\delta = 0.20$  counts many small moves into and out of the signed-balance band and every score falls to about  $0.90$  on it. The horizon rows here require at least ten signed records in the window and a claim last year of at least  $t + H - 1$ , a different eligibility rule from the fixed-horizon rows of [Appendix H](#) (ten non-null records), which is why the cohorts differ.

## J Numeric Effect-Reversal Endpoint: A Negative Scope Result

Defining an event as a sign change of the DerSimonian-Laird pooled log-ratio effect between the pre-cutoff estimates and those inside a ten-year window, which uses no direction label, Compact-7 reaches AUROC  $0.426$  [ $0.343, 0.512$ ] pooled over all domains (587 units, 90 events), against  $0.643$  [ $0.562, 0.718$ ] for the unfitted magnitude  $-|\mu_{\text{pre}}|$  ( $\Delta -0.217$  [ $-0.356, -0.071$ ]); the hypothesised verdict is not supported in any domain. Only 7.5% of records carry a parseable ratio with a confidence interval, so the endpoint rests on 587 units. Compact-7 does not predict generic pooled-effect reversal; it predicts later change of the categorical direction, which is what it was built for, and this falsification boundary is retained deliberately.

| Claim                                        | Usually cited source    | Year             | Archive from  | Eligible cutoffs before | Source archive | in Endpoint positive (earliest cutoff) |
|----------------------------------------------|-------------------------|------------------|---------------|-------------------------|----------------|----------------------------------------|
| hormone replacement → cardiovascular disease | WHI trial               | 2002             | 1998          | 0                       | no             | no                                     |
| hormone replacement → breast cancer          | WHI trial               | 2002             | 1998          | 0                       | no             | yes (2022)                             |
| beta-carotene → cancer                       | ATBC, CARET trials      | 1994, 1996       | 2012 (capped) | 0                       | no             | unobservable                           |
| vitamin E → cardiovascular disease           | HOPE, HPS trials        | 2000, 2002       | 1998          | 0                       | no             | no                                     |
| selenium → prostate cancer                   | SELECT trial            | 2009             | 1983          | 2                       | yes            | no                                     |
| folic acid → colorectal cancer               | Cole et al. trial       | 2007             | 1981          | 3                       | yes            | yes (1990)                             |
| niacin → cardiovascular disease              | AIM-HIGH, THRIVE trials | HPS2- 2011, 2014 | 1980          | 4                       | yes            | no                                     |
| dietary cholesterol → cardiovascular disease | 2015 DGAC report        | 2015             | 1984          | 4                       | n/a            | no                                     |

Table 34: Coverage of selected historical cases: the source usually cited for the reversal (WHI: [Writing Group for the Women’s Health Initiative Investigators, 2002](#); ATBC: [The Alpha-Tocopherol, Beta Carotene Cancer Prevention Study Group, 1994](#); CARET: [Omenn et al., 1996](#); HOPE: [Yusuf et al., 2000](#); HPS: [Heart Protection Study Collaborative Group, 2002](#); SELECT: [Lippman et al., 2009](#); Cole et al., 2007; AIM-HIGH: [AIM-HIGH Investigators, 2011](#); HPS2-THRIVE: [HPS2-THRIVE Collaborative Group, 2014](#); DGAC: [Dietary Guidelines Advisory Committee, 2015](#)) and its year, the first year of the archived stream, the number of eligible cutoffs before the pivotal year, whether the cited source’s PMID is among the archived records (the 2015 guidance has no single PMID), and whether the subsequent-stream endpoint is positive at any cutoff under the primary annotation (earliest such cutoff in parentheses); eligible cutoffs are the claim’s future-eligible benchmark cutoffs strictly before the pivotal year. Three of the seven observable cases, and the unobservable one, have no eligible cutoff before the pivotal year, so no forecast could have been posed for them. The cutoff is the forecast origin, not the date of change, so no detection lag is defined; [Prasad et al. \(2013\)](#) is the general reference for reversal as a phenomenon, not the source of these dates.

| Units kept                    | $n$   | Events | Rate  | AUROC [95% CI]       | AUPRC | $1 -  p_t $ |
|-------------------------------|-------|--------|-------|----------------------|-------|-------------|
| All future-eligible           | 1,012 | 71     | 0.070 | 0.946 [0.918, 0.969] | 0.527 | 0.881       |
| $ p_t  \geq 0.05$             | 976   | 52     | 0.053 | 0.945 [0.913, 0.969] | 0.456 | 0.855       |
| $ p_t  \geq 0.10$             | 949   | 43     | 0.045 | 0.951 [0.919, 0.976] | 0.477 | 0.842       |
| $ p_t  \geq 0.15$             | 921   | 36     | 0.039 | 0.949 [0.916, 0.976] | 0.433 | 0.833       |
| $ p_t  \geq 0.20$             | 904   | 30     | 0.033 | 0.940 [0.899, 0.974] | 0.364 | 0.811       |
| $ p_t  \geq 0.25$             | 860   | 24     | 0.028 | 0.963 [0.934, 0.989] | 0.400 | 0.804       |
| $ p_t  \geq 0.30$             | 816   | 17     | 0.021 | 0.959 [0.926, 0.991] | 0.340 | 0.765       |
| No zero-direction units       | 1,001 | 60     | 0.060 | 0.938 [0.907, 0.964] | 0.417 | 0.862       |
| $n_{\text{signed}}^+ \geq 10$ | 1,012 | 71     | 0.070 | 0.946 [0.918, 0.969] | 0.527 | 0.881       |
| $n_{\text{signed}}^+ \geq 20$ | 1,000 | 71     | 0.071 | 0.945 [0.917, 0.970] | 0.524 | 0.881       |

Table 35: Target-margin, tie-exclusion and signed-post-evidence robustness. Compact-7 is refitted leave-one-claim-out inside each retained sample; the last column is the AUROC of the unfitted score  $1 - |p_t|$  on the same units. Pre-cutoff margin filtering removes the eight pre-cutoff zero-direction units but does not necessarily remove the three post-cutoff ties: two zero-direction units remain at  $|p_t| \geq 0.05$ , one from  $|p_t| \geq 0.10$  through  $|p_t| \geq 0.20$ , and none at  $|p_t| \geq 0.25$ . The surviving post-cutoff ties are two cutoffs of one claim (butter and cardiovascular disease, 2011 with  $|p_t| = 0.21$  and 2022 with  $|p_t| = 0.08$ ), both with a HARMFUL pre-cutoff sign, so the margin-filtered rows are unchanged under the alternative PROTECTIVE tie mapping, as is the zero-direction-free row; this equivalence does not apply to the unfiltered row or to the signed-post-evidence rows, which retain all eleven zero-direction events (63 events under the alternative mapping). The signed-post-evidence rows require at least  $m$  signed (non-null, non-unclear) post-cutoff records to define the target; the minimum over the future-eligible sample is 11 and the median 230, so  $m = 10$  excludes nothing.

## K Agreement Definition Audit and Sensitivity

The shipped agreement  $a$  is computed by `agg_verdict`: with  $n_N$  NULL records among  $n$  resolved records,  $a = n_N/n$  when  $n_N/n \geq 0.5$  (the NULL-dominant branch) and otherwise the majority share among the signed records, which equals  $(1 + |p_t|)/2$ . Rebuilding the direction block from the record streams reproduces every shipped direction column exactly (maximum absolute difference 0). Of the 1,078 units, 1,035 are on the signed branch, where the identity holds on every unit (Pearson 1.000 with  $|p_t|$ ), 42 are on the NULL branch, where  $a$  equals the NULL share, and one has fewer than eight signed records and  $a = 0$ ; 22 units have a NULL share in  $[0.45, 0.50)$  and 19 in  $[0.50, 0.55)$ . Ten pairs of consecutive cutoffs of one claim cross the branch, with a mean absolute jump in  $a$  of 0.168 (0.046 under definition B); for example `late_night_eating_obesity` moves from  $a = 0.50$  at 2013 to 0.857 at 2015 as its NULL share moves from 0.50 to 0.44. Over all units the correlation of  $a$  with  $|p_t|$  is 0.944 (Spearman 0.959).

Four definitions were then substituted upstream of both models, with everything else rebuilt: A the shipped implementation; B the maximum share among PROTECTIVE/HARMFUL/NULL; C the signed-only share  $(1 + |p_t|)/2$  with no branch; D one minus the normalised entropy of the three-way

| Endpoint          | Units | Events | Pos. claims | Prev. | Compact-7 AUROC | [95% CI]       | AUPRC | 2-var AUROC | $1 -  p_t $ | Claim AUROC |
|-------------------|-------|--------|-------------|-------|-----------------|----------------|-------|-------------|-------------|-------------|
| R_sub (primary)   | 1012  | 71     | 23          | .070  | .946            | [0.918, 0.969] | .527  | .948        | .881        | .974        |
| R_cum             | 1012  | 34     | 15          | .034  | .921            | [0.862, 0.984] | .424  | .941        | .932        | .976        |
| R3_sub delta=0.10 | 1012  | 100    | 29          | .099  | .952            | [0.925, 0.973] | .597  | .946        | .889        | .980        |
| R3_cum delta=0.10 | 1012  | 58     | 19          | .057  | .944            | [0.908, 0.974] | .479  | .946        | .881        | .966        |
| R3_sub delta=0.20 | 1012  | 117    | 37          | .116  | .896            | [0.857, 0.929] | .438  | .892        | .854        | .902        |
| R3_cum delta=0.20 | 1012  | 72     | 24          | .071  | .902            | [0.850, 0.946] | .383  | .909        | .853        | .923        |
| R_sub 3y          | 821   | 56     | 26          | .068  | .907            | [0.871, 0.940] | .390  | .913        | .913        | .943        |
| R_cum 3y          | 821   | 14     | 12          | .017  | .974            | [0.944, 0.996] | .436  | .974        | .992        | .966        |
| R_sub 5y          | 777   | 50     | 24          | .064  | .930            | [0.897, 0.957] | .421  | .929        | .908        | .948        |
| R_cum 5y          | 777   | 17     | 14          | .022  | .939            | [0.873, 0.981] | .239  | .967        | .991        | .967        |
| R_sub 10y         | 526   | 39     | 19          | .074  | .916            | [0.870, 0.955] | .479  | .934        | .899        | .948        |
| R_cum 10y         | 526   | 20     | 13          | .038  | .937            | [0.882, 0.982] | .431  | .956        | .971        | .980        |

Table 36: Subsequent-evidence endpoint ( $R_{\text{sub}}$ , the primary target), cumulative updated-direction endpoint ( $R_{\text{cum}}$ ), three-state endpoints with a no-predominant-direction band  $|p| < \delta$ , and fixed-horizon variants. Non-horizon variants use the 1,012 future-eligible source units; fixed-horizon rows use their stated horizon-specific eligibility. Models are fitted leave-one-claim-out within each evaluated sample.

distribution; and C+null, which adds the NULL share as an explicit third input of the two-variable model. Table 37 reports the results with paired claim-clustered differences under the primary annotation. B and D are clearly worse; C is marginally worse ( $-0.023$ , upper limit  $0.002$ ) and C+null recovers most of the gap ( $-0.015$ , interval covering zero). That primary-annotation comparison is not a justification for A across annotations: under the second annotation the substitution of C for A repairs the shipped score’s loss ( $0.758$  to  $0.940$  on common support; Table 14), and on the primary annotation it removes the established gain over the margin rule. A is retained as the archived reference specification because every protocol cell, control and re-annotation was computed on it; C is the matched diagnostic intervention, reported alongside A and never merged with it into one method row, and C+null is the recommended agreement input for a subsequent version. On histories with fewer than eight signed records, where  $p_t$  is gated to 0, definition A returns  $a = 0$  (the archived fallback); the rebuild script for definition C applies the same gate and returns  $a = 0$  on those units rather than the formula value  $(1 + 0)/2 = 0.5$  (verified against the frozen builder with no mismatches), so that the two definitions differ only on the NULL-dominant branch (one unit under the primary annotation, ten under the second, fourteen under intersection labels are affected by the gate). The manuscript describes A’s two-branch form and its dependence on  $|p_t|$  exactly instead of treating direction and agreement as independent dimensions.

## L Semantic-Orientation Audit

Deterministic rules over the claim identifier, exposure term and outcome term assign each of the 165 claims to zero or more orientation categories: deficiency exposure (12 claims; deficiency, malnutrition, intolerance); therapeutic or preventive use (24; prevention, treatment, therapy, remission, symptom, response, replacement, probiotic, multivitamin and similar terms); restriction or inversion (15; skipping, replacing, low-fat, energy deficit, restricted, fasting, late-night, alternate-day, low intake, free diet); benefit-valence outcome (27; weight loss, cognitive function, remission, growth, diversity, fertility, bioavailability, absorption, response, control, prevention, neurodevelopment, metabolism, haemoglobin); and non-disease outcome term (19; duration, intake, risk, progression, and the benefit-valence terms). The rules are the whole method; no claim is re-labelled by hand, and the per-claim categories, reasons and affected record counts are released. The target is invariant to flipping the sign of every label of a claim, so canonicalising a claim’s orientation cannot change any event; the predictors are not invariant, because a score with a signed  $p$  term changes under  $p \rightarrow -p$  (on the signed branch a logit  $b_0 + b_1p + b_2|p|$  changes by  $-2b_1p$ ), so claim-specific recoding can alter predictions and fitted results. A mixture of orientations within a claim cannot be resolved without reading records. The sensitivity is therefore exclusion: the core tier removes the 53 claims in the four categories named by the review (deficiency, therapeutic or preventive, restriction or inversion, non-disease outcome term), and the strict tier also removes benefit-valence outcomes (63 claims). Table 38 shows that removing

| Def.     | 2-var AUROC | [95% CI]       | AUPRC | $\Delta$ vs A | [95% CI]         | Compact-7 AUROC |
|----------|-------------|----------------|-------|---------------|------------------|-----------------|
| A        | .948        | [0.921, 0.971] | .535  |               |                  | .946            |
| B        | .890        | [0.830, 0.939] | .387  | -0.058        | [-0.114, -0.010] | .899            |
| C        | .925        | [0.890, 0.959] | .500  | -0.023        | [-0.050, 0.002]  | .925            |
| D        | .849        | [0.758, 0.922] | .349  | -0.099        | [-0.191, -0.026] | .858            |
| C+null   | .932        | [0.901, 0.960] | .477  | -0.015        | [-0.036, 0.006]  |                 |
| p_t only | .766        | [0.696, 0.826] | .128  |               |                  |                 |
| 1-lp_tl  | .881        | [0.824, 0.936] | .468  |               |                  |                 |

Table 37: Agreement-definition sensitivity on the 1,012 future-eligible units. A: shipped implementation; B: max share of PROTECTIVE/HARMFUL/NULL; C: signed-only majority share  $(1 + |p_t|)/2$ ; D: normalised-entropy consensus; C+null adds the NULL share as a third input. Paired differences on identical claim resamples.

a third to two-fifths of the benchmark leaves the benchmark signal unchanged; the events lost with the flagged claims (20 and 26 of 71) are not the source of the result. The retained claims are lexically filtered, not an adjudicated orientation-clean cohort.

## M Complete Dependence Analyses

**Cutoff designs and claim-level scoring.** Table 39 gives every cutoff design, scored both with the all-cutoff model’s out-of-fold predictions and after refitting inside the design, together with claim-level scoring and the publication-disjoint protocol. These sensitivities reuse the same events and are not independent confirmations.

**Positive-claim influence and event concentration.** Dropping each of the 23 positive claims in turn and refitting changes AUROC by between  $-0.0037$  (eggs  $\rightarrow$  cardiovascular events, 6 events) and  $+0.0052$  (vitamin B12  $\rightarrow$  cancer, 7 events) and AUPRC by at most 0.054; the four heaviest claims hold 35% of the 71 events and eleven claims contribute three or more.

**PMID-overlap-disjoint evaluation.** Of the 96,091 distinct identifiers, 12,411 (12.9%) appear in two or more claims’ histories, and 25.6% of the 112,453 associations use such a shared identifier (2.3 claims per shared identifier on average). Leave-one-claim-out removes the held-out claim but not its publications from the training claims. The disjoint protocol therefore, for each held-out claim, removes every identifier of its labelled history from every other claim, rebuilds the Compact-7 features, the target and the future eligibility of every training unit from the reduced histories (a rebuild that reproduces

the benchmark exactly when nothing is removed), refits Compact-7, and scores the held-out claim’s unchanged units. The rebuilt training sets lose on average 0.25% of their associations (at most 2.3%) and 0.6 units per fold; the disjoint protocol gives AUROC 0.946 [0.918, 0.969] and AUPRC 0.525 [0.377, 0.666], a paired change of  $-0.0001$   $[-0.001, 0.001]$  and  $-0.002$   $[-0.007, 0.002]$  against standard leave-one-claim-out. The per-fold removal is small by construction: each held-out claim’s shared publications are a small fraction of the other 164 claims’ histories. The claim graph in which two claims are linked by any shared identifier has a single connected component containing all 165 claims, so a leave-one-component-out split is degenerate and is not reported; instead, a stricter unique-publication check removes every one of the 12,411 shared identifiers (25.6% of all associations) from every claim, rebuilds features, targets and eligibility for all units, and runs standard leave-one-claim-out on what remains: 880 future-eligible units in 150 claims with 57 events in 20 claims, AUROC 0.925 [0.886, 0.960] and AUPRC 0.384 [0.255, 0.621]. The signal therefore does not depend on publications that the claims share. Repeated estimates from the same cohorts published under different identifiers remain a possible dependence that no identifier-level protocol removes.

## N Full Feature and Ablation Results

**Learned models.** Table 40 gives every learned model on the same units, folds and resamples, together with the 12 rules scored for later change (Table 41). On the 145-claim subset the same picture holds (Compact-7 0.950; random forest 0.944, elastic net 0.944, gradient boosting 0.933, GAM

| Benchmark                 | Units | Claims | Events | Compact-7 AUROC | [95% CI]       | 2-var AUROC | Claim AUROC |
|---------------------------|-------|--------|--------|-----------------|----------------|-------------|-------------|
| full benchmark            | 1012  | 160    | 71     | .946            | [0.918, 0.969] | .948        | .974        |
| orientation-clean, core   | 681   | 107    | 51     | .945            | [0.909, 0.977] | .953        | .985        |
| orientation-clean, strict | 618   | 97     | 45     | .952            | [0.915, 0.984] | .964        | .990        |

Table 38: Semantic-orientation sensitivity: the benchmark with every claim flagged by the deterministic orientation rules removed (core: deficiency, therapeutic or preventive, restriction or inversion, non-disease outcome term; strict: core plus benefit-valence outcomes), models refitted leave-one-claim-out inside the retained claims.

| Design                                    | Units | Events | AUROC, all-cutoff model | AUROC, refit         |
|-------------------------------------------|-------|--------|-------------------------|----------------------|
| All cutoffs                               | 1,012 | 71     | 0.946 [0.918, 0.969]    | –                    |
| Claim level, maximum score over cutoffs   | 160   | 23     | 0.974 [0.950, 0.993]    | –                    |
| PMID-overlap-disjoint leave-one-claim-out | 1,012 | 71     | –                       | 0.946 [0.918, 0.969] |
| One random cutoff per claim               | 160   | 10.8   | 0.945 [0.908, 0.976]    | 0.921 [0.850, 0.966] |
| Earliest eligible cutoff                  | 160   | 17     | 0.965 [0.933, 0.990]    | 0.943 [0.896, 0.982] |
| Middle eligible cutoff                    | 160   | 11     | 0.961 [0.913, 0.995]    | 0.932 [0.870, 0.983] |
| Latest eligible cutoff                    | 160   | 12     | 0.971 [0.941, 0.992]    | 0.964 [0.925, 0.990] |
| Equalised cutoff count*                   | 160   | 10.7   | 0.944 [0.906, 0.971]    | 0.916 [0.845, 0.964] |
| Seven cutoffs, claims with fewer dropped  | 770   | 59     | 0.940                   | 0.937                |
| Cutoff positions 1-2                      | 318   | 25     | 0.938 [0.908, 0.965]    | –                    |
| Cutoff positions 3+                       | 694   | 46     | 0.949 [0.918, 0.976]    | –                    |

Table 39: Cutoff designs, claim-level scoring and the publication-disjoint protocol. \*Mean over 500 (random) or 200 (equalised) draws; the bracket is the 2.5 to 97.5 percentile range across draws, a design-variability range rather than a confidence interval. Other brackets are claim-clustered 95% intervals. Claim-level AUPRC is 0.797 [0.617, 0.957]; the publication-disjoint AUPRC is 0.525 [0.377, 0.666]. Leave-one-positive-claim-out refits change AUROC by at most 0.0052 and AUPRC by at most 0.054.

0.934).

**Within Compact-7.** The standardised coefficients of the fit on all 1,012 source units are agreement  $-3.32$ , pooled direction  $+2.97$ ,  $n_{\text{pre}} +1.32$ ,  $n_{\text{directional}} -1.07$ ,  $n_{RR} -0.70$ ,  $\sigma_{\log RR} -0.06$  and  $\mu_{\log RR} +0.05$ . Agreement alone ranks later change at 0.901 and pooled direction alone at 0.766; the two together reach 0.948 [0.921, 0.971], no clear difference from the seven ( $+0.002$  [ $-0.006, 0.011$ ]); the volume pair alone gives 0.603 and the effect-estimate triple 0.635. Removing  $p_t$  from Compact-7 costs 0.045 and leaves 0.900 (Table 42); because the count and effect inputs are correlated with the direction inputs, coefficient magnitudes are not a reliable attribution of their value.

**Adding blocks and nested selection.** The eleven direction variables alone match Compact-7 (0.947); adding the temporal-accrual block changes nothing (0.946) and adding the decline block lowers it (0.916). None of the other 48 variables gives a clear gain when added to the seven; the best, agreement contraction, adds  $+0.003$  [ $-0.008, 0.014$ ]. Greedy selection over the full representation, nested inside the folds, reaches 0.935 [0.902, 0.962] forward (me-

dian 13 variables, chosen in 99% of folds: null share, pooled direction and agreement) and 0.923 [0.883, 0.961] backward, neither clearly different from the seven-variable set.

## O Heterogeneity Analyses

Stratified by pre-cutoff  $I^2$  (units with fewer than three effect estimates, where  $I^2$  is not estimable, form their own stratum), Compact-7 ranks later change at 0.943 [0.899, 0.981] where  $I^2$  is not estimable (167 future-eligible units, 25 events), 0.864 [0.684, 1.000] at  $I^2 \leq 0.50$  (23 units, one event), 0.964 [0.911, 0.996] at  $0.50 < I^2 \leq 0.90$  (222 units, 12 events) and 0.937 [0.901, 0.971] at  $I^2 > 0.90$  (600 units, 33 events); the later-change rate is highest, 0.150, where  $I^2$  cannot be estimated. Three statements should be kept apart. Incremental benefit: the six heterogeneity variables change Compact-7 by  $-0.004$  [ $-0.013, 0.002$ ] AUROC, so they add nothing measurable conditional on the reference inputs. Model-specific performance: fitted on their own they rank later change at 0.486 [0.339, 0.642]. Association:  $I^2$  alone gives 0.385 when oriented so that higher heterogeneity means more expected change, which is 0.615 under the reversed orientation and is therefore evidence of an inverse association in this benchmark, not of zero information; the refitted coefficient of  $I^2$  is consistently negative in every domain (Appendix F). This analysis does not establish that heterogeneity carries no information in general.

## P Rule Comparators, Single Signals and Learned-Model Tuning

Table 43 gives the operational definition of every rule comparator and single signal and Table 44 the tuning grids of the learned models. Every rule reads the common pre-cutoff representation. The rules are simple threshold policies *inspired* by the named evidence-assessment principles (cumulative voting, conclusiveness monitoring, living-review triggers, evidence grading, pooled estimation); they

| Family                   | Method (inputs)                                         | AUROC        | 95% CI         | AUPRC        | 95% CI         | F1           | Claim AUROC  | $\Delta$ AUROC [95% CI] | $\Delta$ AUPRC [95% CI] |
|--------------------------|---------------------------------------------------------|--------------|----------------|--------------|----------------|--------------|--------------|-------------------------|-------------------------|
| Ours                     | <b>Compact-7 (7)</b>                                    | <b>0.946</b> | [0.918, 0.969] | 0.527        | [0.378, 0.669] | 0.494        | <b>0.974</b> | –                       | –                       |
| Same 7 inputs            | Generalised additive model (7)                          | 0.925        | [0.880, 0.964] | 0.491        | [0.329, 0.645] | 0.460        | 0.967        | +0.020 [0.000, 0.050]   | +0.035 [−0.019, 0.111]  |
|                          | Gradient boosting (7)                                   | 0.911        | [0.861, 0.957] | 0.447        | [0.311, 0.612] | 0.483        | 0.974        | +0.035 [0.009, 0.066]   | +0.080 [0.017, 0.134]   |
|                          | Random forest (7)                                       | 0.906        | [0.865, 0.946] | 0.439        | [0.318, 0.581] | 0.413        | 0.969        | +0.040 [0.015, 0.063]   | +0.088 [0.008, 0.170]   |
| All pre-cutoff variables | Random forest                                           | 0.938        | [0.908, 0.965] | <b>0.545</b> | [0.394, 0.688] | 0.545        | 0.969        | +0.007 [−0.011, 0.027]  | −0.018 [−0.121, 0.089]  |
|                          | Elastic net                                             | 0.938        | [0.909, 0.963] | 0.450        | [0.304, 0.605] | 0.517        | 0.932        | +0.008 [−0.013, 0.034]  | +0.077 [−0.032, 0.198]  |
|                          | Gradient boosting (XGBoost)                             | 0.932        | [0.888, 0.967] | 0.543        | [0.359, 0.719] | <b>0.578</b> | 0.971        | +0.014 [−0.013, 0.047]  | −0.016 [−0.125, 0.101]  |
|                          | SVM, RBF kernel                                         | 0.925        | [0.894, 0.955] | 0.458        | [0.287, 0.625] | 0.365        | 0.941        | +0.021 [−0.005, 0.051]  | +0.069 [−0.050, 0.217]  |
|                          | $k$ -nearest neighbours                                 | 0.909        | [0.858, 0.954] | 0.382        | [0.251, 0.546] | 0.497        | 0.929        | +0.037 [−0.004, 0.083]  | +0.145 [0.009, 0.270]   |
|                          | Best single variable                                    | 0.901        | [0.849, 0.948] | 0.509        | [0.320, 0.668] | 0.447        | 0.947        | +0.044 [0.003, 0.087]   | +0.017 [−0.086, 0.149]  |
|                          | Logistic regression, unpenalised                        | 0.897        | [0.847, 0.945] | 0.373        | [0.231, 0.555] | 0.417        | 0.929        | +0.048 [0.005, 0.100]   | +0.153 [0.025, 0.266]   |
|                          | Balanced logistic (the Compact-7 estimator, all inputs) | 0.893        | [0.839, 0.944] | 0.374        | [0.235, 0.549] | 0.443        | 0.929        | +0.053 [0.009, 0.108]   | +0.153 [0.036, 0.262]   |
|                          | Neural network (MLP, 32 units)                          | 0.889        | [0.841, 0.936] | 0.360        | [0.220, 0.549] | 0.397        | 0.928        | +0.057 [0.014, 0.105]   | +0.167 [0.009, 0.302]   |
|                          | Gaussian naive Bayes                                    | 0.778        | [0.671, 0.877] | 0.289        | [0.170, 0.464] | 0.347        | 0.839        | +0.167 [0.071, 0.272]   | +0.237 [0.082, 0.372]   |
|                          | Decision tree                                           | 0.674        | [0.582, 0.775] | 0.315        | [0.214, 0.446] | 0.507        | 0.954        | +0.272 [0.167, 0.369]   | +0.211 [0.055, 0.347]   |
| Nested selection         | Forward selection over all variables (median 13)        | 0.935        | [0.902, 0.962] | 0.481        | [0.323, 0.648] | –            | –            | +0.011 [−0.011, 0.035]  | +0.046 [−0.069, 0.156]  |
|                          | Backward elimination from all variables (median 12)     | 0.923        | [0.883, 0.961] | 0.485        | [0.340, 0.638] | –            | –            | +0.023 [−0.012, 0.065]  | +0.041 [−0.052, 0.139]  |

Table 40: Every learned model for later change on the 1,012 future-eligible units (71 events), leave-one-claim-out with nested tuning (Table 44), claim-clustered 95% intervals from the same 2,000 resamples.  $\Delta$  is Compact-7 minus the model on identical resamples. Bold marks the best value in a column among the learned models listed; the two-variable score (Table 42) has the highest AUROC point estimate of any fitted score, 0.948. A paired interval that covers zero is not a difference. Random forest and gradient boosting on the full representation have the higher AUPRC point estimates and gradient boosting the higher F1; they are retained as valid competitive baselines. Nested selection chooses the variable subset inside each outer fold.

| Family    | Method                           | AUROC        | 95% CI         | AUPRC        | 95% CI         | F1           | Prec. <sub>m</sub> | Claim AUROC  | $\Delta$ AUROC [95% CI] |
|-----------|----------------------------------|--------------|----------------|--------------|----------------|--------------|--------------------|--------------|-------------------------|
| Reference | Compact-7                        | <b>0.946</b> | [0.918, 0.969] | <b>0.527</b> | [0.378, 0.669] | 0.494        | 0.507              | <b>0.974</b> | –                       |
| Rule      | Cumulative-agreement vote        | 0.859        | [0.793, 0.920] | 0.285        | [0.177, 0.403] | 0.422        | 0.408              | 0.924        | +0.086 [0.034, 0.141]   |
|           | Conflict-aware rule              | 0.846        | [0.786, 0.900] | 0.241        | [0.149, 0.346] | 0.198        | 0.325              | 0.939        | +0.100 [0.051, 0.150]   |
|           | Replacement-time retrieval       | 0.808        | [0.754, 0.861] | 0.186        | [0.114, 0.265] | 0.330        | 0.203              | 0.803        | +0.138 [0.088, 0.188]   |
|           | Synthesis-presence workflow      | 0.741        | [0.707, 0.775] | 0.128        | [0.076, 0.183] | 0.227        | 0.128              | 0.668        | +0.205 [0.167, 0.242]   |
|           | Decline-triggered workflow       | 0.730        | [0.672, 0.784] | 0.148        | [0.091, 0.213] | 0.280        | 0.036              | 0.612        | +0.216 [0.168, 0.265]   |
|           | Scoping-review workflow          | 0.704        | [0.627, 0.767] | 0.118        | [0.068, 0.173] | 0.219        | 0.126              | 0.650        | +0.242 [0.184, 0.312]   |
|           | Recency only                     | 0.644        | [0.559, 0.720] | 0.105        | [0.054, 0.168] | 0.214        | 0.131              | 0.665        | +0.301 [0.221, 0.390]   |
|           | Conclusiveness proxy (TSA-style) | 0.620        | [0.527, 0.714] | 0.102        | [0.056, 0.161] | 0.216        | 0.144              | 0.629        | +0.326 [0.234, 0.416]   |
|           | Evidence-grade workflow          | 0.582        | [0.558, 0.607] | 0.083        | [0.049, 0.120] | 0.153        | 0.083              | 0.518        | +0.363 [0.330, 0.396]   |
|           | Volume threshold                 | 0.565        | [0.472, 0.665] | 0.081        | [0.046, 0.127] | 0.156        | 0.091              | 0.544        | +0.381 [0.286, 0.472]   |
|           | Living-review workflow           | 0.564        | [0.513, 0.604] | 0.080        | [0.046, 0.117] | 0.148        | 0.081              | 0.500        | +0.382 [0.343, 0.427]   |
|           | Pooled-estimate workflow         | 0.427        | [0.344, 0.527] | 0.064        | [0.037, 0.096] | 0.133        | 0.047              | 0.502        | +0.518 [0.410, 0.611]   |
| Signal    | Low agreement                    | 0.908        | [0.858, 0.950] | 0.447        | [0.299, 0.599] | 0.478        | 0.423              | 0.945        | +0.038 [0.003, 0.078]   |
|           | Recent dissent                   | 0.883        | [0.831, 0.936] | 0.523        | [0.377, 0.676] | <b>0.566</b> | <b>0.578</b>       | 0.973        | +0.063 [0.006, 0.125]   |
|           | $1 -  p_t $ (direction margin)   | 0.881        | [0.824, 0.936] | 0.468        | [0.359, 0.589] | 0.398        | 0.423              | 0.971        | +0.064 [0.011, 0.122]   |
|           | Absent consensus                 | 0.786        | [0.702, 0.864] | 0.259        | [0.151, 0.365] | 0.463        | 0.356              | 0.880        | +0.160 [0.086, 0.238]   |
|           | Low conclusiveness               | 0.632        | [0.534, 0.733] | 0.112        | [0.060, 0.184] | 0.229        | 0.160              | 0.680        | +0.313 [0.217, 0.410]   |
|           | Low evidence volume              | 0.551        | [0.432, 0.677] | 0.100        | [0.056, 0.175] | 0.164        | 0.155              | 0.511        | +0.394 [0.271, 0.509]   |
|           | Decline signal                   | 0.504        | [0.443, 0.561] | 0.073        | [0.043, 0.112] | 0.147        | 0.042              | 0.496        | +0.442 [0.377, 0.511]   |
|           | Heterogeneity $I^2$              | 0.385        | [0.285, 0.491] | 0.067        | [0.038, 0.107] | 0.131        | 0.042              | 0.617        | +0.561 [0.443, 0.670]   |
|           | Imprecision                      | 0.338        | [0.246, 0.429] | 0.053        | [0.031, 0.081] | 0.131        | 0.014              | 0.448        | +0.608 [0.507, 0.709]   |
|           | Recent accrual share             | 0.330        | [0.241, 0.433] | 0.049        | [0.030, 0.076] | 0.127        | 0.023              | 0.448        | +0.616 [0.506, 0.710]   |
|           | Agreement contraction            | 0.318        | [0.236, 0.411] | 0.052        | [0.031, 0.093] | 0.131        | 0.042              | 0.598        | +0.628 [0.536, 0.710]   |
| Trivial   | Event prevalence (constant)      | 0.500        | –              | 0.070        | –              | –            | 0.070              | 0.500        | –                       |

Table 41: The 12 rule comparators (scored STABLE < STILL-FORMING < UNSTABLE) and the single oriented signals, for later change on the 1,012 future-eligible units with the same folds, threshold rule and claim resamples as Compact-7. F1 is taken at the training-claim threshold; Prec.<sub>m</sub> is the precision when as many units are flagged as there are events; claim AUROC scores a claim by its maximum over cutoffs.

| Predictor                           | AUROC        | AUPRC        | $\Delta$ AUROC vs Compact-7 [95% CI] |
|-------------------------------------|--------------|--------------|--------------------------------------|
| <b>Compact-7 (7)</b>                | <b>0.946</b> | <b>0.527</b> | —                                    |
| Pooled direction + agreement (2)    | 0.948        | 0.535        | +0.002 [−0.006, 0.011]               |
| Direction block alone (11)          | 0.947        | 0.533        | +0.001                               |
| Compact-7 + temporal accrual (11)   | 0.946        | 0.526        | +0.000                               |
| Compact-7 + decline (11)            | 0.916        | 0.494        | −0.030                               |
| Agreement alone (1)                 | 0.901        | 0.436        | −0.045 [−0.093, −0.006]              |
| Compact-7 without $p_t$ (6)         | 0.900        | 0.429        | −0.045                               |
| Temporal accrual alone (4)          | 0.898        | 0.450        | −0.048                               |
| All 55 variables, balanced logistic | 0.893        | 0.374        | −0.053 [−0.108, −0.009]              |
| Pooled direction alone (1)          | 0.766        | 0.128        | −0.179 [−0.249, −0.122]              |
| Decline block alone (4)             | 0.537        | 0.084        | clearly lower                        |
| Heterogeneity block alone (6)       | 0.486        | 0.077        | clearly lower                        |

Table 42: Which pre-cutoff signals carry the prediction, on the 1,012 future-eligible units, leave-one-claim-out. Fitted predictors are class-balanced logistic regressions on the named variables;  $\Delta$  is the predictor minus Compact-7 on identical claim resamples.

are not implementations or evaluations of GRADE, Trial Sequential Analysis, living reviews or meta-analysis, and beating them substantiates no comparison with those methodologies. Thresholding a continuous signal into a few states also discards ranking resolution, so the continuous single signals are the more informative comparators. For ranking, a rule’s state is scored STABLE = 0, STILL-FORMING (or UNASSESSABLE) = 1, UNSTABLE = 2. Learned models are tuned by nested five-fold GroupKFold over the training claims of every outer fold; the setting with the highest AUROC on the pooled inner out-of-fold scores wins, with ties to the most regularised setting, and is refitted on the training claims before the held-out claim is scored. Hard decisions for every method use the F1-maximising threshold on the training claims of the fold.

## Q Prompted LLM Forecasters on Label Digests

**Summary.** On the 1,001-unit tie-free subset, the strongest valid digest prompts reach AUROC 0.841 (Qwen2.5-72B, zero-shot) and 0.843 (Llama-3.1-70B, one-shot), against 0.938 for Compact-7 refitted on the same units (paired differences +0.096 [0.031, 0.172] and +0.095 [0.054, 0.147]). This is not a controlled architecture comparison: the baseline is supervised, prompting is not; named prompts may draw on pretraining knowledge; and smaller models are option-order sensitive (see below). The result is limited to the tested digest-prompting protocol and is not a claim about language models in general. The same models on the calendar cohort, with and without pre-origin abstracts, are in [Appendix R](#).

Each model sees a digest of the claim’s pre-cutoff direction labels: the number of records pub-

lished before the cutoff by label, the span of years, the largest directional share over all years and over the last six years before the cutoff, and year-by-year counts (the earliest eight and the latest twenty years when the history is longer than 28 years). No post-cutoff record enters the prompt. [Figure 7](#) gives the system instructions. The forecast is asked with the claim named (“does {exposure} affect {outcome}?”) and blinded (“does an exposure (name withheld) affect a health outcome (name withheld)?”). The answer is read from the next-token log-probabilities of the option letters, and the option probabilities are a softmax over the letters; the share of next-token probability falling on the letters is recorded for every run (0.95 to 1.00 for seven models, 0.68 to 0.90 for DeepSeek-R1-Distill-8B, and below 0.011 for DeepSeek-R1-Distill-70B, whose rows are therefore invalid and marked <sup>†</sup>). Every task runs twice with the option letters rotated and every reported value averages the two orders’ probabilities. In the one-shot setting one solved example from a unit of a different claim, drawn at random with a fixed seed, is shown as an earlier chat turn with its true answer, so the held-out claim stays unseen. The reasoning distills open a thinking block after the assistant tag, which is closed empty so that the next token is the answer; Qwen3-8B runs with thinking disabled. The 32B and 70B-class models are pre-quantised 4-bit checkpoints. Nothing is fitted. The prompt asks whether the predominant direction will reverse, which is a protective↔harmful reversal; it does not distinguish a no-predominant-direction state, so the stored forecasts are scored on the 1,001-unit tie-free subset ( $p_t \neq 0$  and  $p_t^+ \neq 0$ ), where [Equation \(1\)](#) reduces to exactly that event, against Compact-7 refitted leave-one-claim-out on the same units (0.938 [0.907, 0.964], AUPRC 0.417). No prompt was re-run. [Table 45](#) gives the AUROC and AUPRC of every model and setting on that subset; the paired  $\Delta$ AUROC of the refit Compact-7 against every valid forecast row excludes zero (smallest gap +0.095 [0.054, 0.147], Llama-3.1-70B one-shot named), while the paired  $\Delta$ AUPRC against the strongest forecast, Qwen2.5-72B zero-shot named, is +0.142 [−0.018, 0.287] and covers zero. Scale helps, mostly on the named forecast: Llama-3.1’s zero-shot named AUROC rises from 0.371 at 8B to 0.814 at 70B and Qwen2.5’s from 0.496 at 7B to 0.841 at 72B.

| Comparator                                                                           | Inputs                               | Operational rule                                                                                                               |
|--------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Cumulative-agreement vote                                                            | agreement                            | STABLE if agreement $\geq 0.70$ ; UNSTABLE if agreement $< 0.55$ ; else STILL-FORMING                                          |
| Volume threshold                                                                     | n_pre_t                              | STABLE if at least 150 pre-cutoff records; else STILL-FORMING                                                                  |
| Recency only                                                                         | recency_density                      | STABLE if at least 35% of pre-cutoff records fall in the last five years; else STILL-FORMING                                   |
| Conclusiveness proxy (TSA-style)                                                     | naive_tsa                            | STABLE if accumulated precision reaches half the information-size proxy (naive_tsa $\geq 0.5$ ); else STILL-FORMING            |
| Conflict-aware rule                                                                  | i_squared, dissent_recent, agreement | UNSTABLE if $I^2 > 0.9$ and recent dissent $> 0.4$ ; STABLE if agreement $\geq 0.7$ ; else STILL-FORMING                       |
| Replacement-time retrieval                                                           | agreement_recent                     | STABLE if recent-window agreement $\geq 0.7$ ; else STILL-FORMING                                                              |
| Synthesis-presence workflow                                                          | synth_frac, agreement                | STABLE if syntheses are present ( $\geq 3\%$ of records) and agreement $\geq 0.65$ ; else STILL-FORMING                        |
| Evidence-grade workflow                                                              | n_pre_t, agreement, rct_frac         | STABLE if at least 100 records, agreement $\geq 0.7$ and trial share $\geq 0.2$ ; else STILL-FORMING                           |
| Living-review workflow                                                               | vol_growth, agreement                | STILL-FORMING while evidence volume is growing (vol_growth $> 1.2$ ); else STABLE if agreement $\geq 0.7$ , else STILL-FORMING |
| Scoping-review workflow                                                              | span_years, agreement                | STABLE if the literature spans at least 20 years and agreement $\geq 0.65$ ; else STILL-FORMING                                |
| Pooled-estimate workflow                                                             | n_rr, i_squared                      | UNSTABLE if $I^2 > 0.95$ ; STABLE if at least 10 pooled estimates and $I^2 < 0.75$ ; else STILL-FORMING                        |
| Decline-triggered workflow                                                           | decline_signal, agreement            | UNSTABLE on a decline signal (decline_signal $\geq 0.45$ ); else STABLE if agreement $\geq 0.7$ , else STILL-FORMING           |
| <i>Single signals, oriented so that a higher score means change is more expected</i> |                                      |                                                                                                                                |
| $1 -  p_t $ (direction margin)                                                       | pooled_direction                     | score = $1 -  p_t $                                                                                                            |
| Low agreement                                                                        | agreement                            | score = $1 - \text{agreement}$                                                                                                 |
| Recent dissent                                                                       | dissent_recent                       | score = dissent_recent                                                                                                         |
| Agreement contraction                                                                | agreement_contraction                | score = $1 - \text{agreement\_contraction}$                                                                                    |
| Absent consensus                                                                     | has_consensus                        | score = $1 - \text{has\_consensus}$                                                                                            |
| Heterogeneity $I^2$                                                                  | i_squared                            | score = $I^2$                                                                                                                  |
| Imprecision                                                                          | ci_crosses_null                      | score = share of confidence intervals crossing the null                                                                        |
| Low evidence volume                                                                  | n_pre_t                              | score = $1 - \text{n\_pre\_t}$                                                                                                 |
| Recent accrual share                                                                 | recency_density                      | score = share of records in the last five years                                                                                |
| Low conclusiveness                                                                   | naive_tsa                            | score = $1 - \text{naive\_tsa}$                                                                                                |
| Decline signal                                                                       | decline_signal                       | score = decline_signal                                                                                                         |

Table 43: Operational definitions of the rule comparators and single signals. Variable names refer to the inventory of Table 48.

| Learned model (inputs)           | Setting and nested tuning grid; mode over folds                                           |
|----------------------------------|-------------------------------------------------------------------------------------------|
| Generalised additive model (7)   | one spline term per Compact-7 input, 10 splines; $\lambda \in \{100, 10, 1, 0.1\}$        |
| Random forest (7)                | 400 trees; min_samples_leaf $\in \{5, 3, 1\}$                                             |
| Gradient boosting, XGBoost (7)   | learning rate 0.05; max_depth $\in \{2, 3\} \times \text{n\_estimators} \in \{100, 300\}$ |
| Random forest (all)              | 400 trees; min_samples_leaf $\in \{5, 3, 1\}$                                             |
| Elastic net (all)                | $C \in \{0.01, 0.1, 1, 10\} \times \text{l1\_ratio} \in \{0.2, 0.5, 0.8\}$                |
| Gradient boosting, XGBoost (all) | learning rate 0.05; max_depth $\in \{2, 3\} \times \text{n\_estimators} \in \{100, 300\}$ |
| SVM, RBF kernel (all)            | $C \in \{0.1, 1, 10\} \times \gamma \in \{\text{scale}, 0.01\}$ , balanced weights        |
| k-nearest neighbours (all)       | $k \in \{5, 15, 45\}$ , distance weights                                                  |
| Neural network (all)             | one hidden layer of 32 units, $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ , early stopping |
| Decision tree (all)              | max_depth $\in \{2, 3, 4\}$                                                               |
| Logistic regression (all)        | unpenalised, standardised inputs                                                          |
| Balanced logistic (all)          | the Compact-7 estimator on all pre-cutoff variables                                       |
| Gaussian naive Bayes (all)       | no tuning                                                                                 |
| Best single variable             | the single oriented variable with the highest AU-ROC on the training claims               |

Table 44: Learned baselines, standardised inputs, tuned per outer fold by nested five-fold GroupKFold over the training claims. “7” means the seven Compact-7 inputs, “all” the full pre-cutoff representation. Implementations: scikit-learn (Pedregosa et al., 2011), XGBoost (Chen and Guestrin, 2016), pyGAM (Servén and Brummitt, 2018).

**Option-order sensitivity.** The 7B to 8B models are dominated by letter-position preference: their  $P(\text{reverse})$  under the two option orders correlates

FORECAST, SYSTEM: You forecast how a body of scientific evidence will develop after a historical moment. You see only studies published before the cutoff year. When the studies published after the cutoff are pooled, will the predominant direction of their protective and harmful findings be the opposite of the predominant direction before the cutoff? Answer with exactly one letter: A if the predominant direction will reverse; B if the predominant direction will stay the same. Answer with the letter only.  
 USER: {claim, named or blinded} \n Cutoff year: {t} \n \n {digest} \n \n Will the predominant direction reverse after {t}?

Figure 7: The forecasting prompt, shown with the first option order.

at  $r = -0.27$  to  $0.05$  zero-shot (Qwen3-8B scores  $0.21$  under one order and  $0.78$  under the other). The 32B and 70B-class models are steadier: Qwen2.5-72B’s forecasts correlate at  $r = 0.89$  (named) and  $0.85$  (blinded) across orders, and Qwen2.5-32B’s at  $r = 0.68$ . Llama-3.1-70B’s zero-shot forecasts anticorrelate across orders ( $r = -0.45, -0.61$ ). Averaging the two orders removes the position bias but not the instability.

| Model                                              | Zero-shot                |             |                          |             | One-shot                 |             |                          |             |
|----------------------------------------------------|--------------------------|-------------|--------------------------|-------------|--------------------------|-------------|--------------------------|-------------|
|                                                    | Claim named              |             | Claim blinded            |             | Claim named              |             | Claim blinded            |             |
|                                                    | AUROC                    | AUPRC       | AUROC                    | AUPRC       | AUROC                    | AUPRC       | AUROC                    | AUPRC       |
| Llama-3.1-8B-Instruct                              | .371 [.284, .452]        | .045        | .393 [.325, .459]        | .048        | .636 [.564, .706]        | .087        | .598 [.525, .671]        | .073        |
| Qwen2.5-7B-Instruct                                | .496 [.423, .583]        | .059        | .622 [.518, .719]        | .083        | .663 [.596, .726]        | .121        | .576 [.513, .638]        | .077        |
| Mistral-7B-Instruct-v0.3                           | .604 [.462, .724]        | .080        | .660 [.559, .748]        | .101        | .528 [.451, .605]        | .070        | .500 [.426, .574]        | .062        |
| Qwen3-8B (thinking off)                            | .675 [.546, .793]        | .121        | .702 [.590, .794]        | .121        | .633 [.563, .700]        | .079        | .588 [.513, .654]        | .070        |
| DeepSeek-R1-Distill-Llama-8B                       | .388 [.301, .475]        | .049        | .309 [.234, .392]        | .041        | .461 [.361, .555]        | .057        | .401 [.312, .491]        | .051        |
| Qwen2.5-32B-Instruct (4-bit)                       | .790 [.714, .851]        | .153        | <b>.805</b> [.735, .860] | .173        | .804 [.758, .847]        | .180        | .694 [.621, .760]        | .108        |
| Llama-3.1-70B-Instruct (4-bit)                     | .814 [.756, .858]        | .170        | .793 [.731, .845]        | .164        | <b>.843</b> [.790, .886] | .184        | <b>.829</b> [.774, .874] | .173        |
| Qwen2.5-72B-Instruct (4-bit)                       | <b>.841</b> [.764, .911] | .275        | .732 [.605, .827]        | .162        | .770 [.705, .831]        | .134        | .619 [.561, .677]        | .073        |
| DeepSeek-R1-Distill-Llama-70B (4-bit) <sup>†</sup> | n/a                      | n/a         | n/a                      | n/a         | n/a                      | n/a         | n/a                      | n/a         |
| Compact-7, refit on the tie-free units             | <b>.938</b> [.907, .964] | <b>.417</b> | <b>.938</b> [.907, .964] | <b>.417</b> | <b>.938</b> [.907, .964] | <b>.417</b> | <b>.938</b> [.907, .964] | <b>.417</b> |

Table 45: Direct reversal forecasts on the 1,001-unit tie-free subset (60 events, prevalence 0.060), where the target is a protective↔harmful reversal: AUROC [claim-clustered 95% CI] and AUPRC, every value averaged over two option orders. Compact-7 is refitted leave-one-claim-out on the same units and never sees the claim name, so its value is the same in every column. <sup>†</sup>Invalid read-out (0.02 to 1.0% of the next-token probability on an option letter); its values are unavailable rather than meaningful scores. Conclusions are bounded to this digest-prompting setup.

| Horizon | Margin | Text | Text + comp. | Comp. | Text added [95% CI]  |
|---------|--------|------|--------------|-------|----------------------|
| 3 y     | .907   | .606 | .746         | .913  | -.166 [-.253, -.078] |
| 5 y     | .897   | .646 | .781         | .920  | -.139 [-.265, -.040] |
| 10 y    | .867   | .608 | .758         | .876  | -.118 [-.194, -.045] |

Table 46: Embedding forecasters on the calendar cohort (AUROC). “Comp.” is the signed two-variable score; “Text added” is text + composition minus composition, paired over claims.

## R Text-Reading Forecasters on the Calendar Cohort

Both families use only records published before the origin and are evaluated in the calendar cells of Table 2; trained models use claim-disjoint units whose outcomes are observable at the origin.

**Abstract embeddings.** Each record’s title and abstract (the title alone when no abstract is available) is embedded once with a biomedical sentence encoder (S-PubMedBert-MS-MARCO, 768 dimensions, L2-normalised). A unit’s text features are the mean embedding of all its pre-origin records, the mean of its 20 most recent pre-origin records and the difference between the two, reduced to 64 dimensions by principal components fitted on the training units only. A class-balanced  $L_2$  logistic regression is fitted per origin on text alone, on text with the two composition inputs, and on the composition inputs alone (Table 46). Text alone ranks later change at 0.606-0.646, and adding it to the composition inputs lowers AUROC by an established margin at every horizon.

**Language models.** Each prompt states the exposure, the outcome, the origin year and the pre-origin counts of HARMFUL, PROTECTIVE and

NULL records with the current majority direction (the digest). The abstract condition adds the titles and abstracts of the eight most recent pre-origin records, each truncated to its first 1,000 characters, so that every prompt fits the context window without selection by content. The masked condition replaces the exposure and outcome names with placeholders in the prompt and in the abstracts. The model answers a two-option question (the predominant direction reverses or stays) under both option orders, and the score is the probability of the reversal option averaged over the two orders. At 7-8B (Llama-3.1-8B and Qwen2.5-7B) every cell lies between 0.478 and 0.579 and is established below the margin rule in every cell; the per-cell values are released with the benchmark. At 70B and 72B (Table 47) no configuration establishes superiority over the margin rule; the only positive point difference, Qwen at ten years (0.875 against 0.867), is +0.008 with an interval that spans zero, and the remaining differences are negative point estimates whose intervals are released with the per-unit scores. Abstracts add +0.013 to +0.023 when the claim is named and +0.085 to +0.104 when it is masked. Naming the claim is worth +0.119 to +0.129 on the digest alone but +0.037, +0.019 and −0.004 once abstracts are supplied (established only at three years), and its value is not concentrated at early origins, as recall of later literature would predict. Because post-origin pretraining can give these models information unavailable at the historical origin, these runs are not clean prospective tests; nevertheless, none establishes a gain over the margin rule, and this bounds the conclusion to the tested digest and eight-abstract

| Model       | Input             | 3 y  | 5 y  | 10 y |
|-------------|-------------------|------|------|------|
| Llama 70B   | digest, named     | .811 | .837 | .819 |
| Llama 70B   | abstracts, named  | .825 | .850 | .842 |
| Llama 70B   | abstracts, masked | .788 | .831 | .846 |
| Qwen 72B    | digest, named     | .831 | .864 | .875 |
| Margin rule | –                 | .907 | .897 | .867 |

Table 47: Prompted language models on the calendar cohort (AUROC).

prompts, not to text-reading methods in general.

## S Reproducibility

The artefacts contain the claim list with its fixed query for every claim, the retrieved PubMed identifiers with publication years and the retrieval audit, the extractor-assigned direction labels with their confidences (Llama-3.1-8B-Instruct, pinned revision, bfloat16, closed four-label read-out), the claim-cutoff benchmark with its pre-cutoff columns and post-cutoff fields under the tie convention of Equation (1), the 145-claim subset benchmark, the two external benchmarks as claim-cutoff units, the stored option probabilities of every language-model run with the model revision, settings and the complete prompts of Figures 2 and 7, every result table as CSV and JSON with a metadata block recording the sample, split, bootstrap (claim-clustered percentile, 2,000 resamples, seed 3) and decision rules, and the code: one library holding the feature builder, the leave-one-claim-out fitting, the metrics and the bootstrap; one script per experiment; and one launcher that regenerates every table. Randomness enters only through the seeded bootstrap resamplers, the seeded tree and network learners, and the seeded random draws of the cutoff designs and one-shot examples. Every numbered analysis checks its own consistency with the system at run time: the benchmark rebuilds byte-identically from the records and labels, the PMID-disjoint rebuild reproduces every unit when nothing is removed, and every additional experiment asserts that its Compact-7 scores equal those of the system. The audit analyses (numeric anchor against both annotations, by era and by anchor scale with the signed-anchor confusion matrices, agreement audit with the branch-free refits, the common-support intervention, its branch slices and the branch membership mask, the exact pair decomposition with per-pair contributions, cumulative and three-state endpoints, orientation audit, two-variable transfer, calendar cohort, stationary

nulls with the partition grid, invariance audit and Null D draws, wild cluster, clustered-ROC and re-fitting bootstraps, fold-level coefficients and slopes, class-by-bin and documented-reversal tabulations, and second annotation) are one script each over a frozen snapshot of the released benchmark, records and code, with a manifest recording input hashes, seeds, library versions and output files; each asserts at run time that its rebuild of the shipped columns is exact. Abstract text is not redistributed.

### Independent rebuild and reproducibility levels.

The release contains per-unit predictions and outcomes for every reported metric and interval, association-level labels from both pipelines (primary first-token scores and second-pipeline label probabilities) with identifiers and years, intersection labels, the calendar-cohort census, the external unit files, the adjudication frame and the protocol code. An independent rebuild from the released labels reproduces every unit and event count (1,012 units with 71, 52 and 55 events; calendar cohort 756/58, 625/45 and 492/35), matches the frozen feature builder on all 1,012 units and reproduces the reported AUROCs (0.948, 0.857, 0.898; 0.913, 0.920, 0.876). Three reproducibility claims are distinct and the release supports them to different degrees: reproduction of stored metrics (fully, from the per-unit files); reconstruction from released labels (counts, pooled directions, agreement under every definition, every endpoint, the stationary probabilities and the null controls); and re-execution of text-dependent methods, which requires the original abstracts, is possible for the source and calendar cells by retrieving the released PMIDs, and is not possible for the external corpora, which ship as unit-level label counts without identifiers. Two evaluation tracks follow from this last distinction: label-feature methods are eligible for every check of Section 4, text-reading methods for checks 1-4 only, since the external corpora ship without document inputs; text-reading methods can reach Tier A and a source-only Tier C marked as such, never Tier B, and an unavailable check is reported as unavailable, not as passed. The artefacts are supplied through the anonymous repository named in the main text (Appendix C.3 lists the rebuild levels it supports); the exact retrieval timestamp was not recorded but is bounded as described, and the candidate-list provenance is not reconstructible. The feature inventory that follows is a data dictionary for the released columns.

Table 48: Inventory of the 55 pre-cutoff variables, with the block used in the ablation and the operational definition of each column.  $R$  denotes the pre-cutoff records (with a parsed year and an extractor-assigned direction),  $t$  the cutoff year, and  $\text{eff}$  the set of effect estimates recovered from 95% confidence intervals in the title and abstract.  $\dagger$  and  $\ddagger$  mark inputs of a descriptive state rule that is released with the data but not used in this paper;  $*$  marks one of the seven Compact-7 inputs. All sentinels are 0.0 unless stated.

| Variable                           | Block         | Definition                                                                                                                                                                                                                                                                                                                   |
|------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pooled_direction <sup>*</sup>      | direction     | Operational pooled direction $p_t$ : the raw mean $(h - l)/(h + l)$ over signed pre-cutoff annotations (PROTECTIVE = -1, HARMFUL = +1; NULL excluded, UNCLEAR omitted) when at least eight signed records are present; otherwise 0 (the gate takes precedence). The ungated raw mean is $r_t$ for a non-empty signed sample. |
| agreement <sup>†*</sup>            | direction     | Consensus share over all resolved directions: if the null share is $\geq 0.50$ it is that null share, else $\max(p, k - p)/k$ over the $k$ signed votes ( $p$ = harmful); this is $a$ .                                                                                                                                      |
| agreement_recent <sup>†</sup>      | direction     | The same share restricted to records with year $\geq t - 6$ ; falls back to agreement when fewer than 4 signed recent votes exist; this is $a_r$ .                                                                                                                                                                           |
| dissent_recent <sup>†</sup>        | direction     | Fraction of signed votes with year $\geq t - 6$ disagreeing with the reference sign (the consensus verdict, or the overall majority sign when the verdict is 0); 0.0 under the same $< 4$ fallback; this is $\delta_r$ .                                                                                                     |
| agreement_contraction <sup>†</sup> | direction     | Majority share of the later half of the signed-vote sequence minus that of the earlier half, split at $\lfloor k/2 \rfloor$ in record order; this is $\gamma$ .                                                                                                                                                              |
| direction_trajectory               | direction     | OLS slope of the signed direction on publication year; 0.0 if fewer than 4 points or zero year range.                                                                                                                                                                                                                        |
| null_frac                          | direction     | Share of resolved directions that are NULL.                                                                                                                                                                                                                                                                                  |
| n_directional <sup>‡*</sup>        | direction     | Count of pre-cutoff records with a resolved (non-UNCLEAR) direction.                                                                                                                                                                                                                                                         |
| n_nonnull <sup>‡</sup>             | direction     | Count of pre-cutoff records with a signed (non-NULL) direction; the direction block is zeroed when this is $< 8$ .                                                                                                                                                                                                           |
| consensus_verdict                  | direction     | Sign of the consensus $\in \{-1, 0, +1\}$ ; 0 when the null share is $\geq 0.50$ or agreement $< 0.60$ .                                                                                                                                                                                                                     |
| has_consensus                      | direction     | $1[\text{agreement} \geq 0.60 \wedge \text{consensus\_verdict} \neq 0]$ .                                                                                                                                                                                                                                                    |
| tau2_DL                            | heterogeneity | DerSimonian-Laird $\hat{\tau}^2 = \max(0, (Q - (k - 1))/C)$ , $C = \sum w - \sum w^2 / \sum w$ , $w = 1/v$ , on the log-ratio or difference scale; 0.0 with fewer than 3 estimates.                                                                                                                                          |
| tau2_REML                          | heterogeneity | Iterative REML $\hat{\tau}^2$ (40 iterations, tolerance $10^{-8}$ ), initialised at tau2_DL.                                                                                                                                                                                                                                 |
| tau2_Hedges                        | heterogeneity | $\max(0, \text{Var}(y; \text{dof}=1) - \bar{v})$ , the method-of-moments estimator.                                                                                                                                                                                                                                          |
| mean_tau2                          | heterogeneity | Arithmetic mean of the three $\hat{\tau}^2$ estimates.                                                                                                                                                                                                                                                                       |
| i_squared                          | heterogeneity | $\max(0, (Q - (k - 1))/Q)$ from the same $Q$ ; 0.0 when $Q \leq 0$ or fewer than 3 estimates.                                                                                                                                                                                                                                |
| spread                             | heterogeneity | 90th minus 10th percentile of the effect estimates $y$ .                                                                                                                                                                                                                                                                     |
| decline_signal                     | decline       | $(\text{early} - \text{late})/\text{early}$ , where early and late are the mean $ y $ over the first and last $k = \max(2, \lfloor n/3 \rfloor)$ estimates ordered by year; needs at least 6 estimates.                                                                                                                      |
| early_magnitude                    | decline       | Mean $ y $ over the earliest $k$ effect estimates by year.                                                                                                                                                                                                                                                                   |
| late_magnitude                     | decline       | Mean $ y $ over the latest $k$ effect estimates by year.                                                                                                                                                                                                                                                                     |
| shrink_ratio                       | decline       | late/early.                                                                                                                                                                                                                                                                                                                  |
| synth_frac                         | design        | Fraction of pre-cutoff texts matching the evidence-synthesis pattern ( <i>meta-analysis*</i> , <i>systematic review</i> , <i>pooled analysis*</i> ).                                                                                                                                                                         |
| rct_frac                           | design        | Fraction matching the trial pattern ( <i>randomi[sz]ed</i> , <i>double-blind</i> , <i>placebo-controlled</i> , <i>clinical trial</i> ).                                                                                                                                                                                      |
| cohort_frac                        | design        | Fraction matching the observational pattern ( <i>cohort</i> , <i>prospective</i> , <i>longitudinal</i> , <i>follow-up</i> ).                                                                                                                                                                                                 |
| synth_trend                        | design        | OLS slope of the synthesis indicator on publication year.                                                                                                                                                                                                                                                                    |
| rct_trend                          | design        | OLS slope of the trial indicator on publication year.                                                                                                                                                                                                                                                                        |
| synth_recent                       | design        | Mean synthesis indicator over records with year $\geq t - 5$ ; 0.0 when that window is empty.                                                                                                                                                                                                                                |
| design_upgrade_frac                | design        | Mean synthesis indicator on year $\geq t - 5$ minus the mean on year $< t - 5$ , each term 0 when its window is empty.                                                                                                                                                                                                       |
| confounding_trend                  | design        | OLS slope on year of the adjustment indicator ( <i>adjust*</i> , <i>multivariable</i> , <i>confound*</i> , <i>propensity</i> , <i>covariate</i> ).                                                                                                                                                                           |
| n_pre_t <sup>*</sup>               | volume        | Number of pre-cutoff records $ R $ ; at least 40 by construction of the cutoff grid.                                                                                                                                                                                                                                         |
| n_records                          | volume        | $ R $ again; numerically identical to n_pre_t in every row.                                                                                                                                                                                                                                                                  |
| vol_last5                          | volume        | Count of pre-cutoff records with year $\geq t - 5$ .                                                                                                                                                                                                                                                                         |
| vol_growth                         | volume        | Add-one smoothed ratio $( \{y \geq t-5\}  + 1)/( \{t-10 \leq y < t-5\}  + 1)$ .                                                                                                                                                                                                                                              |
| volume_ratio                       | volume        | $( R  - h)/\max(h, 1)$ with $h = \lfloor  R /2 \rfloor$ : later-half to earlier-half record count.                                                                                                                                                                                                                           |

continued on next page

Table 48 continued from previous page

| Variable            | Block   | Definition                                                                                                                                                                                                                                                |
|---------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recency_density     | volume  | Fraction of pre-cutoff records with $\text{year} \geq t - 5$ .                                                                                                                                                                                            |
| accrual_rate_5yr    | volume  | Records per year over the final five pre-cutoff years, $ \{y \geq t - 5\} /5$ .                                                                                                                                                                           |
| span_years          | derived | $\max(\text{year}) - \min(\text{year})$ over the pre-cutoff records.                                                                                                                                                                                      |
| n_rr*               | derived | Number of usable effect estimates $ \text{eff} $ (confidence-interval pairs with $se > 10^{-6}$ and $ y  < 10$ ).                                                                                                                                         |
| n_rr_studies        | derived | $ \text{eff} $ again; identical to n_rr.                                                                                                                                                                                                                  |
| decline_sufficient  | derived | Gate flag for the decline block; constant 0.0 in the modelled benchmark.                                                                                                                                                                                  |
| naive_tsa           | derived | $\min(1, \sum_{\text{eff}} se^{-2}/400)$ : a naive trial-sequential information-fraction proxy; 0.0 when no estimate exists.                                                                                                                              |
| n_large_recent      | derived | Count of records with $\text{year} \geq t - 5$ whose text reports a large enrolment (a five-or-more-digit or comma-grouped count followed by <i>participants</i> , <i>subjects</i> , <i>patients</i> , <i>individuals</i> , <i>women</i> or <i>men</i> ). |
| n_ci                | derived | Number of 95% confidence-interval matches across all pre-cutoff texts, counted before the validity filter, so $n_{\text{ci}} \geq n_{\text{rr}}$ .                                                                                                        |
| ci_rate             | derived | $n_{\text{ci}}/\max( R , 1)$ : reported intervals per pre-cutoff record.                                                                                                                                                                                  |
| ci_width_mean       | derived | Mean of $2 \times 1.96 se$ over the effect estimates.                                                                                                                                                                                                     |
| ci_width_sd         | derived | Population standard deviation of the same interval widths.                                                                                                                                                                                                |
| ci_crosses_null     | derived | Fraction of effect estimates with $ y  < 1.96 se$ , i.e. intervals covering the null.                                                                                                                                                                     |
| val_mean*           | derived | Mean of the effect estimates $y$ on their working scale.                                                                                                                                                                                                  |
| val_sd*             | derived | Population standard deviation of $y$ .                                                                                                                                                                                                                    |
| val_spread          | derived | A copy of spread; identical in every row.                                                                                                                                                                                                                 |
| val_cv              | derived | $\text{val\_sd}/\max( \text{val\_mean} , 10^{-6})$ .                                                                                                                                                                                                      |
| n_numeric           | derived | $ \text{eff} $ once more; identical to n_rr.                                                                                                                                                                                                              |
| numeric_rate        | derived | Fraction of pre-cutoff records yielding at least one usable effect estimate.                                                                                                                                                                              |
| numeric_rate_recent | derived | The same fraction restricted to records with $\text{year} \geq t - 5$ ; 0.0 when that window is empty.                                                                                                                                                    |
| numeric_trend       | derived | OLS slope on year of the per-record indicator “this abstract yields an effect estimate”.                                                                                                                                                                  |